FORM 1C 2390 - 74

GEK-41053A

INSTRUCTIONS

MK II MS 7001

TURBINE CONTROL PANEL

REV. 1	REV 4	REV 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L M Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. A01 SH NO A00

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REV. 3	MADE BY L M Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. A02 SH. NO. A01

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REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. A04 SH. NO. A03

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REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 A03 SH. NO. A02 CONT. ON SH.

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REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. C01 SH. NO. B02

FORM 1C 2390 2-72

CARD PIN NOMENCLATURE

(THE FOLLOWING LIST WILL BE OF ASSISTANCE IN "READING" THE CARD SYMBOLS. THE ABBREVIATIONS USED ARE INTENDED AS CRYPTIC NOTATIONS AND ARE NOT NECESSARILY RIGOROUS.)

AL	-ALARM
ALM	-ALARM
ALR	-ACCELERATION LIMIT RELEASE
AB	-AMBIENT BIAS
ACC	-ACCELERATION
ANN	-ANNUNCIATOR
ACC	-ACCELERATION CONTROL
BOV	-BLOW OFF VALVE
C	-COUNTER OR CARRY
CALIB	-CALIBRATE
CL	-CLEAR
COMP	-COMPARE
DA	-DIFFERENTIAL ALARM
DN	-DOWN
DO	-DROP OUT
EXP	-EXPONENTIAL
ESTOP	-EMERGENCY STOP
FB	-FEEDBACK
FFN	-FEEDBACK, FUEL FLOW
FG	-FULLY ON GAS
FL	-FULLY ON LIQUID
FLT	-FAULT
FS	-FAST START
FS5	-FRAME SIZE 5
G	-GAS
GND	-GROUND
HP	-HIGH PRESSURE TURBINE
HR	-HIGH RANGE
INC	-INCREASE
INH	-INHIBIT
INN	-INPUT, NEGATIVE
INP	-INPUT, POSITIVE
ISO	-ISOCHRONOUS
LIQ	-LIQUID
LP	-LOW PRESSURE TURBINE
LSB	-LEAST SIGNIFICANT BIT
LVDT	-LINEAR VARIABLE DISPLACEMENT TRANSFORMER
M	-METER
MIN	-MINIMUM
MPU	-MAGNETIC PICK-UP
MSB	-MOST SIGNIFICANT BIT
MT	-MINIMUM TEST
N	-SPEED
NC	-SPEED CONTROL/NOZZLE CONTROL
NDO	-NORMAL DROP OUT
NHP	-SPEED, HIGH PRESSURE TURBINE
NLP	-SPEED, LOW PRESSURE TURBINE
NOZ	-NOZZLE
OC	-INVERSE COUNTER INPUT
OFBF	-INVERSE FEEDBACK FAULT
OFLT	-INVERSE FAULT
OR-EXO	-"OR" EXPANDER
OS-	-OVERSPEED
OTA,B	-OVERTEMPERATURE A, B
OUTA,B	-ALARM OUTPUT A, B
P	-PURGE
PCD	-PRESSURE, COMPRESSOR DISCHARGE
PCDN	-PRESSURE, COMPRESSOR DISCHARGE NORMALIZED
PK	-PEAK
PKOP	-PEAK OPERATION
PR/A	-PULSE RATE TO ANALOG
PR	-PEAK RESERVE
PRST	-PRESET
PROT	-PEAK RESERVE OVERTEMPERATURE
PU	-PICK UP
PUL	-PULSE
RA,B	-RANGE A, B, ETC.
RD	-RELAY DRIVER
REF	-REFERENCE
RM	-RUN MODE
RST	-RESET
SB	-SHUTDOWN BIAS
SC	-START-UP CONTROL
SCA,B	-SERVO, CURRENT LOOP A, B, ETC
SER	-SERVO
SFT	-SYSTEM FAULT TRIP
SIG	-SIGNAL
SJ	-SUMMING JUNCTION
SM	-START-UP MODE
SRT	-SERTO-TRIP
SP	-SET POINT
SRV	-SPEED RATIO VALVE
ST	-STEER
TC	-THERMOCOUPLE
TC ADJ	-TIME CONSTANT ADJUST
TP	-TIE POINT
TP	-TEST POINT
TX	-AVERAGE EXHAUST TEMPERATURE
UFD	-UNDER FREQUENCY DROP OUT
US	-UNDERSPEED
VCE	-VARIABLE CONTROL VOLTAGE
VXDR	-VOLTAGE, TRANSDUCER

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CARD PIN NOMENCLATURE
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 1 SH. NO. 001

SECTION A

INTRODUCTION

THE MKII MS7001 TURBINE CONTROL PANEL, ALONG WITH SENSORS AND OUTPUT SERVO VALVES, CONTROLS THE START-UP, PROTECTION AND OPERATION OF THE GENERAL ELECTRIC HEAVY-DUTY GAS TURBINE. THE PRIMARY FUNCTIONS PROVIDED BY THE SPEEDTRONIC* MKII ARE SHOWN IN FIGURE A-1. THE CONTROL SYSTEM CONSISTS OF THREE MAJOR LOOPS. THESE ARE START-UP, SPEED AND TEMPERATURE. THESE LOOPS ARE ILLUSTRATED IN FIGURE A-2. THE LOOPS ARE TIED TOGETHER INTO A MINIMUM VALUE GATE SO THAT THE LOOP CALLING FOR THE LEAST AMOUNT OF FUEL IS IN COMMAND. THE CONTROL SYSTEM IS DESIGNED TO PROVIDE MAXIMUM RUNNING RELIABILITY. REDUNDANT SENSORS ARE FREQUENTLY USED TO SUPPLY FEEDBACK SIGNALS. RELIABILITY IS FURTHER ENHANCED BY "REDUNDANCY BY ASSOCIATION". FOR EXAMPLE; IF THE SYSTEM IS OPERATING ON SPEED CONTROL, THE TEMPERATURE CONTROL WILL BACK IT UP.

SEQUENTIAL CIRCUITRY IS PROVIDED TO INTERROGATE THE TURBINE, GENERATOR, STARTING MEANS AND AUXILIARIES TO DETERMINE IF ALL SYSTEMS ARE NORMAL. THE RETURN SIGNALS FROM THESE DEVICES SERVE AS PERMISSIVES TO OPERATE THE TURBINE. THE SEQUENTIAL CIRCUITRY ALSO GENERATES LOGIC SIGNALS TO FACILITATE STARTING AND STOPPING THE TURBINE IN A PRESCRIBED MANNER. THESE LOGIC SIGNALS INCLUDE SPEED SIGNALS, DIGITAL SETPOINT CONTROL, LOAD CAPACITY SELECTION, STARTING MEANS CONTROL, TIMERS AND OTHERS.

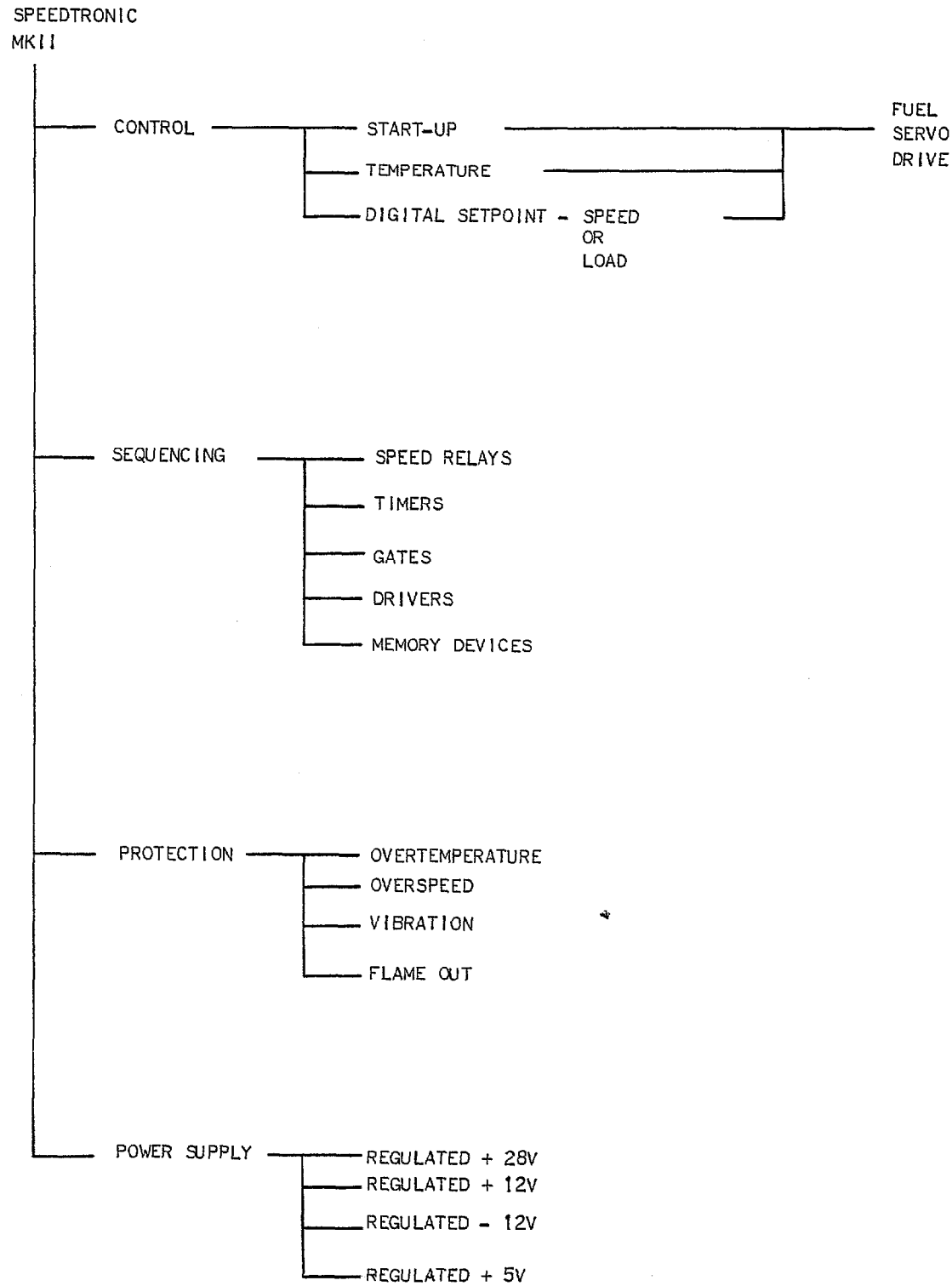
PROTECTION CIRCUITRY IS PROVIDED TO STOP THE TURBINE WHEN CRITICAL PARAMETERS ARE EXCEEDED OR CRITICAL EQUIPMENT FAILS. ALARM ANNUNCIATION IS PROVIDED TO INDICATE WHEN CRITICAL LEVELS ARE BEING APPROACHED AND TO INDICATE ABNORMAL EQUIPMENT OPERATION.

THE CONTROL PANEL CONTAINS ITS OWN HIGHLY RELIABLE SOLID STATE POWER SUPPLY. THE POWER SUPPLY CONSISTS OF A D-C TO D-C CONVERTER AND FINAL REGULATORS FOR P28, P12, P5.3, AND N12 VOLTS. THE CONVERTER OPERATES SATISFACTORILY WITH AN INPUT VOLTAGE OF 95 TO 140 VDC.

THE INTENT OF THESE INSTRUCTIONS IS FOR USE IN CONJUNCTION WITH THE (1) TURBINE CONTROL SPECIFICATION, (2) CONTROL PANEL ELEMENTARY DIAGRAM, (3) SCHEMATIC PIPING DIAGRAM, AND (4) OPERATING INSTRUCTIONS. THESE INSTRUCTIONS ARE TYPICAL AND DO NOT PURPORT TO COVER ALL DETAILS OR VARIATIONS THAT MAY ARISE ON A PARTICULAR JOB.

* TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS R.E.M	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SECTION A INTRODUCTION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 2 SH. NO. 1



PRIMARY FUNCTIONS OF THE
SPEEDTRONIC MKII
FIG. A-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM A-1	
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		PRIMARY FUNCTIONS	
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 3	SH. NO. 2

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REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED 4-9-75			FIRST MADE FOR			
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 801	SH. NO. A04

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C-3	SYSTEM START-UP CONTROL	43
C-4	SSKC CARD	44
C-5	SPEED/LOAD CONTROL	46
C-6	DIGITAL SETPOINT CARDS	47
C-7	SPEED CONTROL CARDS	50
C-8	ACCELERATION PORTION OF SSZB	51
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REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. B02 SH. NO. B01

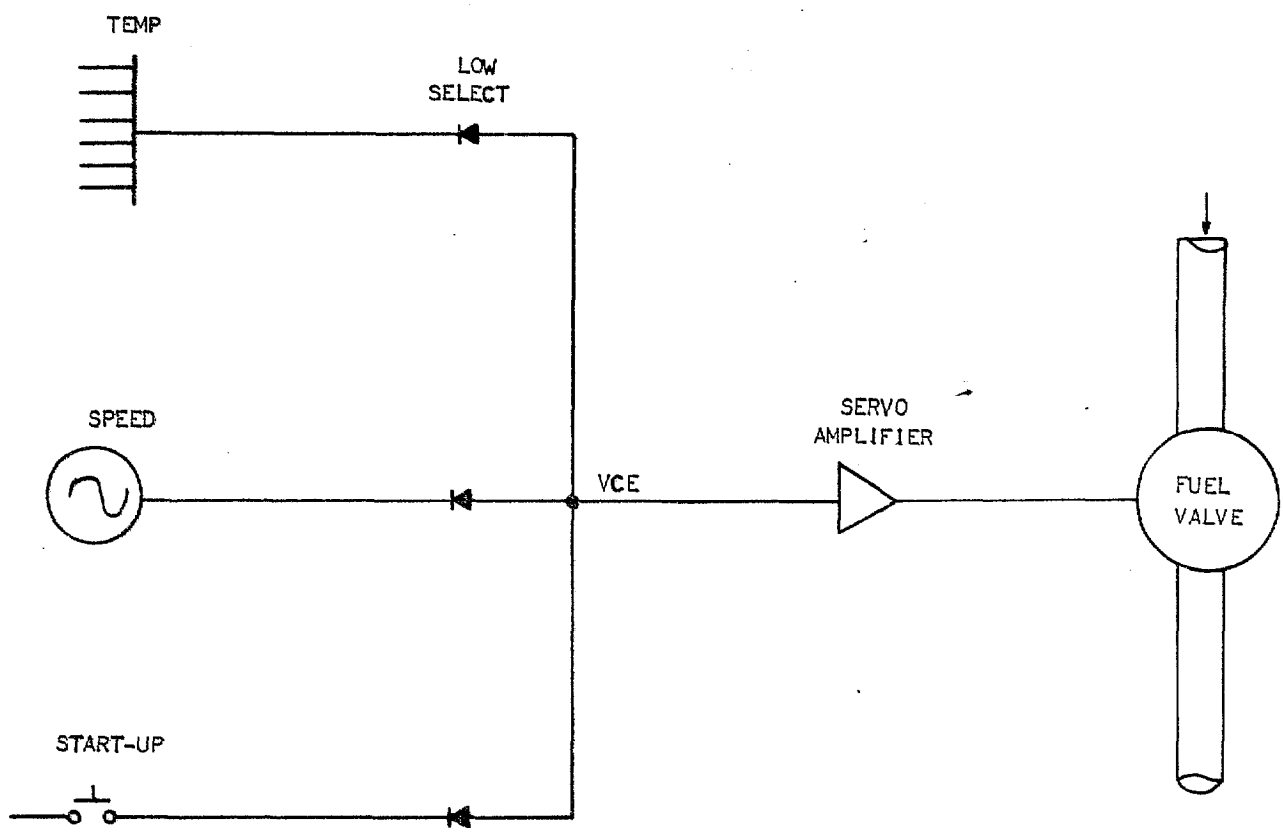
PUBLICATIONS AND REPAIR NOTES

THE INTENT OF THIS INSTRUCTION BOOK IS FOR USE IN CONJUNCTION WITH THE (1) TURBINE CONTROL SPECIFICATION, (2) CONTROL PANEL ELEMENTARY DIAGRAM, (3) SCHEMATIC PIPING DIAGRAM (4) OPERATING INSTRUCTIONS, AND (5) TURBINE CONTROL ELEMENTARY READING AID. THIS INSTRUCTION BOOK IS TYPICAL AND DOES NOT PURPORT TO COVER ALL DETAILS OR VARIATIONS THAT MAY ARISE ON A PARTICULAR APPLICATION.

WHEN TROUBLESHOOTING THE TURBINE CONTROL PANEL, THE TEST POINTS PROVIDED ON THE CALIBRATOR (IF AVAILABLE) AND CARD FRONTS SHOULD BE USED WHENEVER POSSIBLE TO AVOID DAMAGE TO THE CARDS THAT OFTEN RESULTS WHEN CONNECTING TEST PROBES TO THE PINS ON BACK OF THE SPEEDTRONIC PAGE. IF IT IS ABSOLUTELY NECESSARY TO TROUBLE-SHOOT ON THE BACK OF THE PAGE, INSULATED TEST PROBES SHOULD BE USED. RECOMMENDED PROCEDURES CONCERNING REMOVING OR REPLACING CARDS WITH POWER APPLIED IS GIVEN IN TECHNICAL BULLETIN 060 TB1-01-001.

FAILED CARDS AND MODULES SHOULD BE RETURNED TO THE GENERAL ELECTRIC COMPANY FOR REPAIRS. NO ATTEMPT SHOULD BE MADE TO REPAIR THE CARD IN THE FIELD. THE PROCEDURE FOR RETURNING CARDS AND MODULES CAN BE OBTAINED FROM YOUR NEAREST GENERAL ELECTRIC REPRESENTATIVE.

FORM IC 2390 2-72



MAJOR INPUT LOOPS

FIG. - A-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM A-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		MAJOR INPUT LOADS
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 4 SH. NO. 3

SECTION B

SEQUENTIAL CONTROL

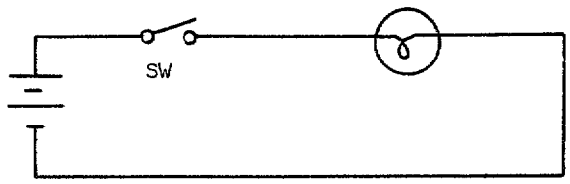
B-1 LOGIC INTERPRETATION

A BRIEF DISCUSSION OF LOGIC SIGNALS IS PRESENTED TO AID IN UNDERSTANDING THE SEQUENTIAL CIRCUITRY. A COMPARISON IS MADE BETWEEN RELAY LOGIC AND SOLID STATE LOGIC TO HELP THOSE FAMILIAR WITH RELAY LOGIC TO READ SOLID STATE LOGIC.

A LOGIC SIGNAL HAS ONLY TWO POSSIBLE STATES. THESE STATES HAVE VARIOUS NAMES IN DIFFERENT LITERATURE. IN THIS INSTRUCTION BOOK THE TWO STATES ARE DEFINED AS "ONE" ("1") AND "ZERO" ("0"). A "1" INDICATES THAT SOME EVENT HAS OCCURRED OR IS OCCURRING. A "0" INDICATES THAT SOME EVENT HAS NOT OCCURRED OR IS NOT OCCURRING. IN THE MKII SYSTEM A "1" STATE IS USUALLY REPRESENTED BY +5 VOLTS AND A "0" STATE IS USUALLY REPRESENTED BY ZERO VOLTS.

IF THE STATE OF SOME EVENT IS DESIGNATED BY USING THE VARIABLE A, THEN WHEN A EQUALS "1" THE EVENT IS OCCURRING OR HAS OCCURRED AND WHEN A EQUALS "0" THE EVENT IS NOT OCCURRING OR HAS NOT OCCURRED. FOR EXAMPLE, IN THE CIRCUIT BELOW THE EVENT IS DEFINED AS THE ILLUMINATION OF THE LIGHT, AND THE VARIABLE A IS USED TO REPRESENT THE STATE OF THE EVENT. WHEN THE SWITCH IS CLOSED, A IS EQUAL TO "1" (LAMP ILLUMINATED) AND WHEN THE SWITCH IS OPEN, A IS EQUAL TO "0" (LAMP IS OUT). IF A IS A "0" WHEN THE LAMP IS ON THEN IT WOULD BE AN INVERSE FUNCTION AND HAVE A BAR OVER IT. (REFER TO PAGE 7).

FOR LOGIC SYMBOLS COMING INTO THE TURBINE CONTROL PANEL FROM REMOTE LOCATIONS SUCH AS THE TURBINE, OR GENERATOR, A SIGNAL IS UNBARRED IF IT IS A LOGIC 1 WHEN THE DEVICE WHICH GENERATES IT IS AT ITS HIGHEST ENERGY LEVEL AND A "0" WHEN THE DEVICE IS AT ITS LOWEST ENERGY LEVEL. IF THE SIGNAL IS A "1" FOR THE LOWEST ENERGY LEVEL, ETC., THEN THE SIGNAL IS BARRED (EG. "$\bar{A}$").

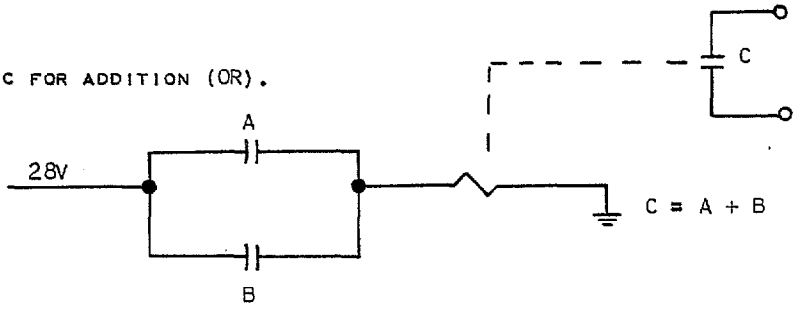


THE MATHEMATICS OF LOGIC SIGNALS IS REFERRED TO AS BOOLEAN ALGEBRA. A VERY BRIEF DISCUSSION OF BOOLEAN ALGEBRA IS GIVEN SUBSEQUENTLY. FOR EACH BOOLEAN OPERATION, BOTH RELAY LOGIC AND MKII LOGIC TO PERFORM THE OPERATION IS GIVEN.

(1) LOGIC ADDITION ("OR" FUNCTION): LOGIC ADDITION IS DESIGNATED BY THE SYMBOL "+". THE ADDITION OPERATION IS DEFINED AS:

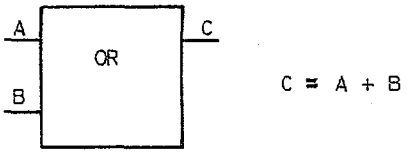
1 + 1 = 1 　　0 + 1 = 1
1 + 0 = 1 　　0 + 0 = 0

(A) RELAY LOGIC FOR ADDITION (OR).



RELAY C WILL BE PICKED UP IF EITHER CONTACT A OR CONTACT B IS CLOSED.

(B) DIRECTO-MATIC II SOLID STATE LOGIC FOR ADDITION (OR).

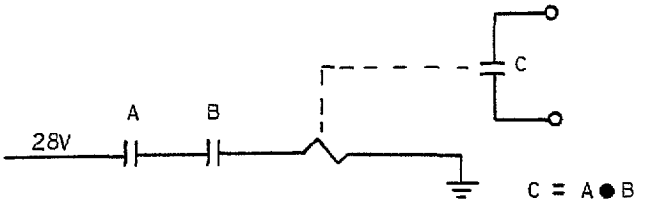


OUTPUT C WILL BE "1" (+5 VOLTS) IF EITHER INPUT A OR B IS A "1" (+5 VOLTS).

(2) LOGIC MULTIPLICATION ("AND" FUNCTION): LOGIC MULTIPLICATION IS DESIGNATED BY THE SYMBOL "●". THE MULTIPLICATION OPERATION IS DEFINED AS:

1●1 = 1 　　0●1 = 0
1●0 = 0 　　0●0 = 0

(A) RELAY IMPLEMENTATION.

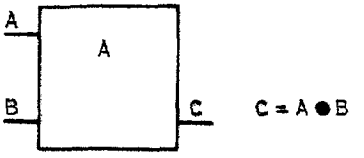


IF A AND B ARE CLOSED, RELAY C WILL BE PICKED UP.

NOTE: THE SYMBOL "●" IS OFTEN OMITTED, I.E. A●B = AB.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-1 LOGIC INTERPRETATION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 7 SH. NO. 6

(B) DIRECTO-MATIC II LOGIC IMPLEMENTATION

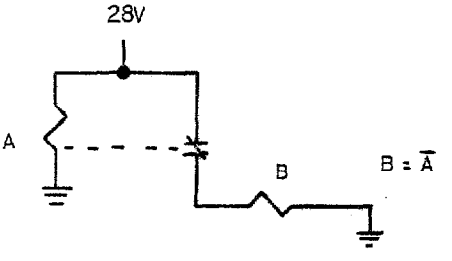


IF A = "1" AND B = "1" THEN C WILL BE "1". FOR ANY OTHER COMBINATION OF A AND B, C IS "0".

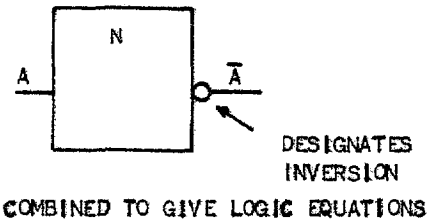
(3) LOGIC NEGATION ("NOT" FUNCTION): LOGIC NEGATION (ALSO CALLED INVERSE OR COMPLEMENT) IS DESIGNATED BY PLACING A BAR ABOVE THE VARIABLE. (FOR EXAMPLE: IF A IS THE VARIABLE, THEN $\overline{A}$ IS THE INVERSE OF A.) THE INVERSE OF A IS DEFINED AS FOLLOWS:

WHEN A = "1", $\overline{A}$ = "0"
WHEN A = "0", $\overline{A}$ = "1"

(A) RELAY IMPLEMENTATION



(B) DIRECTO-MATIC II IMPLEMENTATION



LOGIC OPERATIONS CAN BE COMBINED TO GIVE LOGIC EQUATIONS SUCH AS:

$A = (C \bullet B) + D$

NOTE: THE PRIORITY OF THE OPERATIONS ARE ●, +, NOT. THEREFORE, IF NO PARENTHESIS ARE INCLUDED, ALL THE "●" OPERATIONS ARE COMPLETED FIRST; THEN THE "+" AND THE NEGATION IS LAST. EXAMPLE: $A = C + B \bullet D$ IS UNDERSTOOD TO BE $A = C + (B \bullet D)$.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-1	
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LOGIC INTERPRETATION	
REV. 3	MADE BY L. M. Warley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 8	SH. NO. 7

LOGIC EQUATIONS CONTAINING TERMS SUCH AS $A = \overline{B \bullet D}$ CAN BE SIMPLIFIED USING THE FOLLOWING TECHNIQUE. TO SIMPLIFY AN EXPRESSION SUCH AS $\overline{B \bullet D}$, SUBSTITUTE "+" FOR "$\bullet$" AND COMPLEMENT EACH INDIVIDUAL VARIABLE. FOR EXAMPLE, THE EXPRESSION $\overline{B \bullet D}$ SIMPLIFIES TO $\overline{B} + \overline{D}$. LIKEWISE $\overline{B + D}$ SIMPLIFIES TO $\overline{B} \bullet \overline{D}$. THIS PROCEDURE CAN BE EXTENDED TO MORE COMPLEX EXPRESSIONS. THE FOLLOWING EXAMPLE ILLUSTRATES THE TECHNIQUE USED.

EXAMPLE:

$$E = \overline{\overline{A \bullet B} \bullet (C + D)}$$

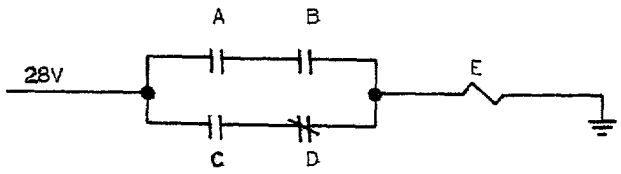
(1) $E = \overline{\overline{A \bullet B} + \overline{(C + D)}}$

(2) $E = A \bullet B + \overline{(C + D)}$

(3) $E = A \bullet B + (\overline{C} \bullet \overline{D})$

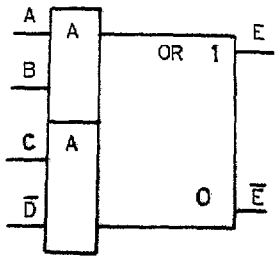
THE IMPLEMENTATION OF LOGIC EQUATIONS USING BOTH RELAY AND DIRECTO-MATIC II LOGIC IS ILLUSTRATED BELOW.

(A) RELAY:



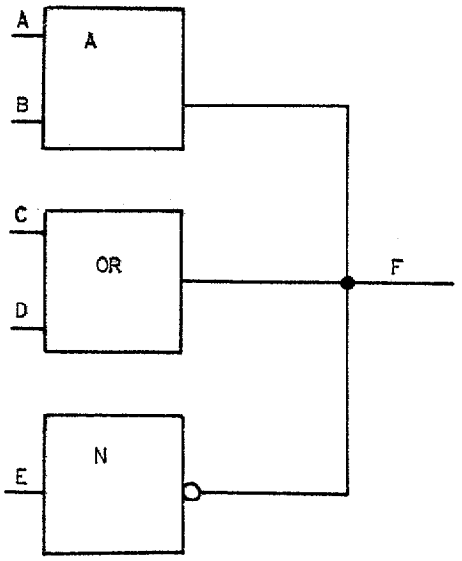
$$E = (A \bullet B) + (C \bullet \overline{D})$$

(B) DIRECTO-MATIC II LOGIC



$$E = A \bullet B + C \bullet \overline{D}$$

THE 1 AND 0 LOCATED INSIDE THE SYMBOL ADJACENT TO OUTPUTS E AND $\overline{E}$ RESPECTIVELY ARE OFTEN USED TO DESIGNATE THE NORMAL AND INVERSE OUTPUTS. THE CIRCUIT IS SAID TO BE IN THE TRUE STATE WHEN THE OUTPUT ADJACENT TO 1 IS A LOGIC ONE. IT IS SAID TO BE IN THE FALSE STATE WHEN THE OUTPUT ADJACENT TO 0 IS A LOGIC ONE. IT SHOULD BE NOTED HERE THAT IN THE CASE WHERE THE OUTPUTS OF MORE THAN ONE LOGIC ELEMENT ARE TIED TOGETHER, A "0" WILL DOMINATE OVER A "1". THE FOLLOWING CIRCUIT ILLUSTRATES THIS.



F IS EQUAL TO "0" IF ANY OF THE FOLLOWING CONDITIONS ARE SATISFIED.

(1) A + B = 0

(2) C ● D = 0

(3) E = 1

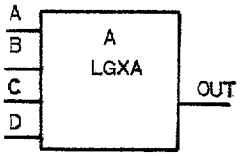
PRECAUTION: A "1" CAN BE FORCED TO A "0" BY JUMPERING THE SIGNAL TO COMMON; HOWEVER, TRYING TO FORCE A "0" TO A "1" BY JUMPERING THE SIGNAL TO +5.3 VOLTS COULD RESULT IN DAMAGE TO THE CIRCUIT.

B-2 DIRECTO-MATIC II LOGIC CARDS AND SYMBOLS

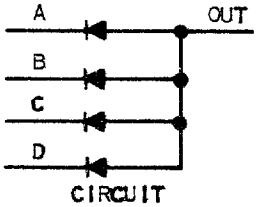
THIS SECTION WILL FAMILIARIZE THE READER WITH DIRECTO-MATIC II LOGIC CARDS. FOR SPECIFIC INFORMATION CONSULT THE CARD APPLICATION DATA BOOK.

NOTE THAT THE PRESENCE OF A " ●" AT THE INPUT TERMINAL OF AN ELEMENT DENOTES THE INPUTS ABILITY TO ACCEPT EITHER NORMAL LOGIC OUTPUTS OR THE OUTPUT OF AN LGXA.

IC3600LGXA - "AND" GATE EXPANDER



9 PER CARD

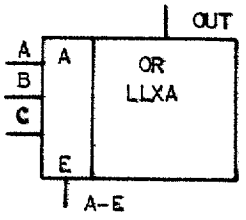


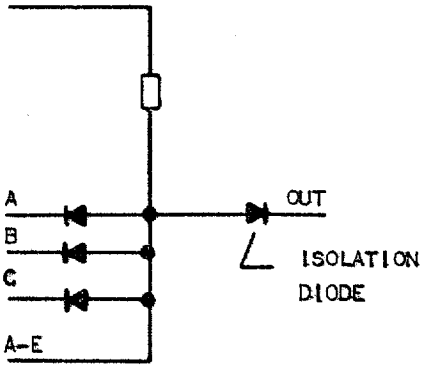
CIRCUIT

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2 LOGIC CARDS AND SYMBOLS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 10 SH. NO. 9

THE LGXA PROVIDES A CLUSTER OF DIODES FOR EXPANDING THE "AND" INPUTS OF MANY DIRECTO-MATIC II LOGIC CARDS. AN ILLUSTRATION OF HOW THE LGXA IS USED IS GIVEN IN THE LGDA DISCUSSION. WHEN THE OUT TERMINAL OF THE LGXA IS CONNECTED TO THE A-E INPUT OF ANOTHER LOGIC ELEMENT, A "0" ON ANY INPUT WILL PULL "OUT" TO "0". A "1" ON ALL INPUTS WILL ALLOW "OUT" TO GO TO A "1".

IC3600LLXA - "OR" LOGIC EXPANDER



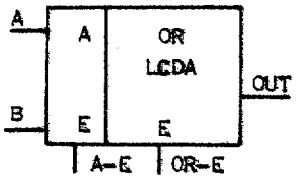


9 CIRCUITS PER CARD

THE LLXA PROVIDES CIRCUITRY FOR INPUT EXPANSION OF LOGIC ELEMENTS HAVING AN "OR-E" INPUT. IF ALL INPUTS ARE "1", THE ISOLATION DIODE WILL BE FORWARD BIASED AND PULL "OR-E" INPUT OF CONNECTED LOGIC ELEMENT TO A "1" WHICH MAKES THE NORMAL OUTPUT OF THAT ELEMENT A "1". IF ANY INPUT IS A "0", THE ISOLATION DIODE WILL BE REVERSED BIASED AND ALLOW THE OUTPUT OF THE CONNECTED ELEMENT TO BE CONTROLLED BY ITS NORMAL INPUTS. AN "A-E" INPUT IS PROVIDED FOR EXPANDING THE "AND" INPUTS USING A LGXA. THE USE OF THE LLXA WILL BE ILLUSTRATED IN THE LGDA DISCUSSION.

NOTE: FOR ALL DIRECTO-MATIC II CARDS, AN UNUSED INPUT LEFT OPEN IS RECOGNIZED AS A "1" EXCEPT FOR OR EXPANDER INPUTS.

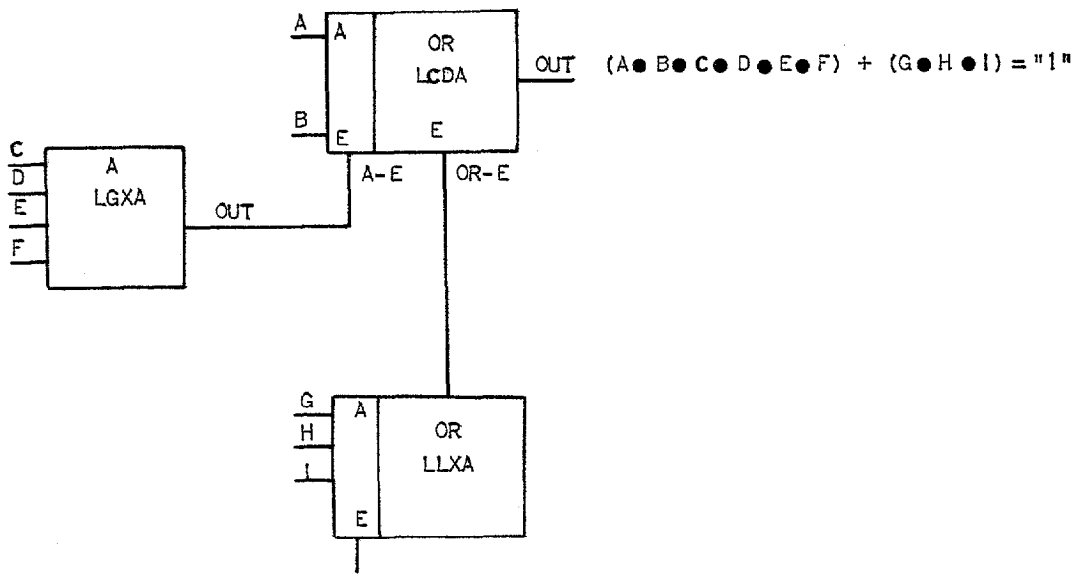
IC3600LGDA - CLOCK DRIVER



9 CIRCUITS PER CARD

THE LGDA CLOCK DRIVER CONTAINS A POWER AMPLIFIER WHICH WILL DRIVE 32 "L" LOGIC LOADS. IN ADDITION TO THE TWO NORMAL "AND" INPUTS, "A-E" AND "OR-E" INPUTS ARE PROVIDED. IF NO CONNECTION IS MADE TO "A-E" OR "OR-E," THE CIRCUIT PERFORMS AS A NORMAL TWO INPUT "AND" GATE WITH POWER AMPLIFIER. AN ILLUSTRATION OF HOW THE "A-E" AND "OR-E" INPUTS COULD BE USED IS ILLUSTRATED AND FOLLOWED BY ITS TRUTH TABLE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS LEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2 LOGIC CARDS AND SYMBOLS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY L M Wesley			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 11 SH. NO. 10



TRUTH TABLE*

A	B	C	D	E	F	G	H	I	OUT
0	0	0	1	1	1	0	1	1	0
X	X	X	1	1	1	1	1	1	1
0	0	1	1	1	1	0	1	1	0
0	1	1	1	1	1	0	1	1	0
1	0	0	1	1	1	0	1	1	0
1	0	1	1	1	1	0	1	1	0
1	1	0	1	1	1	0	1	1	0
1	1	1	1	1	1	0	1	1	1
1	1	1	1	1	1	1	1	1	1

*IN ORDER TO SIMPLIFY TABLE, ASSUME INPUTS H,I,D,E AND F ARE ALWAYS "1". I.E. ONLY INPUTS A,B,C AND G CAN CHANGE STATES.

X EQUALS "DON'T CARE" IF A "1" OR "0".

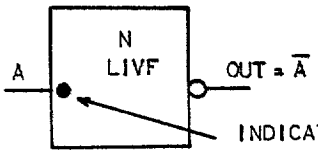
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR
REV. 3	MADE BY L.M. Worley			I.C. NO.

GENERAL ELECTRIC
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM B-2
LOGIC CARDS AND SYMBOLS
GEK-41053
CONT. ON SH. 12 SH. NO. 11

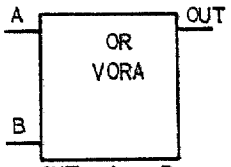
A B C D E F G H J K L

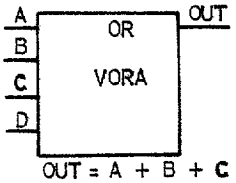
IC3600 LIVF - INVERTER

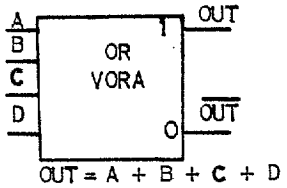


THE LIVF PROVIDES THE FUNCTION OF LOGIC INVERSION. A "1" IN WILL GIVE A "0" OUT AND A "0" IN WILL GIVE A "1" OUT. THE INPUT WILL RECEIVE A HIGH LEVEL INPUT ("0"=1.5 VOLTS MAX. AND "1"=4.75 VOLTS MIN.) WHICH PERMITS EXPANDING THE INPUTS DIRECTLY BY USING AN LGXA. THE SYMBOL "●" ADJACENT TO THE INPUT INDICATES THAT IT IS A HIGH LEVEL INPUT ELEMENT.

IC3600 VORA - LOGIC OR

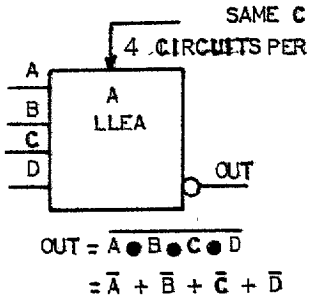


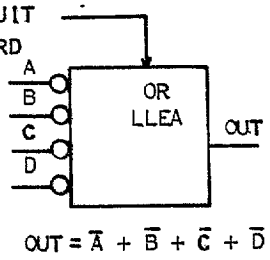


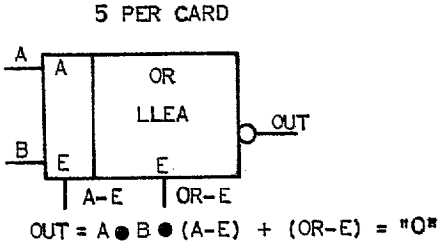


THIS CARD PROVIDES A TRUE "OR" FUNCTION. A "1" ON ANY INPUT WILL GIVE "1" ON "OUT". TWO CIRCUITS PER CARD PROVIDE AN INVERSE OUTPUT, "$\overline{OUT}$". FOR THESE CIRCUITS, A "1" ON ANY INPUT WILL GIVE A "1" ON "OUT" AND A "0" ON "$\overline{OUT}$". IF ALL INPUTS ARE "0", "OUT" IS "0" AND "$\overline{OUT}$" IS "1".

IC3600LLEA - LOGIC ELEMENT



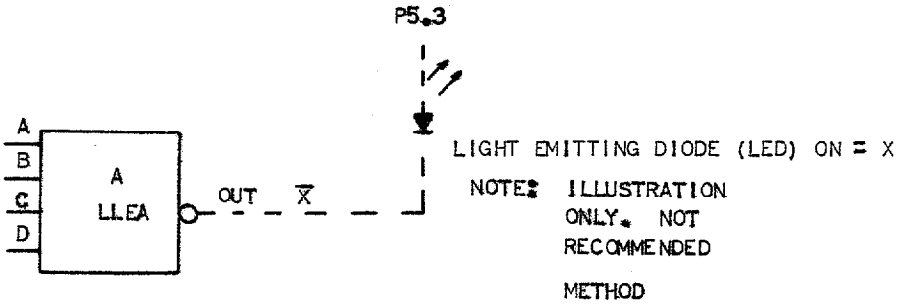




REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L.M. Worley			I.C. NO.		GEK-41053
						CONT. ON SH. 13 SH. NO. 12

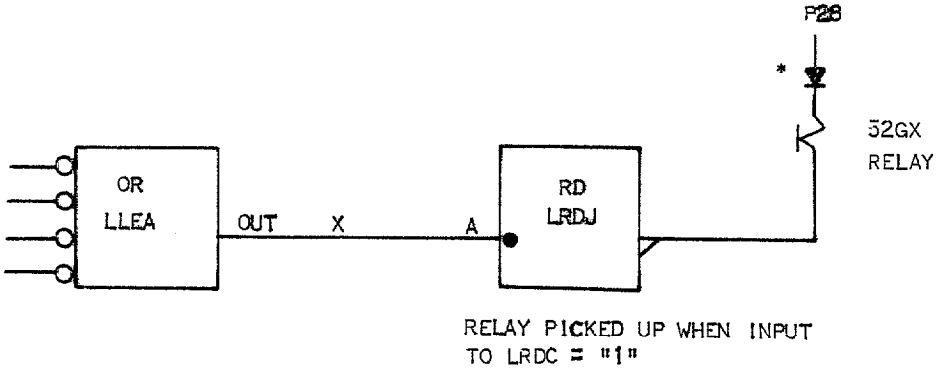
THE LLEA PERFORMS THE NAND (NOT AND) FUNCTION. FIVE CIRCUITS PER CARD HAVE "A-E" AND "OR-E" EXPANDER INPUTS FOR CONNECTION TO LGXA AND LLXA RESPECTIVELY. THE REMAINING 4 CIRCUITS PER CARD ARE TYPICAL NAND GATES. EITHER OF THE TWO LEFT MOST SYMBOLS ABOVE MAY BE USED FOR THESE CIRCUITS. THE SYMBOL USED IS DETERMINED BY THE LOGIC LEVEL DESIRED AT "OUT" TO REPRESENT THE ACTIVE CONDITION OF THE INPUTS. THE DESIRED LOGIC LEVEL IS DEFINED BY THE OUTPUT SIGNAL NOMENCLATURE. THE FOLLOWING EXAMPLES WILL ILLUSTRATE THIS.

EXAMPLE 1:



THE ACTIVE CONDITION IS WHEN THE LED IS ON. IF A "0" IS DESIRED TO REPRESENT THE ACTIVE CONDITION, THE ABOVE SYMBOL IS USED AND THE OUTPUT IS LABELED "$\overline{X}$." THE DESIRED LOGIC LEVEL IS DICTATED BY THE CIRCUIT THAT IS BEING DRIVEN BY THE LLEA.

EXAMPLE 2:



* USED FOR ILLUSTRATION ONLY; NOT NECESSARILY RECOMMENDED METHOD.

THE ACTIVE CONDITION IS CIRCUIT BREAKER 52GX CLOSED. IF A "1" IS DESIRED AT "OUT" TO REPRESENT THE ACTIVE CONDITION AND PICK UP THE RELAY, THIS SYMBOL IS USED AND THE OUTPUT OF THE LLEA LABELED 52GX.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L. M. Worley			I.C. NO.		GEK-41053 CONT. ON SH. 14 SH. NO. 13

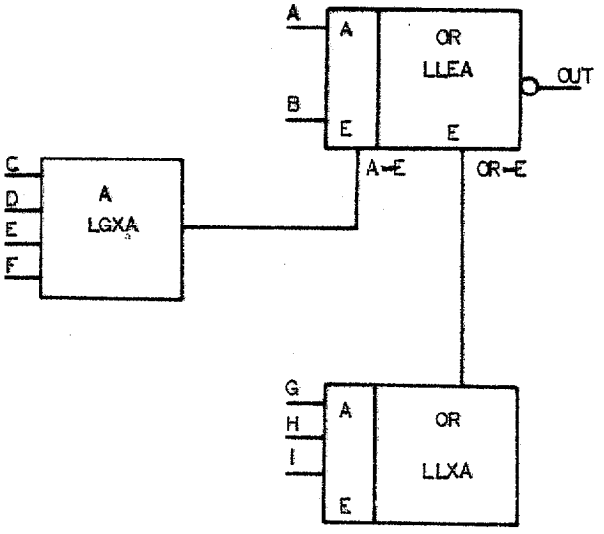
LLEA TRUTH TABLE

FOR CIRCUITS WITHOUT EXPANDER INPUTS

A	B	C	D	OUT
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 15 SH. NO. 14

THE FOLLOWING CIRCUIT ILLUSTRATES THE USE OF THE EXPANDER INPUTS. THE USE OF BOTH "A-E" AND "OR-E" ARE ILLUSTRATED; HOWEVER, EITHER MAY BE USED SEPARATELY.



TRUTH TABLE FOR EXPANDER CIRCUIT

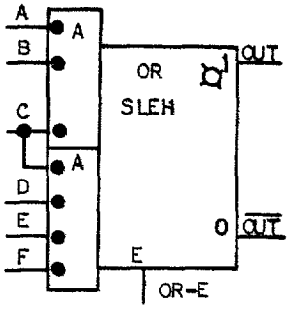
(ASSUME D, E, F, H, I ARE "1" AT ALL TIMES)

A	B	C	G	OUT
0	0	0	0	1
X	X	X	1	0
0	0	1	0	1
0	X	0	0	1
0	1	1	0	1
X	0	0	0	1
X	0	1	0	1
1	1	0	0	1
1	1	1		0

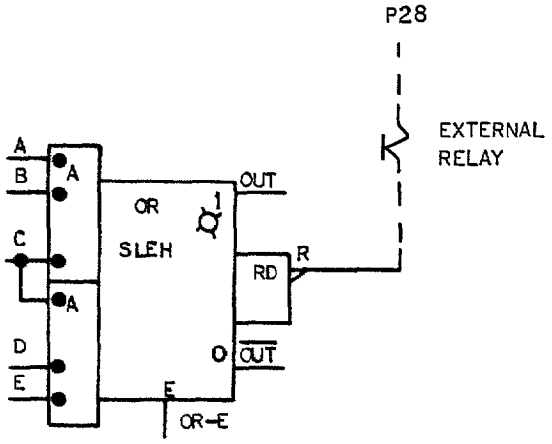
X EQUALS "DON'T CARE" IF "1" OR "0".

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 16 SH. NO. 15

IC3600SLEH - LOGIC ELEMENT



2 CIRCUITS PER CARD



3 CIRCUITS PER CARD

$OUT = A \bullet B \bullet C + D \bullet E \bullet F \bullet C$

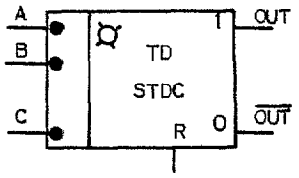
$\overline{OUT} = (\bar{A} + \bar{B} + \bar{C}) \bullet (\bar{D} + \bar{E} + \bar{F} + \bar{C})$

$OUT = A \bullet B \bullet C + D \bullet E \bullet C$

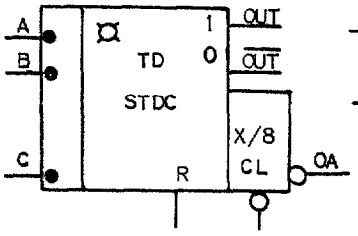
$\overline{OUT} = (\bar{A} + \bar{B} + \bar{C}) \bullet (\bar{D} + \bar{E} + \bar{C})$

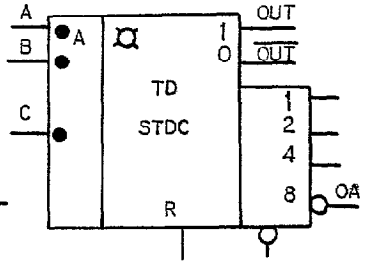
THE BASIC SLEH CIRCUIT CONSISTS OF AN "OR" LOGIC ELEMENT GATED BY TWO "AND" GATES. IF EITHER "AND" GATE HAS ALL "1" INPUTS, OUT WILL BE A "1". AN OR-E EXPANDER INPUT IS PROVIDED FOR USE WITH THE LLXA. THREE CIRCUITS PER SLEH CARD PROVIDE A RELAY DRIVER OUTPUT. THE RELAY DRIVER PROVIDES A CLOSURE TO COMMON WHEN OUT IS A "1". THE INPUTS WILL RECEIVE HIGH LEVEL SIGNALS ("0" = 1.5 VOLTS MAX. AND "1" = 4.75 VOLTS MIN.) WHICH ALLOWS THE DIRECT USE OF LGXA EXPANDER GATES ON THE NORMAL INPUTS.

IC3600STDC - TIME DELAY



3 CIRCUITS PER CARD



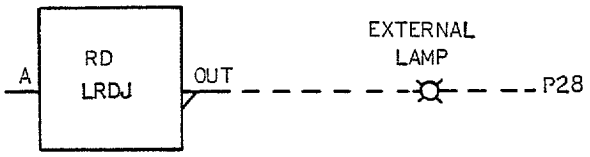


1 CIRCUIT PER CARD

THE STDC CARD PROVIDES A TIME DELAY FUNCTION OF ONE SECOND TO FIVE MINUTES. THE TIMING IS INITIATED BY ALL INPUTS BEING A "1". IF ALL INPUTS REMAIN A "1", OUT BECOMES A "1" AND $\overline{OUT}$ A "0" AFTER THE TIME DELAY SET BY THE ADJUSTMENT POTENTIOMETER. OUT WILL GO FROM A "1" TO A "0" IMMEDIATELY IF ANY INPUT GOES TO A "0". THE LED WILL BE ON FROM THE INSTANT TIMING IS INITIATED UNTIL OUT BECOMES A "1".

THE TIME DELAY PROVIDED IS DIVIDED IN TWO RANGES TO GIVE MORE ADJUSTMENT RESOLUTION. THE SHORT RANGE (PIN R OPEN) IS ADJUSTABLE FROM 1 TO 25 SECONDS. THE LONG RANGE (PIN R TO N12V) IS ADJUSTABLE FROM APPROXIMATELY 20 SECONDS TO 5 MINUTES. THERE ARE FOUR CIRCUITS PER CARD. ONE CIRCUIT PROVIDES AN EIGHT COUNT COUNTER FOR ADDITIONAL TIME DELAY. BY CONNECTING OUT OF THIS CIRCUIT TO ONE OF THE INPUTS, AN OA TO ANOTHER, THE TIMER WILL CYCLE EIGHT TIMES BEFORE STOPPING WITH OA EQUAL "0". THIS WILL PROVIDE A DELAY OF UP TO 40 MINUTES.

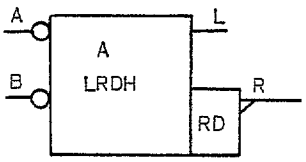
IC3600 LRDJ - RELAY DRIVER



16 CIRCUITS PER CARD

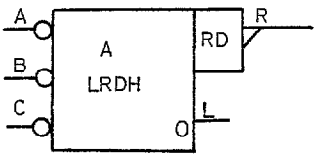
THE LRDJ CIRCUIT PROVIDES A CLOSURE TO COMMON IN RESPONSE TO A "1" INPUT, AN OPEN CIRCUIT IN RESPONSE TO A "0" INPUT. THE CIRCUIT IS INTENDED TO DRIVE A LOAD SUCH AS A RELAY CONNECTED BETWEEN OUT AND POSITIVE SUPPLY VOLTAGE NOT GREATER THAN P28. PRIMARY USE IS FOR A LAMP DRIVER. IT CANNOT BE USED AS A RELAY DRIVER WITHOUT DIODE SUPPRESSION.

IC3600 LRDH - RELAY DRIVER



$L = A \bullet B$

5 PER CARD



$L = \overline{A \bullet B \bullet C}$

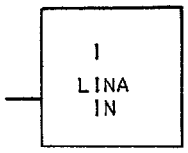
$L = A + B + C$

2 PER CARD

THESE CIRCUITS PROVIDE A CLOSURE TO COMMON FOR A LOAD BETWEEN P28 AND THE "R" OUTPUT IN RESPONSE TO PROPER INPUT LOGIC. IN ADDITION, THEY ALSO PROVIDE A LOGIC OUTPUT SIGNAL AT "L" IN RESPONSE TO THE INPUT LOGIC. CIRCUITS 1 THRU 5 REQUIRE BOTH INPUTS TO BE A LOGIC "0" TO PROVIDE CLOSURE TO COMMON AND TO GENERATE A LOGIC "1" AT THE OUTPUT. CIRCUITS 6 AND 7 ARE SIMILAR TO 1 THRU 5 EXCEPT THAT A 3 INPUT "AND" GATE IS PROVIDED INSTEAD OF A 2-INPUT GATE AND THE LOGIC OUTPUT IS A "0" WHEN CLOSURE TO COMMON IS PROVIDED. DIODE SUPPRESSION IS PROVIDED ON THE CARD AND IS NOT REQUIRED ON THE RELAY.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L. M. Worley			I.C. NO.		GEK-41053
						CONT. ON SH. 18 SH. NO. 17

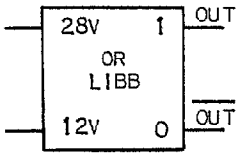
IC3600 LINA - LIGHT INDICATOR



16 CIRCUITS PER CARD

THE LINA INCLUDES A LIGHT AMPLFIER AND AN INDICATING LIGHT. THE LIGHT IS ON WHEN THE INPUT IS A "1". INPUTS FOR UNUSED CIRCUITS MUST BE TIED TO COMMON TO KEEP THE LIGHT OFF.

IC3600 LIBB - INPUT BUFFER



9 CIRCUITS PER CARD

THE LIBB CIRCUIT CONVERTS RELAY CONTACT INPUTS TO LOGIC LEVELS. INPUTS ARE PROVIDED FOR CONTACT CLOSURE TO EITHER 28 VOLTS OR+12 VOLTS. WHEN THE CONTACT TO THE SUPPLY IS CLOSED, OUT IS A "1" AND $\overline{\text{OUT}}$ IS A "0". THE P28V INPUT SHOULD NOT BE USED OFF PAGE. WHEN DESIRED TO USE 28 VOLT INPUT OFF PAGE, THE INPUT FILTER SHOULD BE USED.

NOTE: THE LIBA CARD IS FUNCTIONALLY THE SAME AS THE LIBB BUT HAS DIFFERENT RATINGS AND TIMING. DO NOT INTERCHANGE CARDS.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RCM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LOGIC CARDS AND SYMBOLS
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 19 SH. NO. 18

B-3 ANNUNCIATOR

THE ANNUNCIATOR ON THE MK II MS7001 TURBINE CONTROL PANEL PROVIDES THE CIRCUITRY TO ANNUNCIATE UP TO 60 ALARM CONDITIONS. THE ALARMS ARE LATCHING AND WILL REMAIN ON, EVEN IF THE ALARM CONDITION IS MOMENTARY, UNTIL CLEARED BY A PUSHBUTTON ON THE PANEL. AN ALARM MAY BE ANNUNCIATED BY THE SOUNDING OF A BELL AND THE FLASHING OF A LIGHT.

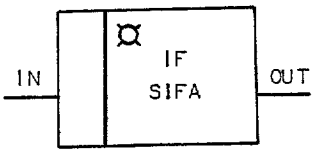
THE ALARM INDICATORS ARE REFERRED TO AS DROPS. SPECIAL FEATURES, WHICH WILL BE DISCUSSED SUBSEQUENTLY, ARE PROVIDED TO HANDLE MULTIPLE ALARMS. A BLOCK DIAGRAM IS GIVEN IN FIGURE B-1. IT CONSISTS OF ONE ANNUNCIATOR MASTER CARD (VANB) AND UP TO 15 ANNUNCIATOR CARDS (VANA). EACH VANA CARD CONTAINS FOUR CIRCUITS; THEREFORE, UP TO 60 DROPS ARE AVAILABLE. THE STANDARD SYSTEM NORMALLY CONSISTS OF 60 DROPS.

THERE ARE THREE OPERATOR PUSHBUTTONS ASSOCIATED WITH THE ANNUNCIATOR. (1) THE "LAMP TEST" PUSHBUTTON PROVIDES A MEANS TO CHECK FOR BURNED OUT BULBS AND FAULTY LAMP DRIVERS. (2) THE "ALARM RESET" PUSHBUTTON WILL SILENCE THE ALARM BELL AND CAUSE THE FLASHING LIGHT TO BECOME STEADY ON. IT ALSO ENABLES AN ADDITIONAL ALARM TO COME ON FLASHING. (3) THE "LAMP RESET" PUSHBUTTON WILL CLEAR THE LIGHT IF THE ALARM CONDITION HAS BEEN REMOVED. AN AUTOMATIC LAMP RESET OPTION IS AVAILABLE TO AUTOMATICALLY RESET THE LAMP WHEN THE ALARM CONDITION IS REMOVED.

THE FLOW CHART (SH. 22) IS INCLUDED TO ILLUSTRATE SOME TYPICAL SEQUENCES THAT MAY OCCUR IN THE CASE OF MULTIPLE ALARM CONDITIONS. ALARMS MUST OCCUR AT LEAST 36 MILLESECONDS APART FOR THE ANNUNCIATOR TO DETERMINE WHICH OCCURRED FIRST. IF THERE IS LESS THAN 36 MILLESECONDS BETWEEN ALARMS, BOTH WILL COME ON FLASHING.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	B-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		ANNUNCIATOR	
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 21	SH. NO. 20

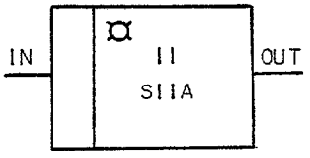
IC3606 SIFA - NON - ISOLATED INPUT FILTER



9 CIRCUITS PER CARD

THE SIFA IS DESIGNED TO RECEIVE A 28 VOLT SIGNAL, FILTER THE SIGNAL, AND PROVIDE DRIVE TO THE 12V INPUT OF THE LIBB CARD. AN LED IS ON WHEN THE INPUT IS AT 28 VOLTS. THIS CIRCUIT IS USED TO RECEIVE OFF PAGE SIGNALS.

IC3606 SIIA - ISOLATED INPUT FILTER

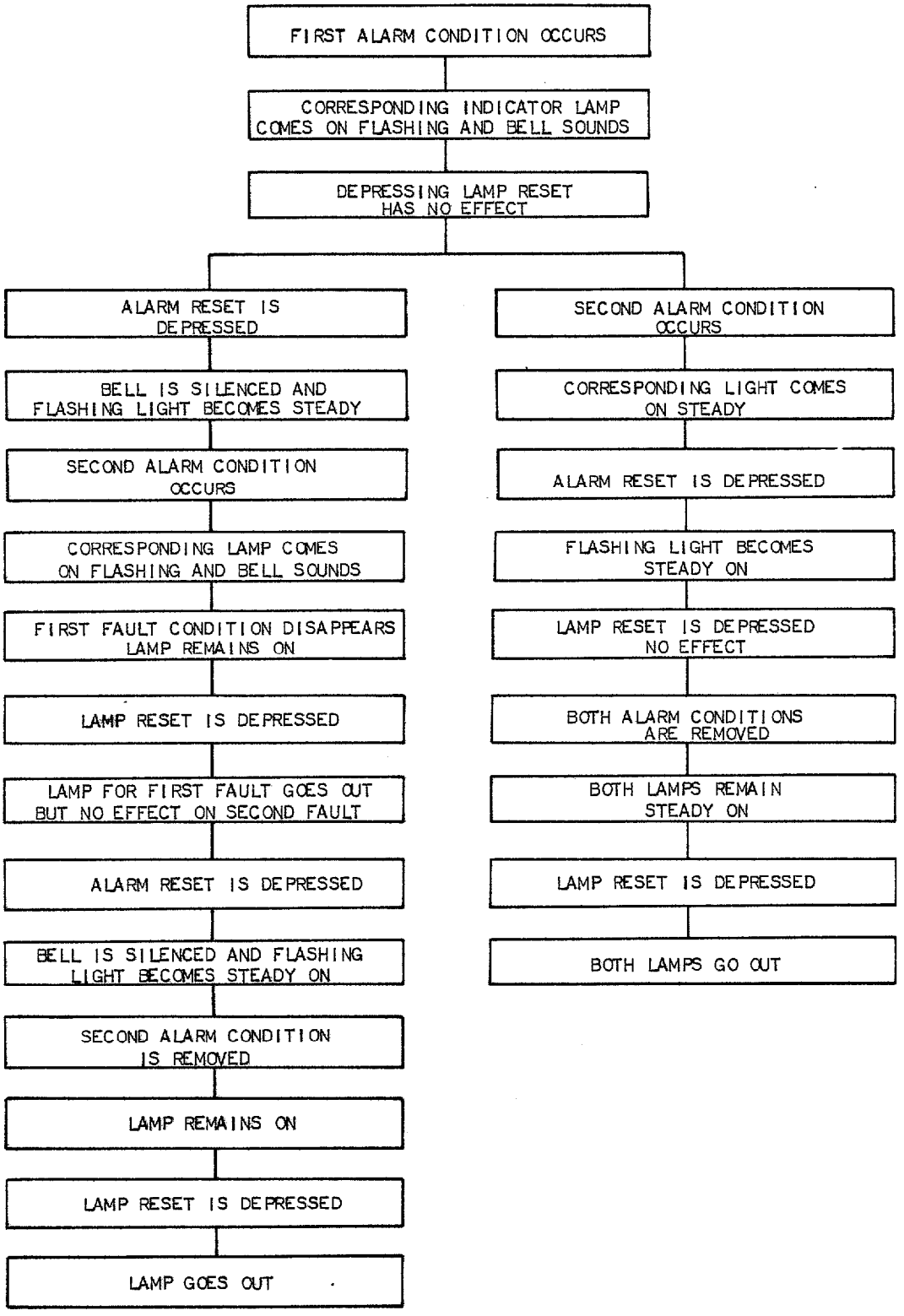


9 CIRCUITS PER CARD

THE SIIA PROVIDES ISOLATION AND FILTERING OF HIGH LEVEL (125 VOLTS DC OR 115 VOLTS AC) SIGNALS. THE OUTPUT IS DESIGNED TO DRIVE THE 12 VOLT INPUT OF THE LIBB CARD.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS LEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2 LOGIC CARDS AND SYMBOLS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 20 SH. NO. 19

FORM 1C 2390 2-72

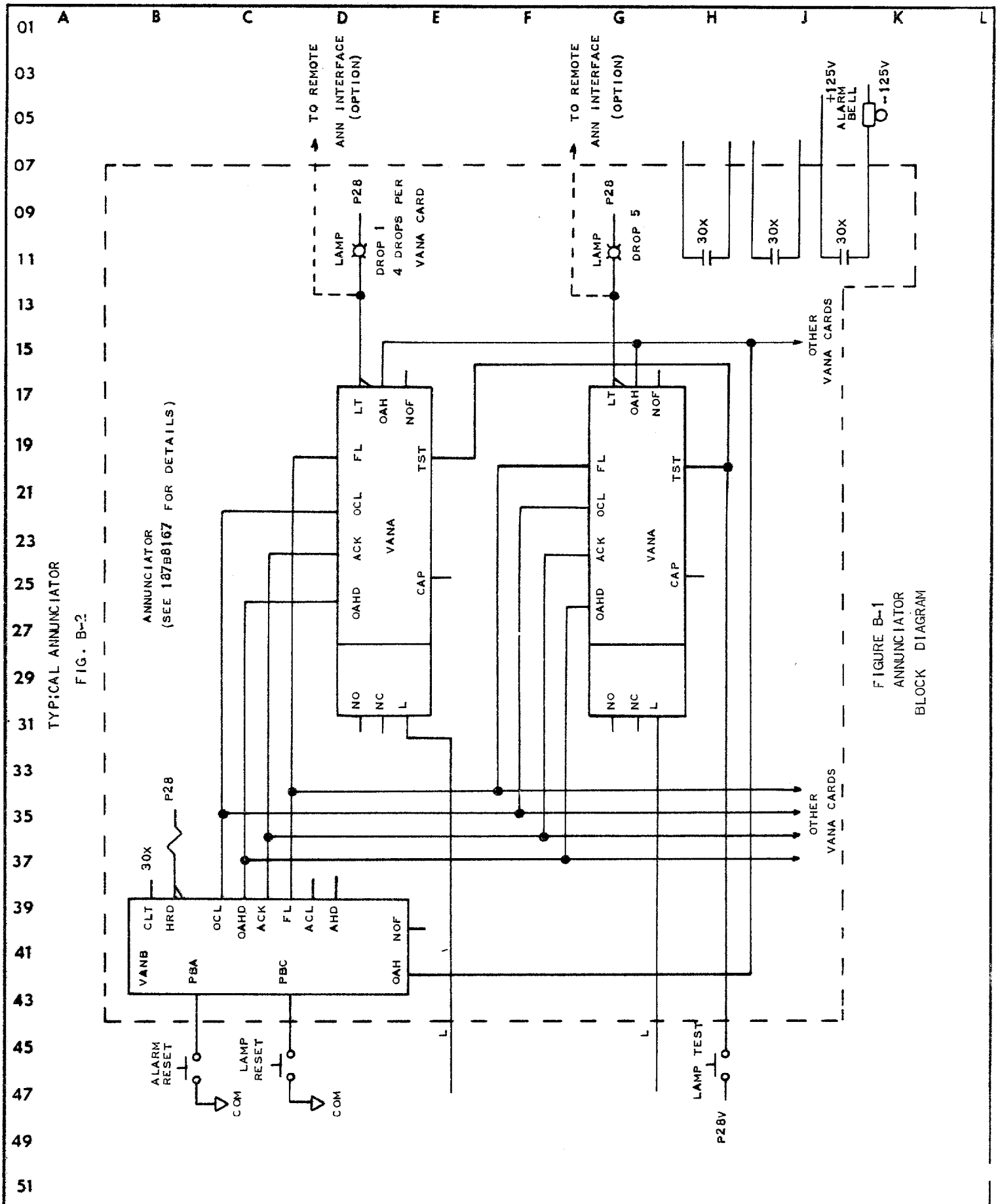


ANNUNCIATOR

OPERATIONAL FLOW CHART

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		ANNUNCIATOR FLOW CHART
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 23 SH. NO. 22

FORM IC 2390 2-72



TYPICAL ANNUNCIATOR
FIG. B-2

FIGURE B-1
ANNUNCIATOR
BLOCK DIAGRAM

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-1 ANNUNCIATOR BLOCK DIAG.
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	GEK-41053
REV. 3	MADE BY L.M. Worley			I.C. NO.		CONT. ON SH. 22 SH. NO. 21

THE VANB CARD PROVIDES THE CONTROL SIGNALS TO THE VANA CARD. (REFER TO FIGURE B-1). IT ALSO CONTAINS THE DRIVER FOR RELAY 30X WHOSE CONTACTS ARE USED TO DRIVE THE ALARM BELL. VANB OUTPUT FL PROVIDES THE FLASHING SIGNAL TO ALL VANA CARDS. IT IS PRESENT AT ALL TIMES.

THE SIGNAL FLOW FOR A TYPICAL ALARM CONDITION WILL BE GIVEN TO AID IN EXPLAINING THE ANNUNCIATOR. REFER TO FIGURE B-1 FOR SIGNAL CONNECTIONS.

ASSUMING THAT NO ALARM CONDITION IS PRESENT, OUTPUT OAHD OF VANB WILL BE A "1". THIS IS A FLASHING PERMISSIVE TO ALL VANA CARDS. NOW ASSUME THAT A FAULT OCCURS (A "0" ON L). ON THE VANA CARD, THE FLASHING SIGNAL APPEARS ON LT AND THE CORRESPONDING INDICATOR LIGHT FLASHES, AND OUTPUT OAH BECOMES A "0". THIS "0" GOES TO VANB INPUT OAH AND PERFORMS TWO FUNCTIONS. IT PLACES A "0" ON HRD AND PICKS UP RELAY 30X WHICH CAUSES THE BELL TO SOUND, AND OUTPUT OAHD BECOMES A "0" WHICH REMOVES THE FLASHING PERMISSIVE SIGNAL. THE FIRST FAULT REMAINS FLASHING BUT ANY MORE FAULTS OCCURRING WILL COME ON STEADY. SUPPOSE THAT THE "ALARM RESET" PUSHBUTTON IS NOW DEPRESSED, THE VANB OUTPUT ACK WILL BECOMES A "1" WHICH ALSO PERFORMS TWO FUNCTIONS IN THE VANA CARD. IT REMOVES THE FLASHING SIGNAL FROM LT AND REPLACES IT WITH A STEADY "0" WHICH MAKES THE LIGHT REMAIN STEADY ON. IT ALSO CAUSES OUTPUT OAH TO BECOMES A "1". THIS "1" ON OAH GOES TO VANB AND SILENCES THE BELL. IT ALSO ALLOWS VANB OUTPUT OAHD TO BECOMES A "1". THIS "1" ON OAHD IS A PERMISSIVE TO ALL OTHER VANA CARDS TO FLASH IF AN ADDITIONAL ALARM OCCURS.

NOW, ASSUME THAT NO OTHER ALARM CONDITION OCCURS, AND THE PRESENT FAULT CONDITION IS CLEARED. DEPRESSING PUSHBUTTON "LAMP RESET" PLACES A "0" ON THE OCL BUS AND THE LT SIGNAL ON VANA BECOMES A "1" WHICH TURNS THE LAMP OFF.

B-4 REMOTE ANNUNCIATOR INTERFACE MODULE (OPTION)

A REMOTE ANNUNCIATOR INTERFACE MODULE IS AVAILABLE TO PROVIDE ALARM INDICATIONS AT A REMOTE LOCATION. THE REMOTE TERMINAL IS COMPLETELY ISOLATED FROM THE TURBINE CONTROL PANEL. THE REMOTE TERMINAL PROVIDES ITS OWN POWER SUPPLY VOLTAGE AND ALARM INDICATION IS VIA RELAY CONTACT CLOSURE IN THE REMOTE ANNUNCIATOR INTERFACE MODULE. REFER TO TURBINE CONTROL PANEL ELEMENTARY SHEETS 18A AND 18B FOR CONNECTION BETWEEN REMOTE INTERFACE MODULE AND LOCAL ANNUNCIATOR. THREE OPERATOR PUSHBUTTONS ARE LOCATED AT THE REMOTE TERMINAL THAT PERFORMS THE SAME FUNCTION AS THE THREE LOCAL PUSHBUTTONS. THE REMOTE PUSHBUTTONS PICK UP RELAYS IN THE INTERFACE MODULE. THE CONTACTS OF THESE RELAYS ARE EITHER PARALLELED OR CONNECTED IN SERIES (WHICHEVER IS APPROPRIATE) WITH THE LOCAL PUSHBUTTONS.

WHEN AN ALARM OCCURS, THE LOCAL ANNUNCIATOR OPERATES AS DISCUSSED PREVIOUSLY. THE OUTPUT OF THE VANA CARD THAT DRIVES THE ALARM LAMP IS ALSO ROUTED TO A CORRESPONDING RELAY IN THE INTERFACE MODULE. THEREFORE, THE RELAY IN THE REMOTE ANNUNCIATOR INTERFACE MODULE IS PICKED UP WHEN THE LAMP IN THE LOCAL ANNUNCIATOR IS ON. THE RELAY IS NORMALLY CONNECTED SUCH THAT IT IS LATCHING AND DOES NOT FOLLOW THE FLASHING LIGHT. HOWEVER, A FLASHING REMOTE ALARM IS AVAILABLE. IN THE CASE WHERE THE RELAY IS LATCHING, WHEN THE "LAMP RESET" PUSHBUTTON IS DEPRESSED THE LATCHING PATH IS REMOVED AND THE RELAY WILL DROP OUT IF THE ALARM CONDITION HAS BEEN REMOVED.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-4 REMOTE ANNUNCIATOR
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 24 SH. NO. 23

B-5 SPEED SENSORS

THE SPEED RELAY CARD RECEIVES THE NHP SIGNAL FROM THE PR/A CONVERTER ON THE SSZB CARD AND PROVIDES LOGIC SIGNALS TO INDICATE THAT THE TURBINE HAS REACHED A CERTAIN SPEED (REFER TO FIGURES B-2 AND B-3).

THE SVSE CARD CONTAINS FIVE SPEED SENSORS WITH GIVEN (FIXED OR ADJUSTABLE) PICKUP AND DROPOUT CHARACTERISTICS. THESE ARE:

- 14HM - MINIMUM SPEED
- 14HA - ACCELERATING SPEED
- 14HS - OPERATING SPEED
- 14HR - ZERO SPEED
- NMR - SPEED MONITOR

THE FIRST THREE SENSORS OPERATE FROM A COMMON INPUT SIGNAL NHP. THE ZERO SPEED SENSOR OPERATES DIRECTLY FROM AN ELECTROMAGNETIC PICKUP. THE SPEED MONITOR RECEIVES A 0 TO 10V SIGNAL AND GIVES A LOGIC OUTPUT AT MR AND $\overline{MR}$ FOR A PRESET INPUT LEVEL.

THE 14HM PICKUP AND DROPOUT POINTS ARE PROGRAMMABLE BY LOGIC INPUTS ON $\overline{4}$, FS AND SJM. A "1" ON $\overline{4}$ REPROGRAMS THE PICKUP POINT TO 2% AND DROPOUT TO 1.8%. RESISTORS CAN BE SWITCHED BETWEEN FS AND SJM TO REPROGRAM THE PICKUP AND DROPOUT. THE DROPOUT OF 14HM IS DEPENDENT ON PICKUP. NO SEPARATE ADJUSTMENT IS PROVIDED. THE 14HM SENSOR OUTPUTS ARE ON PINS HM AND $\overline{HM}$.

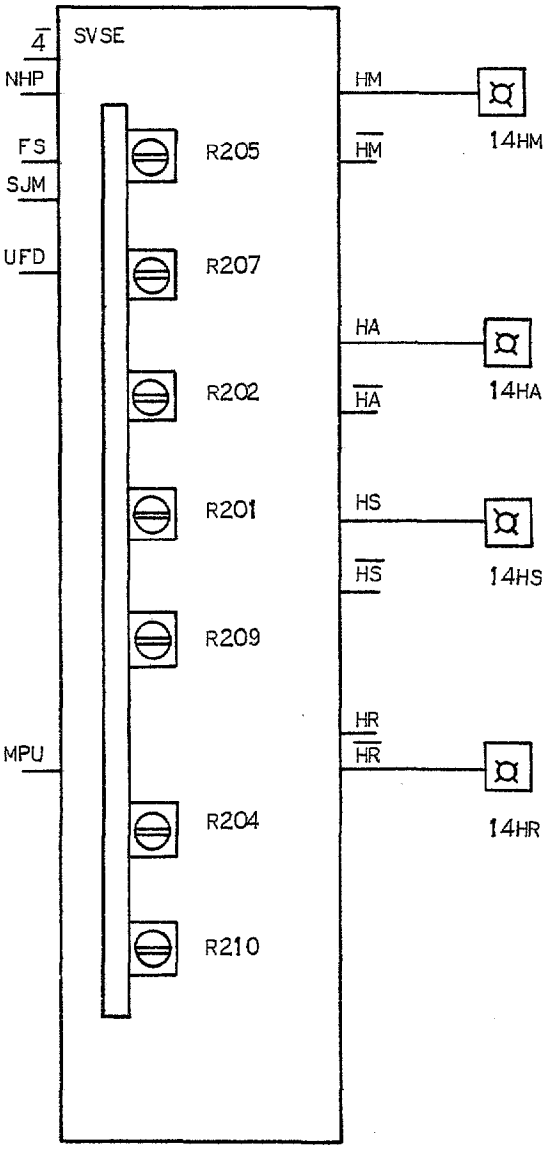
THE 14HA SENSOR LOOKS AT NHP AND PROVIDES LOGIC OUTPUTS ON HA AND $\overline{HA}$ FOR A PRESET LEVEL OF NHP. THE DROPOUT POINT IS DEPENDENT ON THE PICKUP SETTING AND DOES NOT HAVE A SEPARATE ADJUSTMENT. THE DROPOUT IS 75% OF PICKUP VALUE.

THE 14HS SENSOR LOOKS AT NHP AND PROVIDES LOGIC SIGNALS AT HS AND $\overline{HS}$. THE PICKUP AND DROPOUT LEVEL IS ADJUSTABLE. THE DROPOUT IS ALSO PROGRAMMABLE WITH A LOGIC SIGNAL ON INPUT UFD. FOR UFD EQUAL "0" DROPOUT IS 80 TO 98% OF PICKUP. FOR UFD EQUAL "1", DROPOUT IS 90 TO 98% OF PICKUP. POTENTIOMETER R201 (UFDO) IS USED TO SET DROPOUT LEVEL WHEN UFD IS A "1". AFTER R201 IS SET, THEN R209 (HSDO) CAN BE SET FOR NORMAL DROPOUT. (SEE CONTROL SPEC FOR ADJUSTMENT).

THE SPEED MONITOR SENSOR HAS ADJUSTABLE PICKUP RANGES THAT ARE JUMPER SELECTABLE. THE PICKUP RANGES AND JUMPERS ARE:

- 20-60% - JUMPERS RA TO RB
- 50-90% - JUMPERS RC TO RB
- 80-110% - JUMPERS RD TO RB

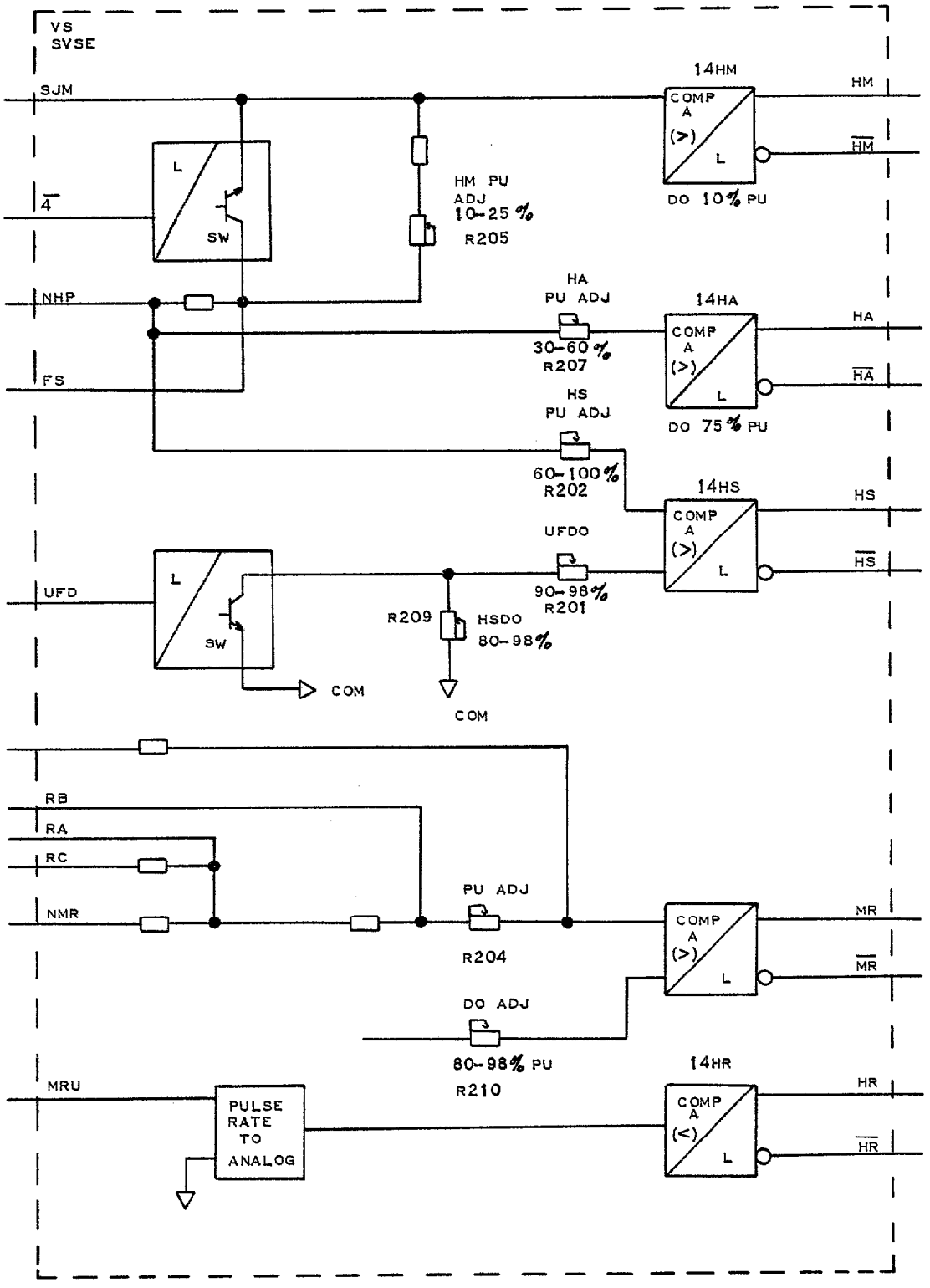
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	B-5
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED SENSORS	
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 25	SH. NO. 24



SPEED SENSORS
ADJUSTMENT LOCATION
FIG. B-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED SENSOR
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 26 SH. NO. 25

FORM 1C 2390 4-74



SVSE CARD
FIG. B-3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM B-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SVSE CARD
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 27 SH. NO. 26

THE 14HR SENSOR RECEIVES A FREQUENCY ON MPU PROPORTIONAL TO RPM AND GENERATES AN ANALOG SIGNAL. THIS ANALOG SIGNAL IS COMPARED TO A FIXED REFERENCE AND GIVES A "1" ON HR IF ANALOG SIGNAL IS LESS THAN REFERENCE. UNLIKE THE OTHER SENSORS, 14HR IS PICKED UP AT ZERO SPEED AND DROPS OUT AS SPEED IS INCREASED.

ADJUSTMENTS

14HM PU - R205 10-25 % SPEED

14HA PU - R207 30-60 % SPEED

14HS PU - R202 60-100 % SPEED

14HS DO - R209 48-98 % SPEED

14HS UFDO - R201 54-98 % SPEED

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	B-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED SENSORS	
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 28	SH. NO. 27

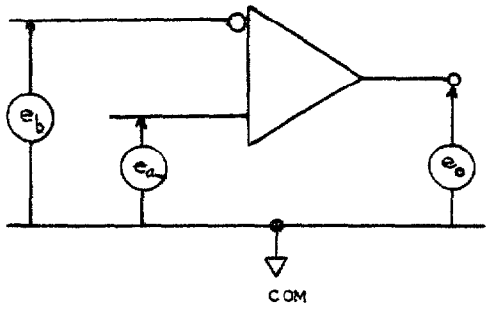
SECTION C

CONTROL

C-1 OPERATIONAL AMPLIFIERS

THE DIRECTO-MATIC II SPEEDTRONIC CONTROL USES THE OPERATIONAL AMPLIFIER (NORMALLY REFERRED TO AS OP AMP) EXTENSIVELY; THEREFORE, TO UNDERSTAND THE CONTROL PANEL, ONE MUST HAVE A BASIC UNDERSTANDING OF THE OP AMP. A BRIEF DISCUSSION OF THE OP AMP ENHANCED WITH SEVERAL EXAMPLES IS GIVEN SUBSEQUENTLY. THE INTENT OF THIS TEXT IS NOT TO PRESENT A RIGOROUS ANALYTICAL DESCRIPTION OF THE OP AMP, BUT AN INTRODUCTION TO THE OP AMP IN SIMPLIFIED TERMS. IT IS HOPED THAT FROM THIS GROUNDWORK THE USER CAN FOLLOW THE TURBINE CONTROL ELEMENTARY AND GAIN AN UNDERSTANDING OF THE SPEEDTRONIC CONTROL.

THE OPERATIONAL AMPLIFIER IS GENERALLY CONSIDERED A HIGH GAIN, HIGH INPUT IMPEDANCE, LOW OUTPUT IMPEDANCE AMPLIFIER CAPABLE OF AMPLIFYING D-C AS WELL AS A-C SIGNALS. THE STANDARD SYMBOL FOR THE OP AMP IS SHOWN BELOW.



THE INPUT WITH THE CIRCLE IS THE INVERTING INPUT AND THE OTHER INPUT IS NON-INVERTING. THE OUTPUT IS

$$e_o = K(e_a - e_b)$$

WHERE K IS THE GAIN OF THE AMPLIFIER. AS MENTIONED PREVIOUSLY, THE OP AMP HAS A VERY HIGH GAIN (TYPICALLY 10^5 IN DIRECTO-MATIC II), THEREFORE, A VERY SMALL DIFFERENCE BETWEEN e_a AND e_b WILL GIVE A LARGE e_o. HOWEVER e_o IS LIMITED BY THE SUPPLY VOLTAGES OF THE OP AMP. WHEN e_o REACHES APPROXIMATELY THE SUPPLY VOLTAGE, IT IS SAID TO BE IN SATURATION. TYPICAL SUPPLY VOLTAGES IN DIRECTO-MATIC II ARE ± 12 VOLTS. USING A +12 VOLT SUPPLY AND A GAIN OF 10^5 THE VALUE OF $e_a - e_b$ FOR SATURATION IS

$$12 = 10^5(e_a - e_b)$$

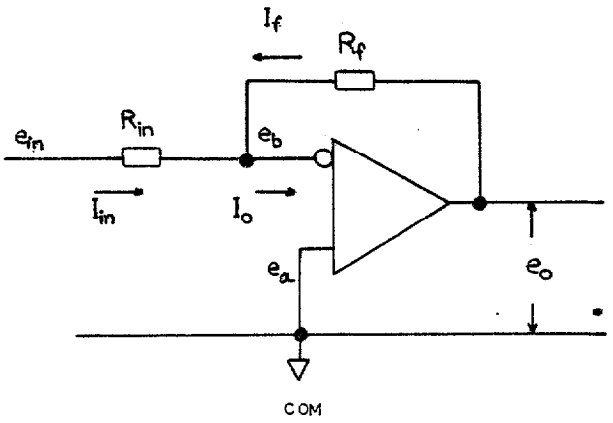
$$(e_a - e_b) = \frac{12}{10^5} = 12 \times 10^{-5}$$

$$(e_a - e_b) = 120 \text{ MICROVOLTS}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1	
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS	
REV. 3	MADE BY L.M. Worley			I.C. NO	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH 29	SH. NO. 28

BECAUSE OF THIS HIGH GAIN THE OP AMP IS PRACTICALLY USELESS WHEN USED ALONE; HOWEVER, WITH NEGATIVE FEEDBACK ITS USES ARE LIMITED ONLY BY THE DESIGNER'S IMAGINATION.

THE MOST COMMON CIRCUIT OF THE OP AMP IS SHOWN BELOW:



IN THIS CASE THE NON-INVERTING INPUT e_a IS TIED TO COMMON. THE USEFUL OPERATING REGION OF THE OP AMP IS WHEN IT IS NOT IN SATURATION; THEREFORE, e_b IS APPROXIMATELY EQUAL TO e_a IN THIS CASE e_a IS TIED TO COMMON; THEREFORE, e_b IS APPROXIMATELY ZERO VOLTS. OPERATING IN THIS MODE e_b IS OFTEN REFERRED TO AS VIRTUAL COMMON. TO DETERMINE e_o AS A FUNCTION OF e_{in}, THE CURRENTS AT e_b ARE SUMMED TOGETHER. THIS JUNCTION IS OFTEN REFERRED TO AS THE SUMMING JUNCTION.

$$I_{in} + I_f - I_o = 0$$

THE OP AMP HAS A VERY HIGH INPUT IMPEDANCE AND I_o CAN BE CONSIDERED ZERO. THEN

$$I_{in} + I_f = 0$$

THE OP AMP HAS A VERY LOW OUTPUT IMPEDANCE AND WILL SUPPLY THE NECESSARY CURRENT TO MAKE THE ABOVE EQUATION TRUE.

I_{in} AND I_f CAN BE FOUND AS FOLLOWS:

$$I_{in} = \frac{e_{in}}{R_{in}} \qquad I_f = \frac{e_o}{R_f}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 30 SH. NO. 29

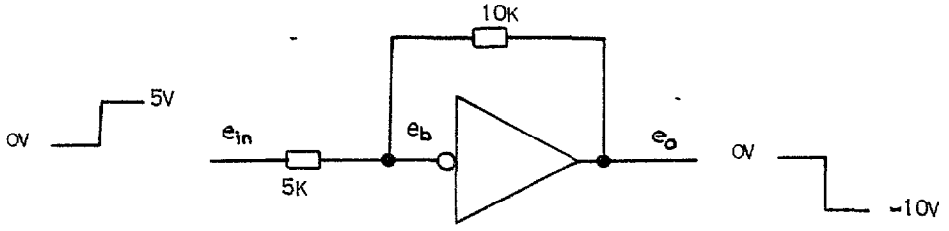
SUBSTITUTING GIVES:

$$\frac{e_{in}}{R_{in}} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\frac{e_{in}}{R_{in}}$$

$$e_o = -e_{in}\frac{R_f}{R_{in}}$$

THIS EQUATION APPLIES AS LONG AS THE OP AMP IS NOT IN SATURATION. IF IT GOES INTO SATURATION, e_b WILL NO LONGER BE AT VIRTUAL COMMON AND THE EQUATION LOSES VALIDITY. TO FURTHER ILLUSTRATE THE INVERTER, THE CIRCUIT IS REPEATED BELOW WITH VALUES ASSIGNED. THE CONNECTION OF e_a TO COMMON ON THE SYMBOL IS OFTEN OMITTED WITH THE UNDERSTANDING THAT IT EXISTS.

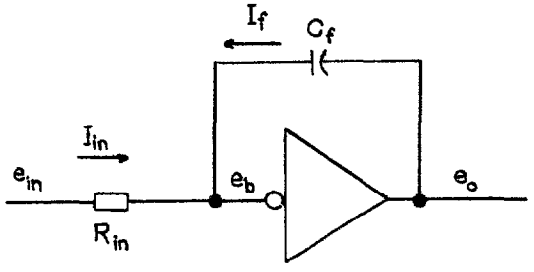


$$e_o = -e_{in}\frac{10K}{5K} = -2e_{in}$$

e_{in} IS A +5V SIGNAL, THEN

e_o = -2(5)V = - 10 VOLTS

THE SECOND OP AMP CIRCUIT DISCUSSED IS THE INTEGRATOR. THE FOLLOWING IS AN INTEGRATOR CIRCUIT:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS	
REV. 3	MADE BY L. M. Darley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH 31	SH. NO. 30

AGAIN, SUMMING THE CURRENTS AT e_b GIVES:

$$I_{in} + I_f = 0 \quad \text{WHERE}$$

$$I_{in} = \frac{e_{in}}{R_{in}} \quad \text{AND}$$

$$I_f = C_f \frac{d(e_o)}{dt}$$

$$\frac{e_{in}}{R_{in}} + C_f \frac{d(e_o)}{dt} = 0$$

$$C_f \frac{d(e_o)}{dt} = -\frac{e_{in}}{R_{in}}$$

$$d(e_o) = -\frac{e_{in}}{R_{in}C_f} dt$$

$$\int d(e_o) = -\int \frac{e_{in}}{R_{in}C_f} dt$$

$$e_o = -\frac{1}{R_{in}C_f}\int e_{in} dt$$

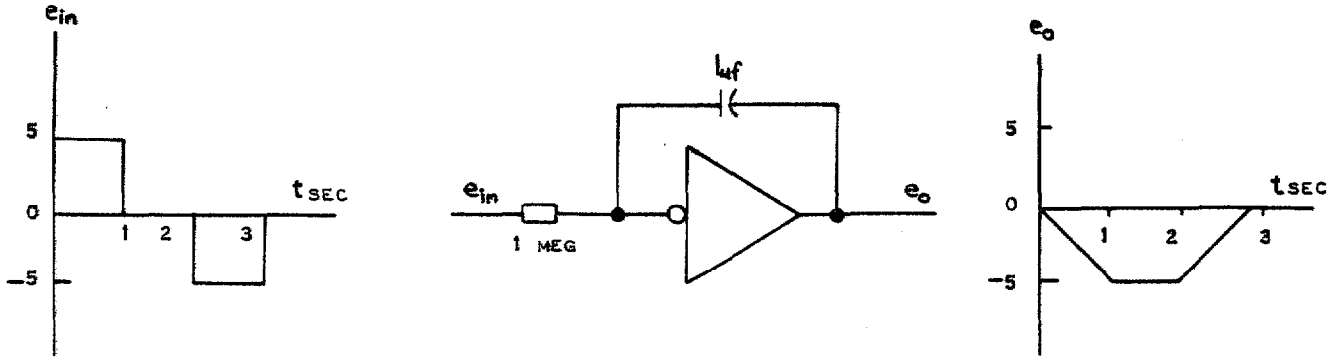
IF e_{in} IS A CONSTANT OR STEP INPUT, THEN

$$e_o = -\frac{e_{in}}{R_{in}C_f} t \quad \text{OR}$$

$$e_o = -\frac{I_{in}}{C_f} t \qquad t = \underline{\text{SECONDS}}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	OPERATIONAL AMPLIFIERS
REV. 3	MADE BY L. M. Worley			I.C. NO.		GEK-41053
						CONT. ON SH. 32 SH. NO. 31

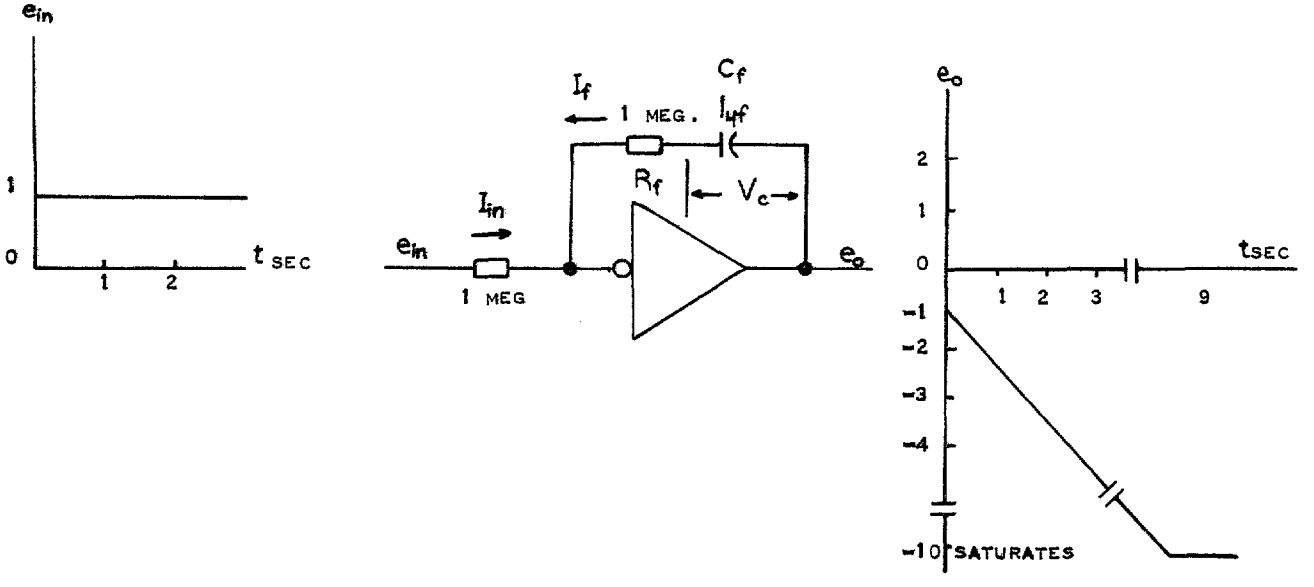
THE INTEGRATOR CIRCUIT IS REDRAWN BELOW WITH VALUES ASSIGNED.



$$e_o = -\frac{5}{(10^6)(10^6)} \qquad t = -5t$$

AS CAN BE SEEN FROM THE CURVES ADJACENT TO THE CIRCUIT, IN RESPONSE TO A STEP INPUT THE OUTPUT INCREASES ON A RAMP. IF THE INPUT RETURNS TO ZERO, THE CAPACITOR REMAINS CHARGED AND e_c REMAINS AT THE LEVEL PRIOR TO INPUT GOING TO ZERO. IF A STEP INPUT OF OPPOSITE POLARITY IS APPLIED, THE CAPACITOR WILL BEGIN DISCHARGING.

FOR THE NEXT CIRCUIT, THE INPUT-OUTPUT RELATIONSHIP WILL NOT BE DEVELOPED MATHEMATICALLY; INSTEAD AN INTUITIVE APPROACH WILL BE PRESENTED. THE CIRCUIT IS OFTEN REFERRED TO AS A PROPORTIONAL PLUS INTEGRAL.



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS	
REV. 3	MADE BY L.M. Warley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 33	SH. NO. 32

TO BEGIN ASSUME THAT e_{in} AND e_o ARE ZERO ($V_c = 0$). NOW ASSUME A STEP INPUT OF 1 VOLT IS APPLIED AT e_{in}. AS BEFORE

$$I_{in} = \frac{e_{in}}{R_{in}} = \frac{1V}{1M} = I_{ua}$$

$$I_f = -I_{in} = -I_{ua}$$

NOW e_o IS EQUAL TO THE VOLTAGE ACROSS R_f AND C_f. THE VOLTAGE ACROSS R_f IS

$$V_t = I_f R_f = -I_{in} R_f$$

$$V_t = -1 \text{ VOLT}$$

THE VOLTAGE ACROSS A CAPACITOR SUBJECTED TO A CONSTANT CURRENT IS

$$V_c = \frac{I}{C} t$$

THEREFORE, IF e_{in} IS A STEP INPUT

$$e_o = -I_{in} R_f - \frac{I_{in}}{C} t$$

AT THE INSTANT THAT THE STEP IS APPLIED ($t = 0$)

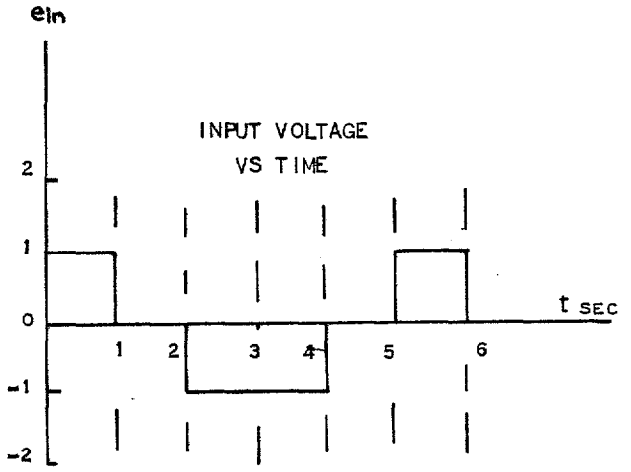
$$e_o = -I_{in} R_f$$

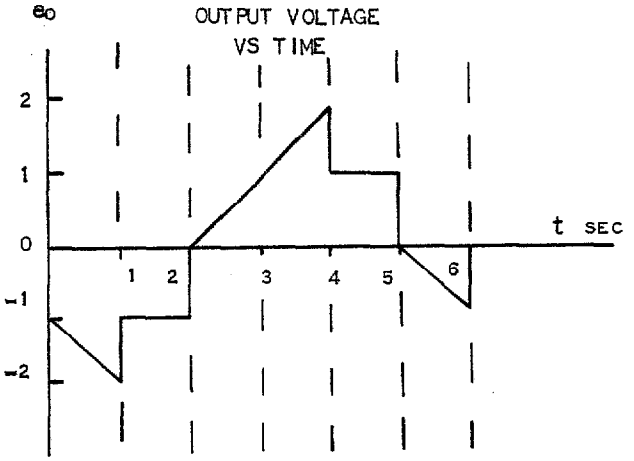
$$e_o = -(I_{ua})(I_{meg})$$

$$e_o = -1 \text{ VOLT}$$

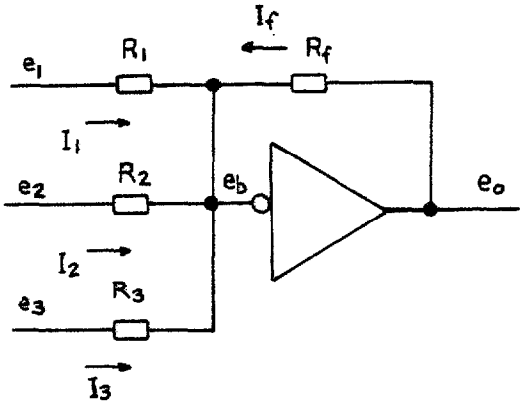
AS TIME INCREASES, THE CAPACITOR CHARGES UP AS SHOWN IN THE GRAPH ADJACENT TO CIRCUIT. FOLLOWING ARE CURVES SHOWING OUTPUT RESPONSE TO A VARYING INPUT VOLTAGE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1	
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS	
REV. 3	MADE BY L. M. Woolley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 34	SH. NO. 33





THE NEXT OP AMP CIRCUIT DISCUSSED IS THE SUMMER. THE CIRCUIT IS SHOWN BELOW:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 35 SH. NO. 34

SUMMING THE CURRENTS AT e_b YIELDS:

$$I_1 + I_2 + I_3 + I_f = 0$$

$$\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e_3}{R_3} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\left(\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e_3}{R_3}\right)$$

$$e_o = -\left(\frac{R_f e_1}{R_1} + \frac{R_f e_2}{R_2} + \frac{R_f e_3}{R_3}\right)$$

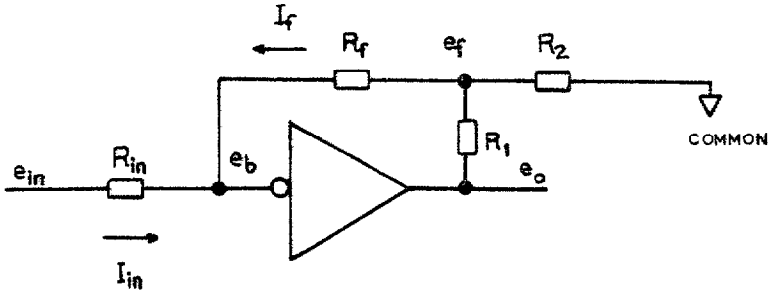
AS IMPLIED BY ITS NAME, THE CIRCUIT SUMS THE INPUTS WITH EACH INPUT WEIGHTED BY THE RATIO OF ITS INPUT RESISTOR TO THE FEEDBACK RESISTOR, R_f .

THE MULTIPLE INPUT CONFIGURATION CAN ALSO BE APPLIED TO THE INTEGRATOR. REFERRING BACK TO THE INTEGRATOR EQUATION OF A STEP INPUT

$$e_o = -\frac{I_{in}t}{C_f}$$

FOR MULTIPLE INPUTS, THE ALGEBRAIC SUM OF THE INPUT CURRENTS IS SUBSTITUTED FOR I_{in} .

OCCASIONALLY THE FEEDBACK VOLTAGE IS NOT DERIVED DIRECTLY FROM e_o .FOLLOWING IS AN EXAMPLE OF SUCH A CIRCUIT:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1 OPERATIONAL AMPLIFIERS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	GEK-41053
REV. 3	MADE BY L.M. Worley			I.C. NO.		CONT. ON SH. 36 SH. NO. 35

AGAIN, SUMMING THE CURRENTS AT e_b GIVES

$$I_{in} + I_f = 0 \text{ (SUMMING JUNCTION EQUATION) WHERE}$$

$$I_{in} = \frac{e_{in}}{R_{in}}$$

$$I_f = \frac{e_f}{R_{in}} \quad (I_f \text{ EQUATION}) \text{ BUT}$$

$$e_f = e_o \left(\frac{\frac{R_f R_2}{R_f + R_2}}{R_1 + \frac{R_f R_2}{R_f + R_2}} \right) \text{ LET}$$

$$R_{eq} = \frac{R_f R_2}{R_f + R_2} \quad \text{THEN}$$

$$e_f = e_o \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

SUBSTITUTING THIS IN ABOVE I_f EQUATION GIVES

$$I_f = \frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

THE SUMMING JUNCTION EQUATION NOW BECOMES

$$\frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right) = \frac{e_{in}}{R_{in}}$$

$$e_o = -e_{in} \frac{R_f}{R_{in}} \left(\frac{R_1 + R_{eq}}{R_{eq}} \right)$$

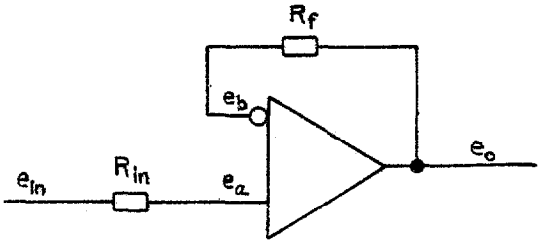
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS	
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 37	SH. NO. 36

UP TO THIS POINT, WE HAVE ONLY DISCUSSED THE INVERTING MODE OF THE OP AMP. BEFORE GOING INTO THE NON-INVERTING MODE, TWO IMPORTANT FEATURES OF THE OP AMP DISCUSSED PREVIOUSLY WILL BE REPEATED.

1. THE OP AMP HAS VERY HIGH INPUT IMPEDANCE; THEREFORE, IT IS SAFE TO ASSUME THAT IT DRAWS ZERO INPUT CURRENT.

2. A VERY SMALL DIFFERENCE BETWEEN e_a AND e_b CAUSES THE OP AMP TO GO INTO SATURATION, THEREFORE, IN THE ACTIVE REGION, IT IS ASSUMED THAT e_a EQUALS e_b.

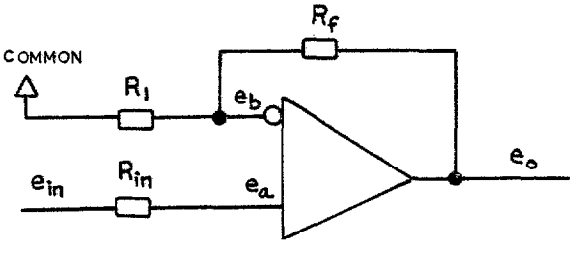
THE FIRST NON-INVERTING OP AMP CIRCUIT TO BE DISCUSSED IS THE VOLTAGE FOLLOWER. THE CIRCUIT IS SHOWN BELOW.



SINCE THE OP AMP INPUT DRAWS NO CURRENT, e_a IS EQUAL TO e_{in}. PER THE FIGURE ON SH. 29 e_b EQUALS e_a AND NO CURRENT IS DRAWN BY THE OP AMP; THEREFORE, e_o EQUALS e_b WHICH IN TURN IS EQUAL TO e_{in}. THE INPUT-OUTPUT EQUATION IS

$$e_o = e_{in}$$

TO OBTAIN A GAIN GREATER THAN ONE, THE FOLLOWING CIRCUIT IS USED:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OPERATIONAL AMPLIFIERS
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 38 SH. NO. 37

AS IN PREVIOUS CIRCUIT e_b EQUALS e_a WHICH EQUALS e_{in} BECAUSE NO CURRENT FLOWS INTO THE OP AMP. e_b IS ALSO EQUAL TO

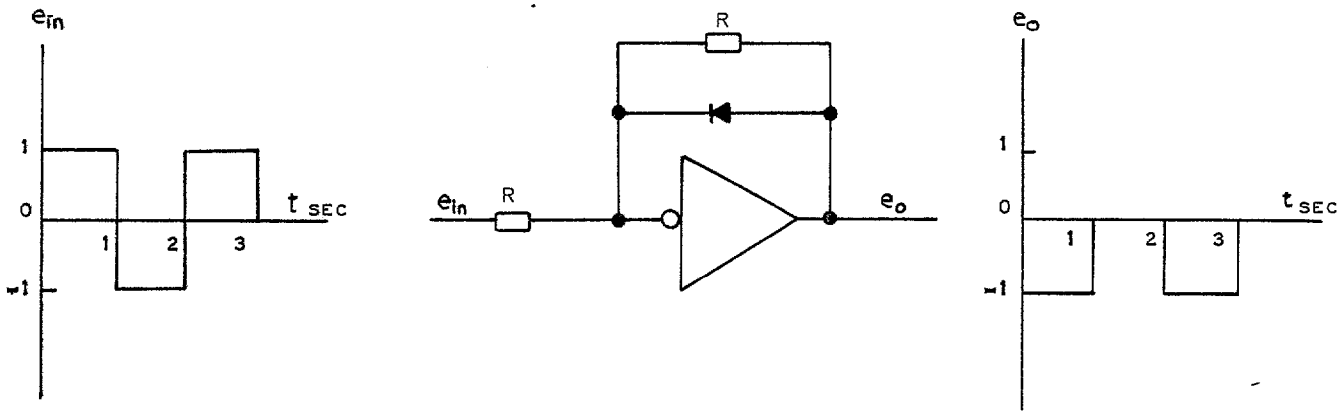
$$e_b = e_o \frac{R_1}{R_1 + R_f} \quad \text{OR}$$

$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_b$$

$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_{in} \qquad e_b = e_{in}$$

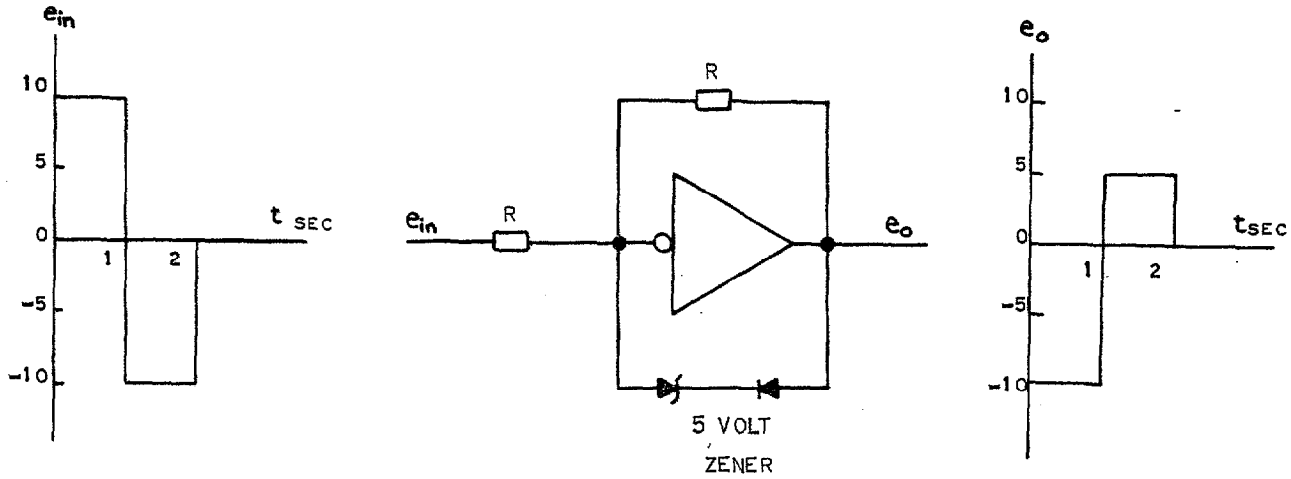
$$e_o = \left(1 + \frac{R_f}{R_1}\right) e_{in}$$

AS SEEN FROM THIS EQUATION THE GAIN OF THIS CIRCUIT IS ALWAYS GREATER THAN ONE. IT IS OFTEN DESIRED TO CLAMP THE OP AMP OUTPUT AT SOME VALUE. THE FOLLOWING CIRCUITS ILLUSTRATE THE USE OF DIODES AND ZENER DIODES TO CLAMP THE OUTPUT.

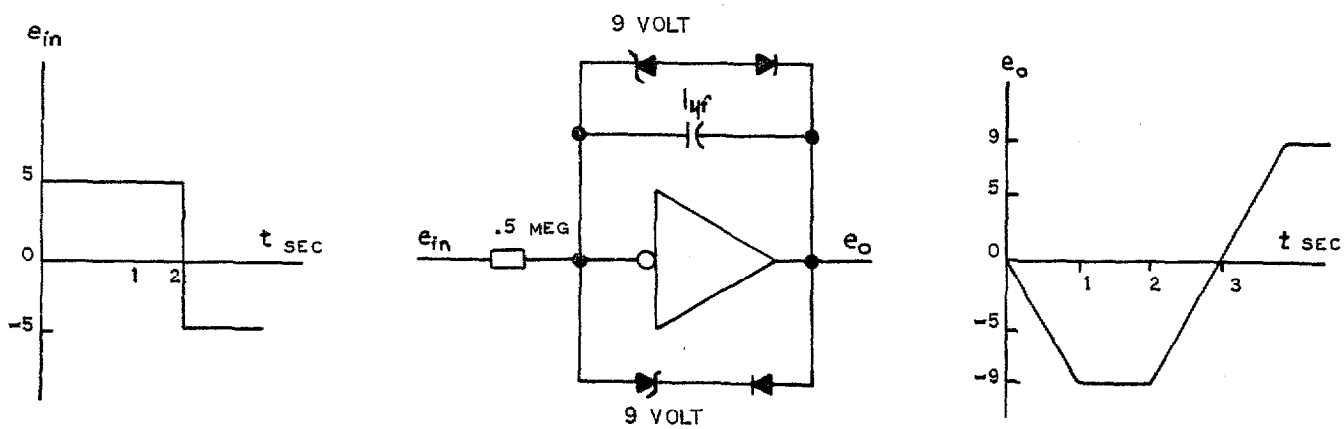


ABOVE CIRCUIT LIMITS POSITIVE SWING TO APPROXIMATELY ZERO.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS LEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1 OPERATIONAL AMPLIFIERS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 39 SH. NO. 38



ABOVE CIRCUIT LIMITS POSITIVE SWING TO APPROXIMATELY 5 VOLTS.



ABOVE CIRCUIT LIMITS POSITIVE AND NEGATIVE SWINGS TO 9 VOLTS: THIS CIRCUIT IS OFTEN USED TO PREVENT OP AMP FROM SATURATING AND SUMMING FUNCTIONS FROM RISING ABOVE COMMON.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1 OPERATIONAL AMPLIFIERS
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 40 SH. NO. 39

C-2 START-UP

THE TURBINE CONTROL PANEL CONTAINS START-UP, SPEED, AND TEMPERATURE CONTROL LOOPS WHICH MODULATE VCE. VCE SERVES AS REFERENCE TO THE FUEL CONTROL LOOPS TO REGULATE THE FUEL FLOW TO THE TURBINE. A TYPICAL CONTROL SYSTEM IS SHOWN IN FIGURE C-1.

THE START-UP CONTROL IS AN OPEN LOOP SYSTEM THAT OPERATES IN CONJUNCTION WITH THE SEQUENTIAL CONTROL CIRCUITRY TO PROVIDE PRESET FUEL LEVELS TO THE TURBINE DURING START-UP. IN ADDITION TO THE ABOVE, IT ALSO SETS THE MAXIMUM AND MINIMUM LIMITS ON VCE AFTER THE START-UP PERIOD IS OVER. A TYPICAL VCE START-UP IS SHOWN IN FIGURE C-2. A TYPICAL START-UP SYSTEM IS SHOWN IN FIGURE C-3.

SSKC CARD - (REFER TO FIGURE C-4)

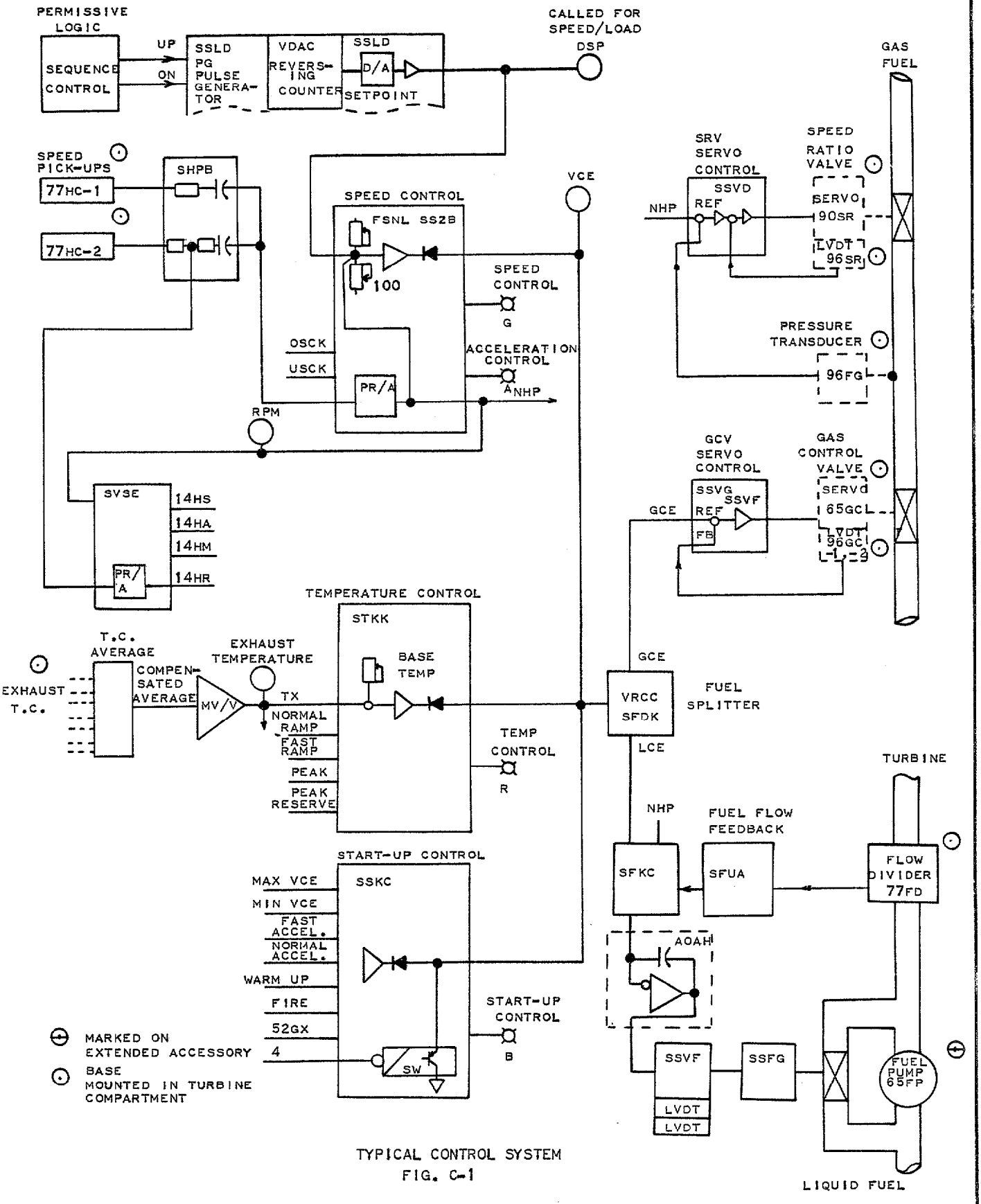
A "1" ON INPUT OFF OR A "1" ON ALL THREE INPUTS FIRE, WU AND ACC WILL CLAMP OUTPUT VCE TO ZERO VOLTS. IF OFF IS A "0" THEN A "0" OCCURRING ON EITHER FIRE, WU OR ACC WILL REMOVE THE CLAMP ON VCE. TO FACILITATE DESCRIBING THE FUNCTION OF THE CARD, ASSUME THAT OFF IS A "0", ALR IS A "0", AND FIRE, WU AND ACC ARE "1". UNDER THESE CONDITIONS, VCE IS CLAMPED TO ZERO VOLTS. NOW ASSUME THAT A "0" OCCURS ON INPUT FIRE. VCE WILL JUMP TO FIRE LEVEL AS SET BY R102. A "0" OCCURRING NOW ON WU WILL CAUSE VCE TO DROP FROM FIRE TO WARM UP LEVEL WHICH IS SET BY R101. VCE WILL REMAIN AT THIS LEVEL UNTIL ACC BECOMES "0" OR IS PULLED DOWN BY SOME OTHER CONTROL LOOP TIED TO VCE. UPON RECEIVING A "0" ON ACC, VCE WILL BE RELEASED TO INCREASE ON AN EXPONENTIAL CURVE UNTIL A LIMIT SET BY R100 IS REACHED. THIS LIMIT ON VCE WILL REMAIN UNTIL INPUT ALR BECOMES A "1". A "1" ON ALR WILL REMOVE THE LIMIT AND ALLOW VCE TO INCREASE TO A MAXIMUM LIMIT SET BY R106.

A MINIMUM VCE CIRCUIT MONITORS VCE AND PROVIDES A NEGATIVE BIAS AT OUTPUT MIN WHEN VCE DECREASES BELOW A MIN LEVEL SET BY R104.

TYPICALLY, THE BIAS OUT OF MIN IS CONNECTED TO THE SPEED AND ACCELERATION CONTROLS AND PREVENTS THEM FROM DRIVING VCE BELOW THE MIN LEVEL. A TEST CIRCUIT IS PROVIDED THAT SUPPLIES A POSITIVE BIAS OUT OF MT WHEN MIN PUSHBUTTON IS DEPRESSED THAT CAN BE USED AS AN INPUT TO THE SPEED CARD TO SIMULATE SPEED FEEDBACK WHICH TELLS THE SPEED CARD TO DECREASE VCE. THE MIN CIRCUIT SHOULD THEN PREVENT VCE FROM GOING BELOW THE PRESET LEVEL.

PUSHBUTTONS ARE PROVIDED FOR TESTING AND ADJUSTING THE PRESET LEVELS. THE MAX PUSHBUTTON HAS A PERMISSIVE INTERLOCK. OPERATOR SWITCH "43" HAS TO BE IN THE OFF OR CRANK POSITION. PUSHBUTTON TCC WILL DECREASE THE EXPONENTIAL RISE TIME.

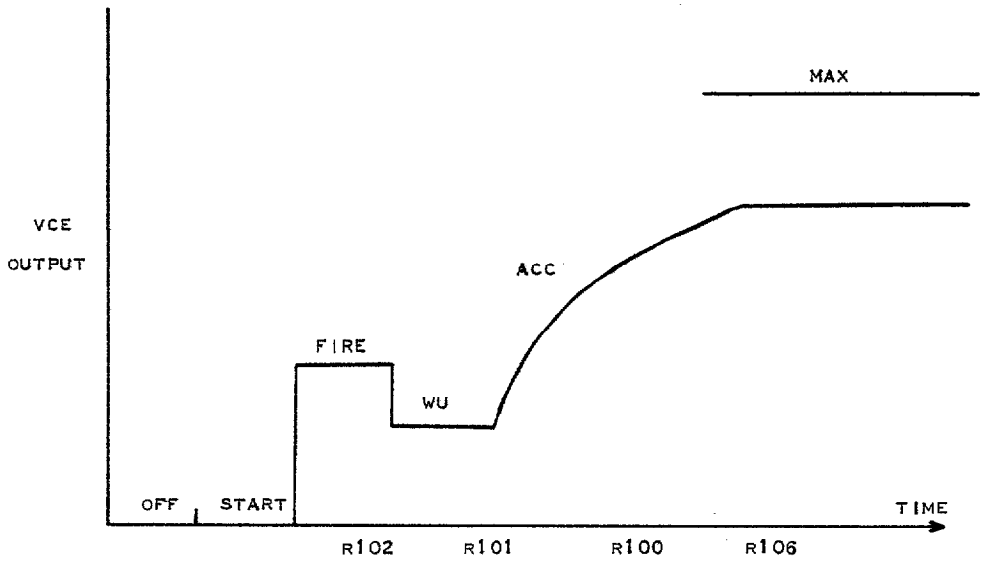
POTENTIOMETER RHI IS PROVIDED FOR MANUAL CONTROL OF VCE. VCE CAN BE ADJUSTED FROM MAX VCE TO ZERO USING RHI IF SOME OTHER CONTROL LOOP IS NOT PULLING VCE DOWN. A LIMIT SWITCH ON THE POTENTIOMETER TIES PINS SW1 AND SW2 TOGETHER WHEN RHI IS NOT FULLY COUNTER-CLOCKWISE.



TYPICAL CONTROL SYSTEM
FIG. C-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TYPICAL CONTROL SYSTEM
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 42 SH. NO. 41

FORM 1C 2390 2-72



OFF	1	0	0	0	0	0
$\overline{\text{FIRE}}$	x	1	0	0	0	0
$\overline{\text{WU}}$	x	1	1	0	0	0
$\overline{\text{ACC}}$	x	1	1	1	0	0
ALR	x	0	0	0	0	1

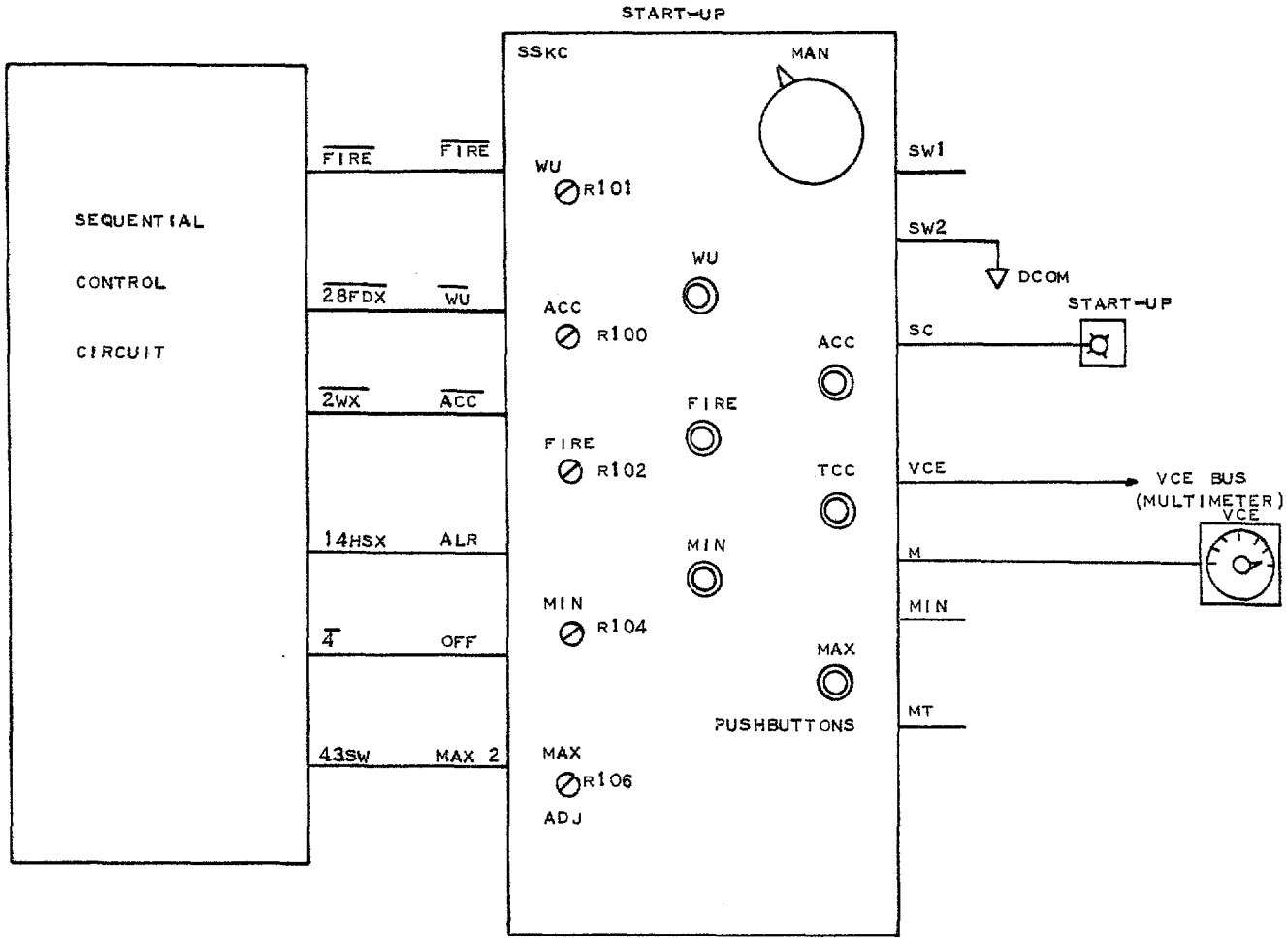
x = DON'T CARE "0" OR "1"

OPEN LOOP START-UP PROFILE

FIG. C-2

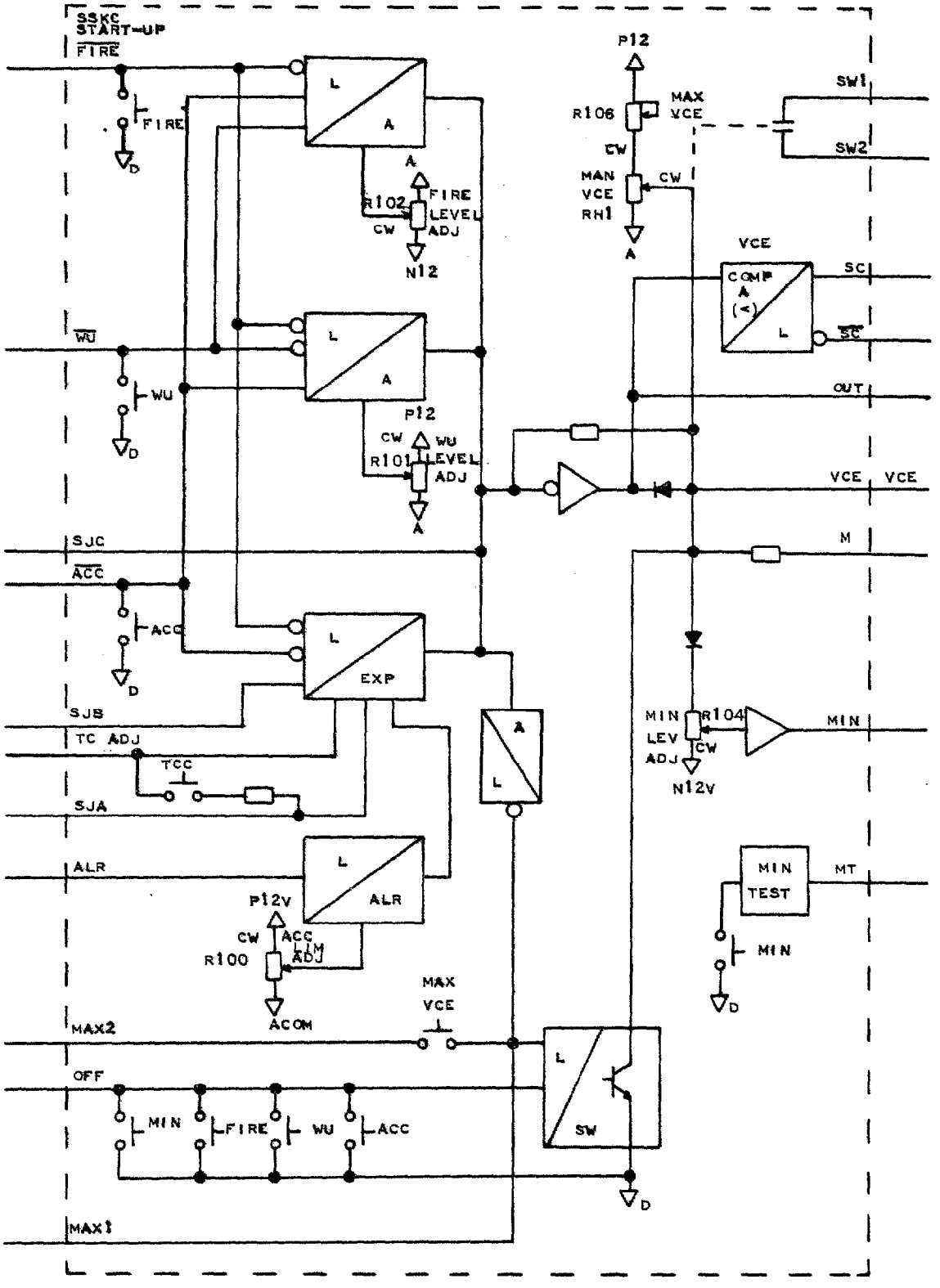
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.
REV. 3	MADE BY L. M. Worley			I.C. NO.	

ELEMENTARY DIAGRAM C-2
OPEN LOAD START-UP
GEK-41053
CONT. ON SH. 43 SH. NO. 42



TYPICAL START-UP CONTROL SYSTEM FIGURE C-3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		START-UP CONTROL SYSTEM
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 44 SH. NO. 43



SSKC CARD

FIG. C-4

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-4
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SSKC CARD
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 45 SH. NO. 44

ADJUSTMENT RANGES

MAX - R106	8 TO 10.5	VOLTS
MIN - R104	2.25 TO 4.00	VOLTS
FIRE - R102	3.25 TO 5.00	VOLTS
WU - R101	2.5 TO 4.0	VOLTS
ACC - R100	3.5 TO 6.5	VOLTS

C-3 DIGITAL SETPOINT

THE DIGITAL SETPOINT PROVIDES A PROGRAMMABLE ANALOG SIGNAL USED AS A CALLED FOR SPEED OR LOAD SIGNAL TO THE SPEED CONTROL CARD. THE SYSTEM CONSISTS OF CARDS, SSLD, SFPB, VDAC, SHPB AND ASSOCIATED CONTROL SIGNALS. (SEE FIGURE C-5).

VDAC CARD - (REFER TO FIGURE C-6)

THE VDAC CARD CONTAINS A TWELVE-BIT REVERSABLE COUNTER, A DIGITAL-TO-ANALOG (D/A) CONVERTER, AND TWELVE LIGHT EMITTING DIODES (LED'S) TO INDICATE THE OUTPUT OF THE COUNTER.

THE COUNTER OUTPUT (DIGITAL SETPOINT) MAY BE PRESET TO 100.3 WHEN THE $\overline{\text{LOAD}}$ INPUT IS A LOGIC ZERO. A LOGIC ZERO ON THE $\overline{\text{ENAB}}$ INPUT PERMITS THE COUNTER TO COUNT. THE DN/UP INPUT STEERS THE COUNTER, WHEN ENABLED, TO COUNT UP (LOGIC 0) OR TO COUNT DOWN (LOGIC ONE). THE INPUT CCP IS A PULSE RATE SIGNAL WHICH ESTABLISHES THE RATE AT WHICH THE COUNTER COUNTS. THE PULSE-RATE SIGNAL IS GENERATED ON THE SSLD CARD.

WHEN THE COUNTER IS EMPTY AND A DOWN COUNT IS PRESENT, OR WHEN THE COUNTER IS FULL AND AN UPCOUNT IS PRESENT, MM1, MM2, AND MM3 WILL ALL BE A LOGIC 1.

SSLD CARD - (REFER TO FIGURE C-6)

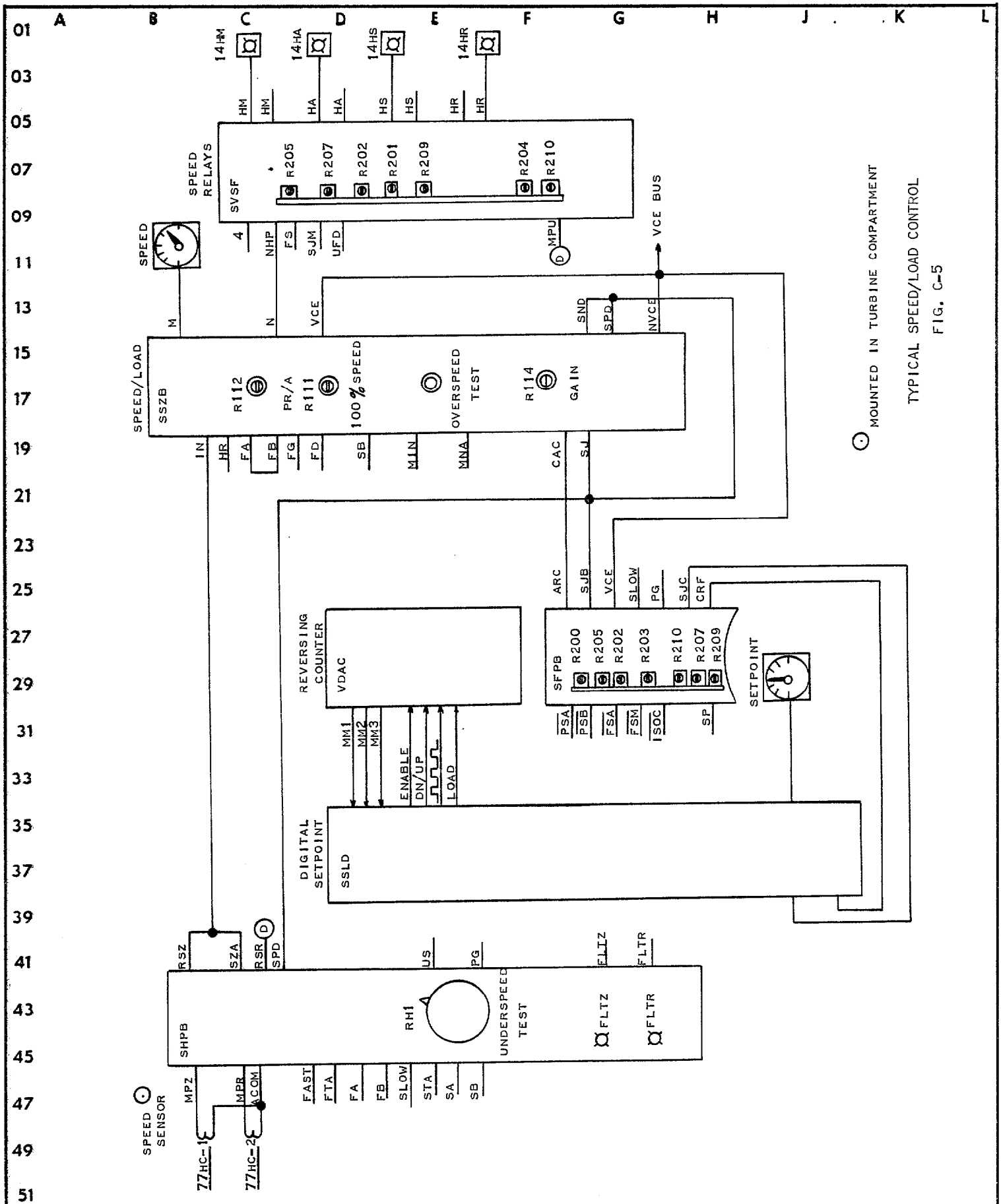
THE SSLD CARD RECEIVES AN ANALOG VOLTAGE FROM THE D/A CONVERTER OF THE VDAC CARD, COMPARES IT WITH AN ANALOG REFERENCE AND GENERATES THE STEERING LOGIC FOR THE UP/DOWN COUNTER ON THE VDAC CARD.

THE SSLD CARD ALSO HAS A PROGRAMMABLE PULSE GENERATOR. THE RESULTING OUTPUT PULSE RATE ESTABLISHES THE COUNTING RATE OF THE VDAC UP/DOWN COUNTER. THE NORMAL FREQUENCY PROVIDES FULL TRAVEL IN 12 MINUTES (4095 COUNTS). THIS BASIC RATE CAN BE DECREASED TO 36(MINUTES OR 48 MINUTES FOR FULL TRAVEL BY JUMPERING TERMINALS TRS OR FE, RESPECTIVELY, TO TERMINAL TW. THE RATE WILL BE MULTIPLIED BY 2,4,6, OR 8 BY LOGIC SIGNALS AT $\overline{\text{HLF}}$, $\overline{\text{QTR}}$, $\overline{\text{SXT}}$, OR AT, A "0" ON PST PROVIDES FULL TRAVEL IN APPROXIMATELY 1 SECOND.

THE SET POINT PRESET CIRCUIT ALLOWS PRESETTING THE VDAC COUNTER TO ANY TWO PRE-SELECTED VALUES.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RBM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		DIGITAL SETPOINT
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 46 SH. NO. 45

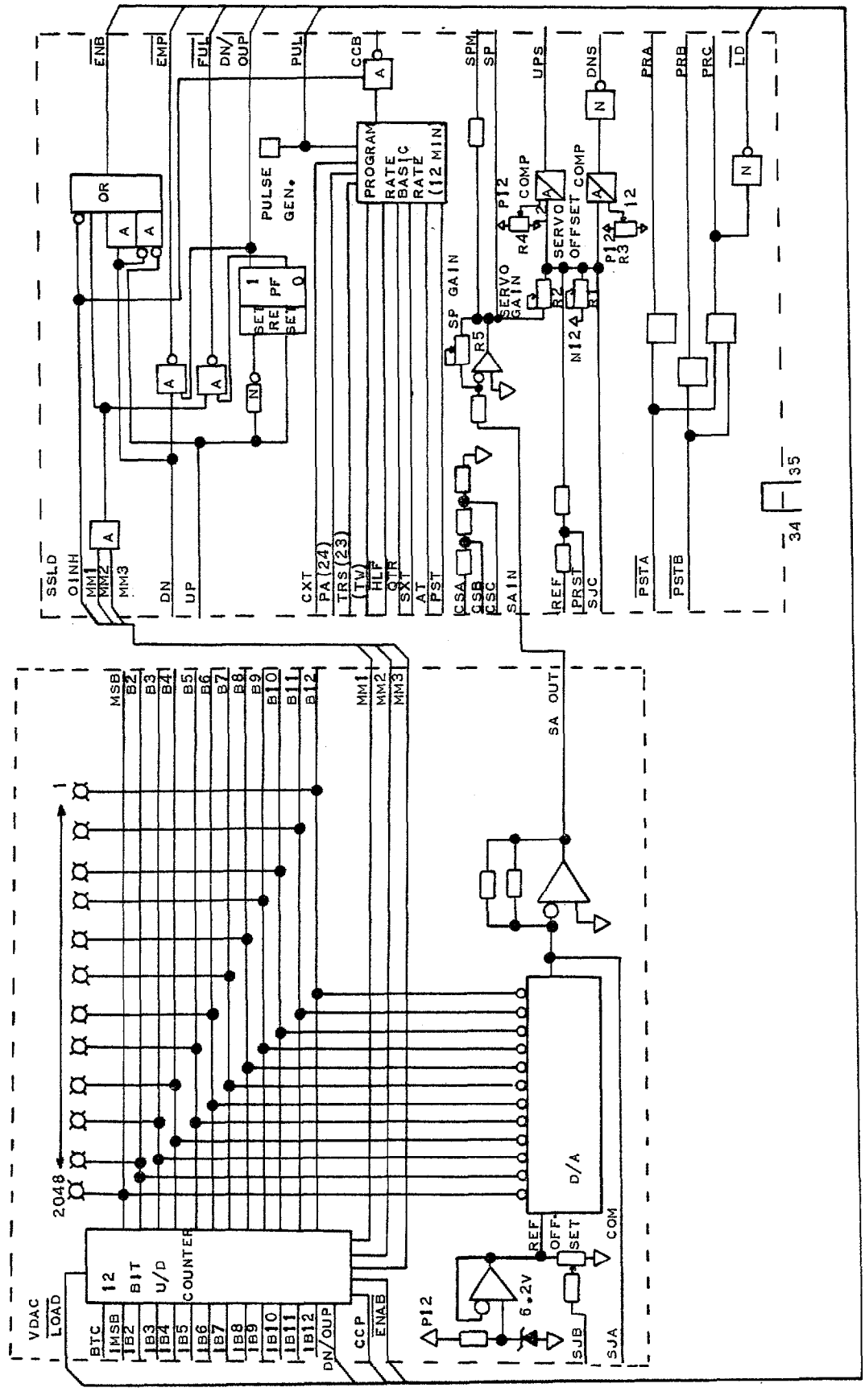
FORM 1C 2390 4-74



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.
REV. 3	MADE BY L. M. Worley			I.C. NO.	

ELEMENTARY DIAGRAM C-5
SPEED/LOAD CONTROL
GEK-41053
CONT. ON SH. 47 SH. NO. 46

FORM 1C 2390 4-74



DIGITAL SETPOINT CARDS

FIG. C-6

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-6
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		DIGITAL SETPOINT CARDS
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 48 SH. NO. 47

C-4 SPEED/LOAD CONTROL

THE SPEED CONTROL IS A CLOSED LOOP STSTEM AND MODULATES VCE TO CONTROL THE SPEED, LOAD AND ACCELERATION OF THE TURBINE. VARIOUS METHODS OF CONTROLLING THE SPEED AND LOAD ARE AVAILABLE. THE CONTROL SPECIFICATION AND CONTROL PANEL ELEMENTARY FOR A PARTICULAR JOB WILL REFLECT THE METHOD USED. A TYPICAL SPEED-LOAD CONTROL CONSISTS OF CARDS SHPB, SFPB, AND SSZB IN CONJUNCTION WITH THE DIGITAL SETPOINT. (SEE FIGURE C-5)

SHPB CARD - (REFER TO FIGURE C-7)

THE SHPB CARD CONTAINS SEVERAL CIRCUITS USED IN THE SPEED CONTROL LOOP PLUS CIRCUITRY USED ELSEWHERE. ONLY THE CIRCUITRY ASSOCIATED WITH THE SPEED CONTROL IS COVERED AT THIS TIME.

THE SHPB CARD PROVIDES A CROSS-COUPLING NETWORK SO THAT THE SPEED FEEDBACK SIGNAL FROM TWO MAGNETIC PICK-UP HEADS CAN BE COMBINED SUCH THAT A FAILURE ON EITHER OF THE SENSORS WILL NOT RESULT IN A LOSS OF THE FEEDBACK SIGNAL. OUTPUTS RSZ AND SZA ARE TIED TOGETHER FOR THIS FUNCTION. PULSES ON INPUT MPR ALSO RESULTS IN PULSES OUT OF RSR WHICH IS USED FOR ZERO SPEED INDICATION.

SHORT AND OPEN DETECTORS ARE PROVIDED FOR EACH SENSOR. A SHORT OR OPEN BETWEEN MPZ AND ACOM RESULTS IN A "1" ON FLTZ AND ILLUMINATES LIGHT FLTZ. A SHORT OR OPEN BETWEEN MPR AND ACOM GIVES A "1" ON FLTR AND ILLUMINATES LIGHT FLTR.

THE SHPB CARD ALSO CONTAINS AN UNDER-SPEED TEST CIRCUIT WHICH, WHEN ACTIVATED BY A LIMIT SWITCH ON POTENTIOMETER RH1, WILL SUPPLY A POSITIVE ADJUSTABLE CURRENT OUT OF PIN SPD. THIS SIGNAL MAY BE CONNECTED TO THE SPEED CONTROL SUMMING JUNCTION TO SIMULATE SPEED FEEDBACK AND REDUCE VCE. OUTPUT US WILL BE A "1" WHEN THE CIRCUIT IS ACTIVATED.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-4 SPEED/LOAD CONTROL
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 49 SH. NO. 48

ADJUSTMENT RANGES

US - RH1 - APPROXIMATELY 0 TO 0.45 MA.

SFPB CARD - (REFER TO FIGURE C-7)

THIS CARD PROVIDES TWO FUNCTIONS TO THE SPEED CONTROL CARD. FIRST, IT PROVIDES THE NECESSARY RESISTOR NETWORK TO OPERATE THE SYSTEM IN THE DROOP MODE. THE RESISTOR NETWORK MAY BE CONNECTED IN SUCH A WAY THAT IT IS ALWAYS IN THE CIRCUIT OR IT CAN BE SWITCHED IN AND OUT OF THE CIRCUIT. IF THE RESISTOR NETWORK IS TO BE PERMANENTLY CONNECTED IN A CIRCUIT, OUTPUT SJA IS USED. IF THE SWITCH FUNCTION IS DESIRED, SJB IS USED. WHEN SJB IS USED, $\overline{\text{TSOC}}$ MUST BE A "1" TO HAVE THE NETWORK SWITCHED IN. SECOND, THE SFPB CARD ALSO PROVIDES A SIGNAL THAT MAY BE USED TO REPROGRAM THE ACCELERATION RATE WHEN APPLICABLE. A "0" ON FSA WILL GIVE A FIXED BIAS OUT OF ARA.

ADJUSTMENTS

FSNL - R207
DROOP - R204

SSZB CARD - (REFER TO FIGURE C-7)

THE SSZB CARD CONTAINS A PULSE RATE TO ANALOG (PR/A) CONVERTER WHICH RECEIVES AN ALTERNATING SIGNAL AT INPUT IN AND PROVIDES A 0 TO 11 VOLT SIGNAL AT OUTPUT N. THIS SPEED FEEDBACK SIGNAL IS SUMMED WITH THE REFERENCE SIGNAL AND OTHER INPUT SIGNALS TO MODULATE NVCE TO OBTAIN THE DESIRED SPEED AND / OR LOAD. JUMPERS BETWEEN FA, FB, AND FC PROGRAM THE PR/A CONVERTER TO GIVE 0 TO 11 VOLTS FOR SPEED MACHINES FROM 0 TO 3600 RPM.

A "0" ON SB PUTS A POSITIVE BIAS ON THE SJ AND FORCES NVCE TO MIN LEVEL IF PIN SDJ IS CONNECTED TO SJ.

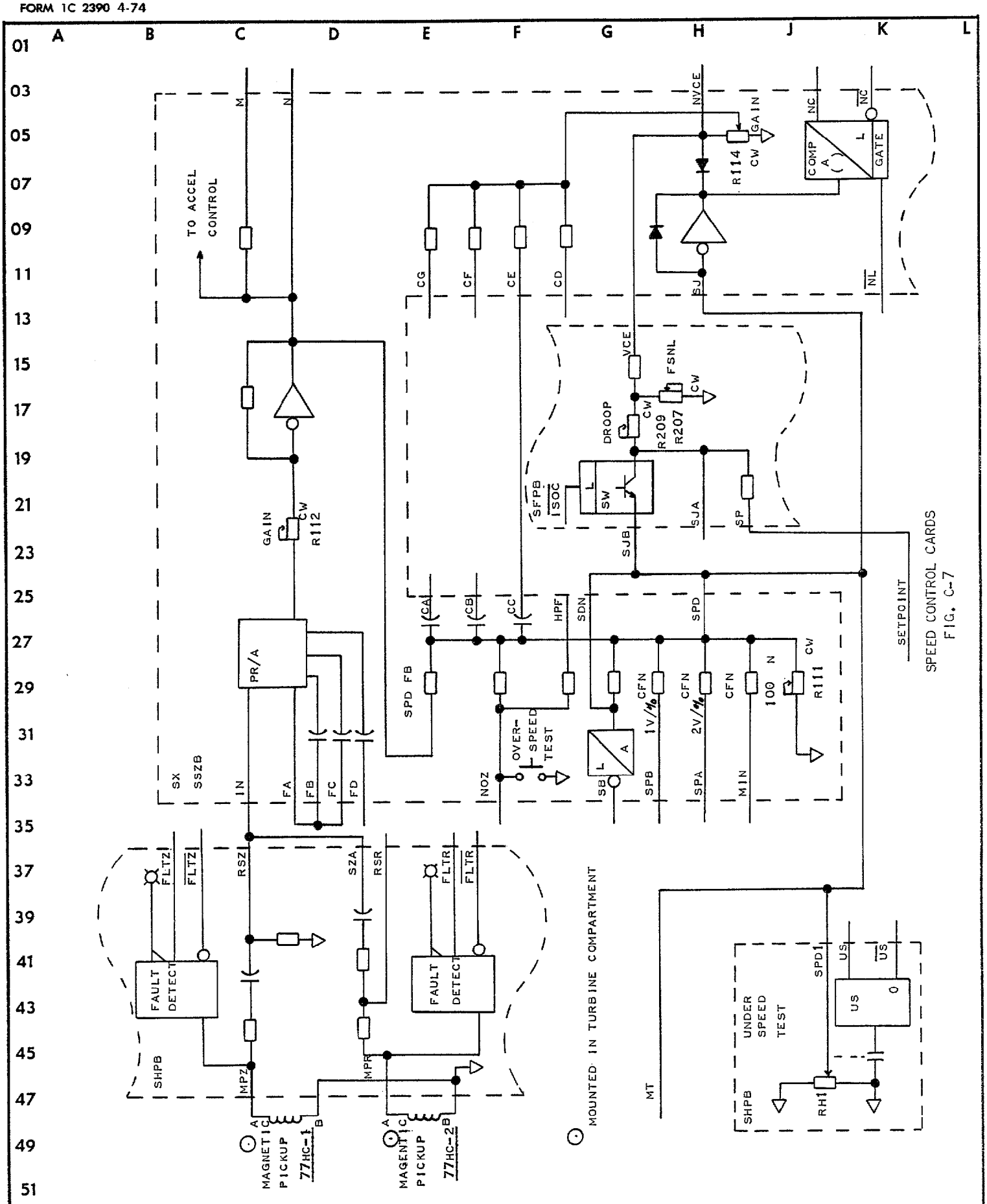
AN OVERSPEED TEST PUSHBUTTON, WHICH WILL FEED A NEGATIVE BIAS INTO SJ AND INCREASE NVCE IS PROVIDED FOR OVERSPEED TESTING.

JUMPERS ARE MADE BETWEEN PIN CC AND CE FOR THE DESIRED TRANSFER FUNCTION. POTENTIOMETER R114 PROVIDES A GAIN ADJUSTMENT. INPUTS SPA AND SPB PROVIDE INPUTS FOR THE DIGITAL SETPOINT (DSO) SIGNAL WHEN THE DROOP RESISTOR NETWORK ON CARD SFPB IS NOT USED (FOR THE MS7001, THE SFPB CARD RESISTOR NETWORK IS USED.) SPA IS USED WHEN A 2 VOLT DSP IS EQUAL TO 1% CHANGE IN SPEED, AND SPB IS USED WHEN 1 VOLT DSP IS EQUAL TO 1% CHANGE IN SPEED.

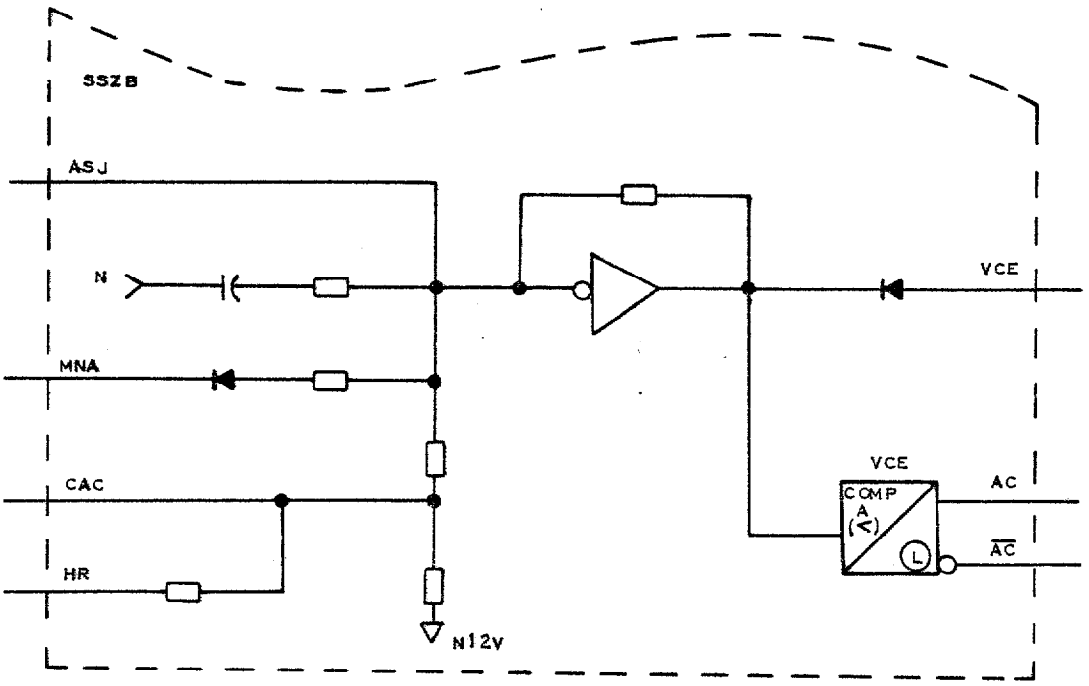
A DIFFERENTIATOR CIRCUIT LOOKS AT THE SPEED FEEDBACK SIGNAL N AND MODULATES VCE TO LIMIT THE ACCELERATIONS OF THE TURBINE. OUTPUT AC IS A "1" WHEN VCE IS UNDER ACCELERATION CONTROL. CONNECTING HR TO ACOM HALFS THE ACCELERATION CONTROL RATE. (SEE FIGURE C-8).

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-4
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED/LOAD CONTROL	
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 50	SH. NO. 49

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REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-7
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED/LOAD CONTROL
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 51 SH. NO. 50



ACCELERATION PORTION OF SSZB

FIG. C-8

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-8
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED/LOAD CONTROL
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 52 SH. NO. 51

ADJUSTMENTS

GAIN (PR/A) - R112
GAIN (100%N) - R114
100%N -R111 - 96% TO 104% RATED SPEED

C-5 TEMPERATURE CONTROL

THE TEMPERATURE FEEDBACK COMES FROM UP TO 12 CHROMEL-ALUMEL THERMOCOUPLES PLACED AT SELECTED POINTS OF THE TURBINE EXHAUST DUCT. THE THERMOCOUPLE SIGNALS ARE AVERAGED AND COLD JUNCTION COMPENSATED AND THEN SUMMED WITH REFERENCE SIGNALS TO MODULATE VCE. THE REFERENCE MAY CONSIST OF SEVERAL HARDWIRED SIGNALS AND TWO SELECTABLE SIGNALS. INCLUDED IN THESE SIGNALS ARE BASE REFERENCE, START-UP SUPPRESSION AMBIENT BIAS (AB), TRANSIENT OVERFIRE, PEAK LOAD (SELECTABLE), AND PEAK RESERVE LOAD (SELECTABLE). COMPENSATED AND UNCOMPENSATED SIGNALS ARE AVAILABLE FOR TEMPERATURE INDICATION ON VARIOUS METERS. A TYPICAL TEMPERATURE CONTROL LOOP CONSISTS OF CARDS STKK, SPOA, SHPB, SFPB AND A TEMPERATURE AVERAGING MODULE. (SEE FIGURES C-9 AND C-10). A THERMOCOUPLE GROUND IS DETECTED AND COMPENSATED FOR BY THE SGDE GARD.

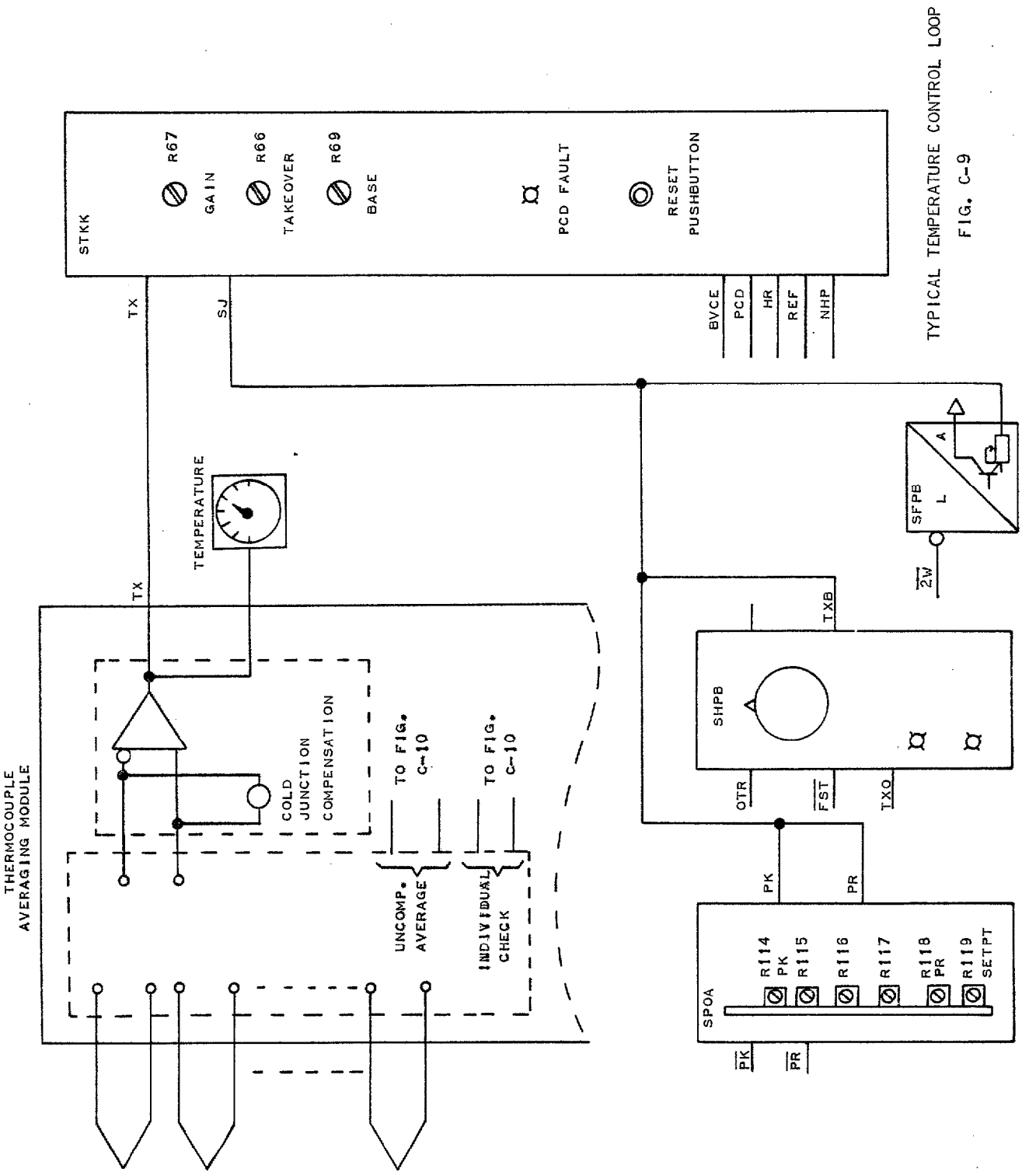
THERMOCOUPLE AVERAGING MODULE

THE INPUTS TO THE AVERAGING CABINET ARE TWELVE CHROMEL-ALUMEL THERMOCOUPLES MOUNTED IN THE TURBINE EXHAUST DUCT. THE TWELVE THERMOCOUPLES ARE CONNECTED IN PARALLEL TO PROVIDE AN AVERAGE SIGNAL. A 50 OHM RESISTOR IS PLACED IN EACH LEAD OF THE THERMOCOUPLES TO SWAMP OUT CIRCULATING CURRENTS DUE TO DIFFERENCES IN RESISTANCE OF THE THERMOCOUPLE WIRES. SWITCHES ARE PROVIDED SO THAT EACH THERMOCOUPLE CAN BE INDIVIDUALLY CHECKED. THE TOGGLE SWITCH CORRESPONDING TO THE THERMOCOUPLE TO BE CHECKED MUST BE PLACED IN THE CHECK POSITION FOR THIS FUNCTION. THE CHECK POSITION OF EACH THERMOCOUPLE SWITCH IS TIED TO A COMMON BUS; THEREFORE, IF MORE THAN ONE SWITCH IS IN THE CHECK POSITION THEIR AVERAGE WILL BE ON THE CHECK BUS. THE THERMOCOUPLE SELECTOR SWITCH MUST BE IN POSITION 21 TO CONNECT THE DIGITAL INDICATOR TO THE CHECK BUS. AN INDIVIDUAL CHECK SIGNAL FOR CONNECTION TO A RECORDER IS AVAILABLE AT +CR AND -CR. A THERMOCOUPLE CAN BE REMOVED FROM THE AVERAGE BY PLACING ITS CORRESPONDING SWITCH IN THE REJECT POSITION.

THE AVERAGING CABINET RECEIVES THE TWELVE THERMOCOUPLE INPUTS FROM THE TURBINE EXHAUST, AVERAGES THESE SIGNALS, COLD JUNCTION COMPENSATES THEM ISOLATES AND AMPLIFIES THEM USING AN STKJ CARD. A RESISTANCE TEMPERATURE DETECTOR IS USED FOR THE COLD JUNCTION COMPENSATION. A POTENTIOMETER IS PROVIDED TO ADJUST THE VOLTAGE DROP ACROSS THE RTD; HOWEVER, THIS IS SET UP IN THE FACTORY AND SHOULD REQUIRE NO FIELD ADJUSTMENT. TWO UNCOMPENSATED AVERAGES ARE PROVIDED. THE UNCOMPENSATED AVERAGE OUT OF +UA AND -UA GOES TO THE THERMOCOUPLE SWITCH POSITION 22. UNCOMPENSATED AVERAGE OUT OF +UA AND -UAR IS AVAILABLE FOR CONNECTING TO A RECORDER. CALIBRATION AND TESTING JACKS ARE ACCESSIBLE AT THE FRONT OF THE MODULE BY REMOVING THE COVER PLATE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-5
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TEMPERATURE CONTROL	
REV. 3	MADE BY L.M. Darley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 53	SH. NO. 52

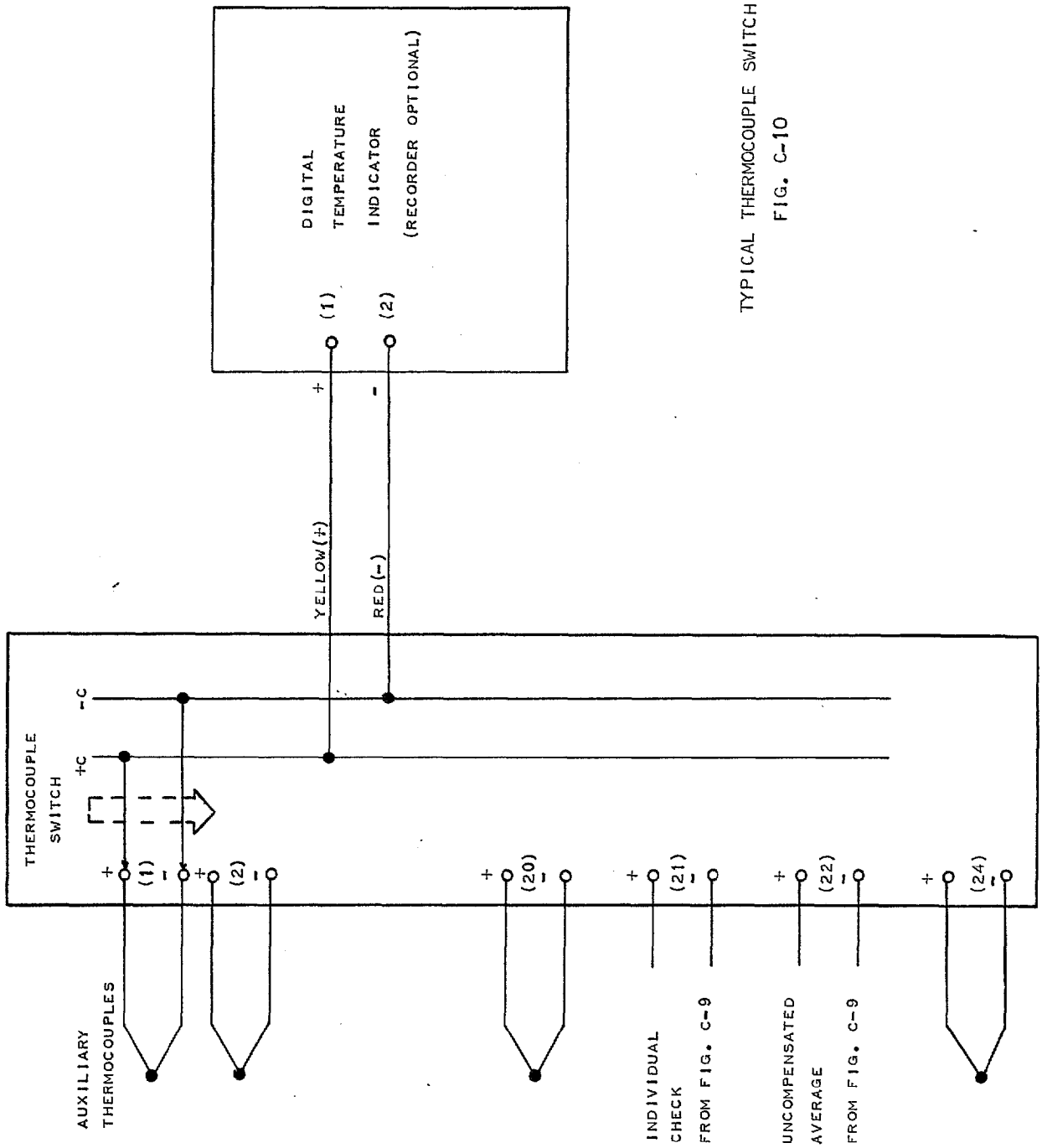
FORM 1C 2390 4-74



TYPICAL TEMPERATURE CONTROL LOOP
FIG. C-9

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RFM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-9
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TEMPERATURE CONTROL
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 54 SH. NO. 53

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TYPICAL THERMOCOUPLE SWITCH
FIG. C-10

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-10
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		THERMOCOUPLE SWITCH
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 55 SH. NO. 54

STKK CARD - (REFER TO FIGURE C-11)

THE STKK CARD RECEIVES THE EXHAUST TEMPERATURE SIGNAL (TX) FROM THE THERMOCOUPLE PROCESSING MODULE. TX IS SUMMED WITH THE REFERENCE SIGNALS AND MODULATES OUTPUT VCE TO CONTROL FUEL FLOW. WHEN TX IS IN CONTROL OF THE VCE BUS, OUTPUT TC IS A LOGIC 1 AND $\overline{TC}$ IS A LOGIC 0.

THE BASE REFERENCE FOR THE CONTROL IS GENERATED ON THE CARD AND IS ADJUSTABLE WITH R69. THE BASE REFERENCE IS ALSO AVAILABLE AT OUTPUT BASE. THE AMBIENT BIAS REFERENCE IS ALSO DEVELOPED ON THE CARD. THE AMBIENT BIAS CIRCUIT CONSISTS OF AN OPERATIONAL AMPLIFIER WITH AN OFFSET AND GAIN ADJUSTMENT. THE INPUT TO THE AMBIENT BIAS CIRCUIT CAN BE A 0 TO 10V SIGNAL ON BVCE, OR A 0 TO 5V SIGNAL ON PCD. A 10-50 MA INPUT SIGNAL MAY BE USED BY CONNECTING TO PCD AND JUMPERING PCD TO CSA/B AND CSA OR A 4-20 MA BY CONNECTING TO PCD AND JUMPERING PCD TO CSA/B AND CSB. THE OUTPUT OF THE AMBIENT BIAS AMPLIFIER IS AVAILABLE AT OUTPUT PIN PCDN. ADDITIONAL REFERENCE SIGNALS MAY BE CONNECTED TO INPUT SJ. TYPICALLY THE START-UP SUPPRESSION, PEAK AND PEAK RESERVE SIGNALS ARE CONNECTED TO SJ. INPUT HR IS AVAILABLE FOR CHANGING THE RANGE OF THE BASE ADJUSTMENT.

ADJUSTMENTS

TAKEOVER - R66 - MV/V AMPLIFIER OFFSET ADJUSTMENT
GAIN - R67 - MV/V AMPLIFIER GAIN ADJUSTMENT
BASE - R69 - BASE REFERENCE ADJUSTMENT

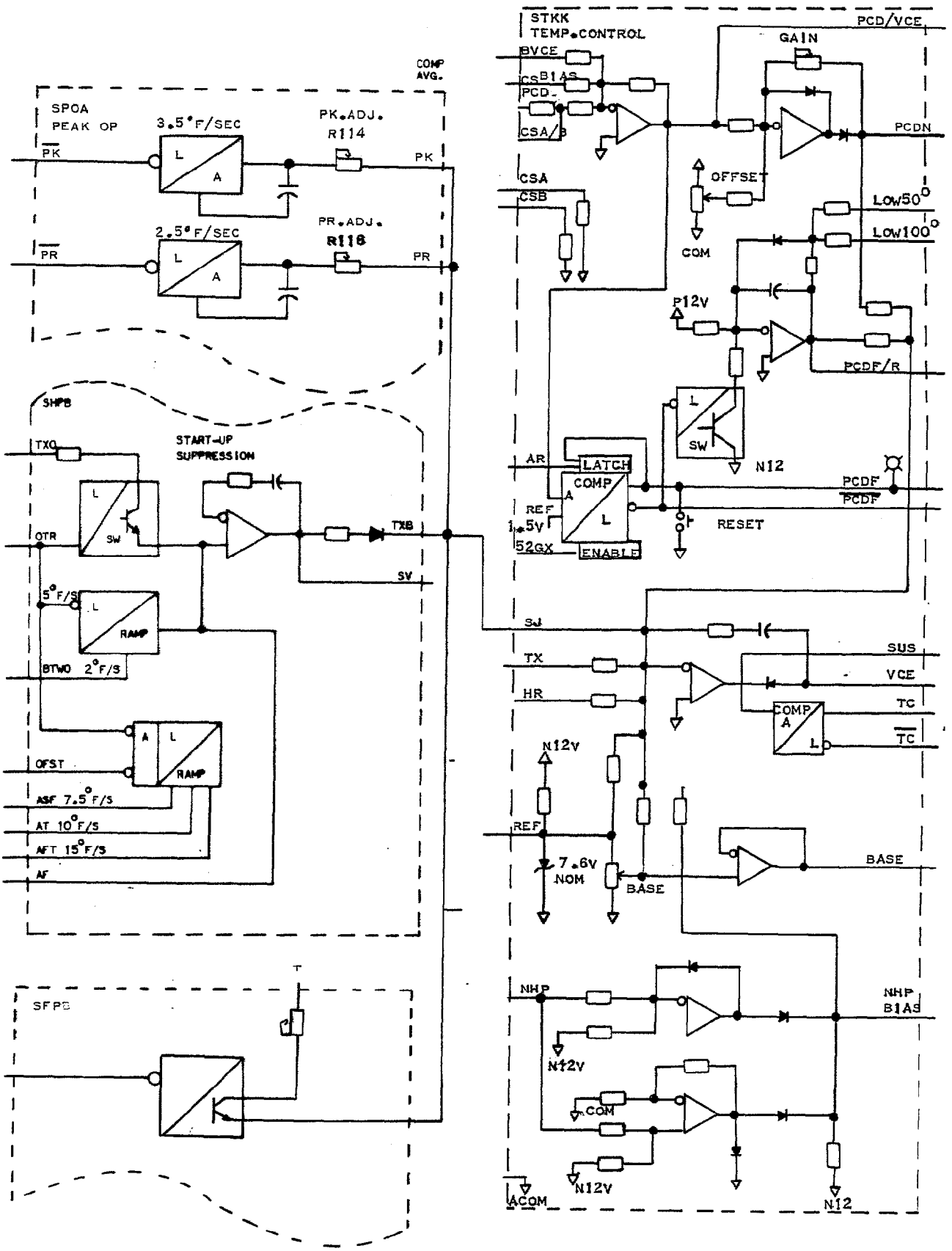
SPOA CARD - (REFER TO FIGURE C-11)

THE SPOA CARD CONTAINS THE CIRCUITRY TO GENERATE THE PEAK AND PEAK RESERVE BIAS FOR THE TEMPERATURE REGULATOR. IF BOTH PEAK AND PEAK RESERVE BIASES ARE DESIRED, OUTPUT PK AND PR ARE TIED TOGETHER AND CONNECTED TO SJ OF STKK. A "0" ON INPUT $\overline{PK}$ WILL RESULT IN A BIAS OUT OF PK THAT IS INCREASED AT A CONTROLLED RATE UNTIL THE LIMIT SET BY R114 IS REACHED. A "0" ON INPUT $\overline{PR}$ (INDEPENDENT OF INPUT $\overline{PK}$) WILL RESULT IN A BIAS OUT OF PR THAT IS INCREASED AT A CONTROLLED RATE UNTIL THE LIMIT SET BY R118 IS REACHED. TYPICALLY THE SEQUENTIAL LOGIC IS DESIGNED TO GENERATE A "0" ON $\overline{PK}$ WHENEVER $\overline{PR}$ IS "0". THIS MAKES PK AND PR ADDITIVE WHEN OPERATING IN THE PEAK RESERVE MODE.

ADJUSTMENTS

PK - R114 - 0-150°F ABOVE BASE
PR - R118 - 0-100°F ABOVE PEAK

FORM 1C 2390 4-74



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-11
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TEMPERATURE CONTROL CARDS
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 57 SH. NO. 56

TEMPERATURE SUPPRESSION DURING WARM-UP (FIGURE C-11)

DURING WARM-UP THE TEMPERATURE CONTROL BIAS FOLLOWS THE WARM-UP TEMPERATURE, WITH A BACK-UP TEMPERATURE ALLOWED. THE FOLLOWER CIRCUIT IS PROVIDED BY THE SHPB CARD. THE MAX WARM-UP TEMPERATURE BIAS IS PROVIDED BY THE SFPB CARD.

AN EXHAUST TEMPERATURE SIGNAL, PROVIDED BY THE STKK CARD, IS FED INTO THE SHPB BIAS CIRCUIT. THIS SIGNAL IS ENABLED DURING WARM-UP ONLY TO BIAS THE TEMPERATURE CONTROL ON THE STKK CARD. THEREFORE, THE START-UP SUPPRESSION BIAS FOLLOWS THE EXHAUST TEMPERATURE. WHEN WARM-UP IS COMPLETED, THE START-UP SUPPRESSION WILL BE REMOVED AT A RATE OF 5°F/SEC. THIS SUPPRESSION REMOVAL WILL START AT THE EXHAUST TEMPERATURE PRESENT AT THE END OF WARM-UP.

AS A BACK-UP, A MAXIMUM BIAS IS PROVIDED DURING WARM-UP BY THE SFPB CARD. SINCE THE TEMPERATURE BIAS SIGNALS ARE OF A MINIMUM SELECT NATURE (ONLY THE SMALLEST CONTROLS THE STKK CARD), THE MAXIMUM WARM-UP BIAS SETS A LIMIT FOR TX DURING WARM-UP.

C-6 FUEL SPLITTER (OPTIONAL)

A FUEL SPLITTER IS PROVIDED ON JOBS THAT HAVE BOTH GAS AND LIQUID FUEL SYSTEMS. THE JOB CONTROL SPECIFICATIONS AND CONTROL ELEMENTARY DIAGRAM WILL DEFINE THE SYSTEM THAT IS USED ON A PARTICULAR JOB.

THE FUEL SPLITTER RECEIVES ITS COMMAND FROM EITHER THE LOCAL SELECTOR SWITCH OR A REMOTE SWITCH. A PURGE PERIOD IS AVAILABLE WHEN GOING FROM FULL GAS TO LIQUID OR FROM FULL LIQUID TO GAS. THE FUEL MIXTURE CAN BE STOPPED AT ANY PROPORTION. THE FUEL SPLITTER CONSISTS OF CARD SFKD AND VRCC, WHICH ARE FUNCTIONALLY DESCRIBED BELOW. SEE FIGURE C-12 FOR A CONNECTION DIAGRAM.

VRCC CARD - (REFER TO FIGURE C-13)

THE VRCC CARD CONTAINS A 10 BIT REVERSIBLE COUNTER. THE RATE OF THE COUNTER IS DETERMINED BY THE FREQUENCY OF THE PULSES COMING IN ON PUL. THE DIRECTION OF THE COUNT IS CONTROLLED BY INPUTS ON ST1 AND STO. A "1" AN ST1 AND A "0" ON STO WILL STEER THE COUNTER UP. A "0" ON ST1 AND A "1" ON STO WILL STEER THE COUNTER DOWN. IF BOTH ST1 AND STO ARE "1", ALL BITS WILL FLIP ON THE NEXT COUNT PULSE. A "0" ON CL WILL CLEAR ALL BITS.

SFKD CARD - (REFER TO FIGURE C-13)

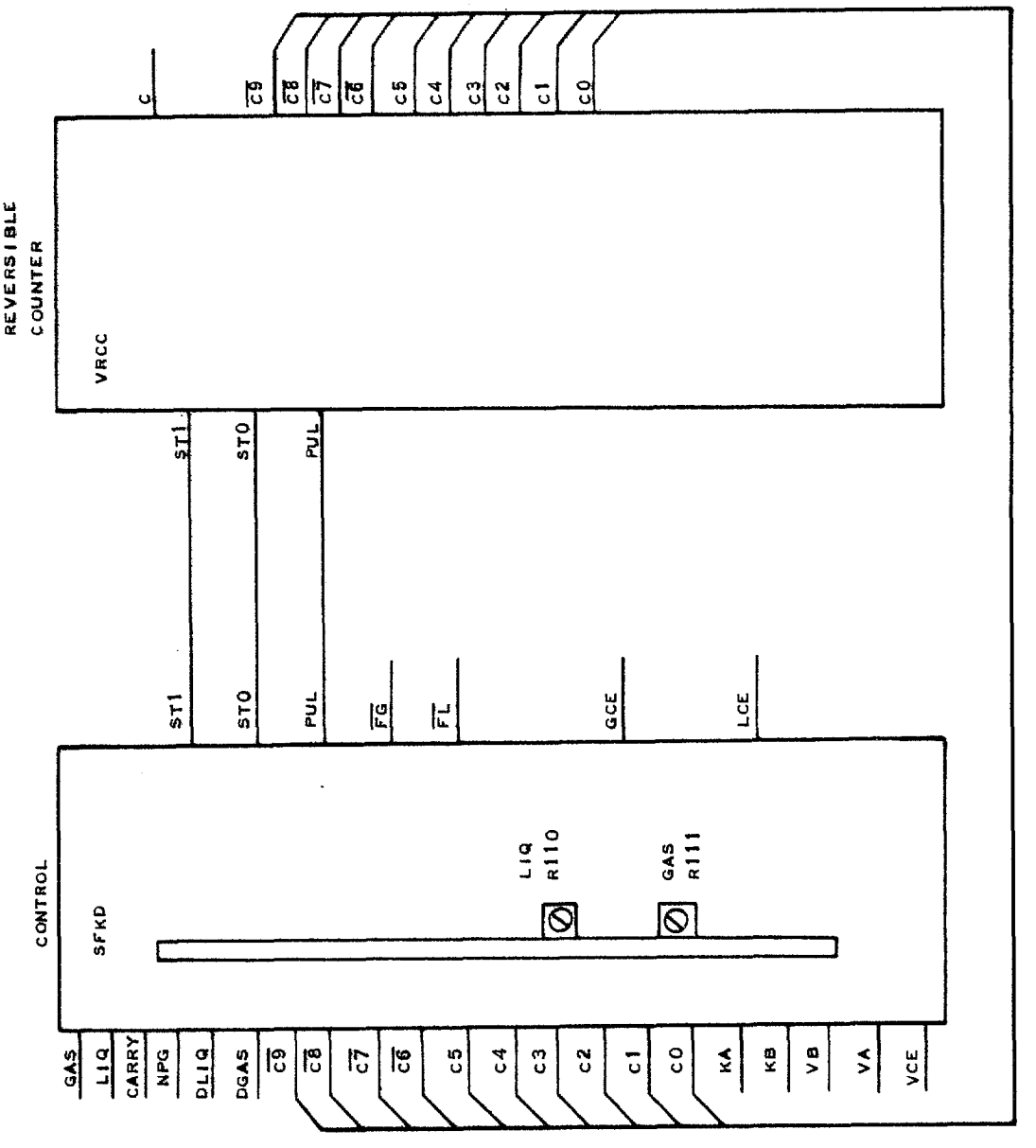
THE SFKD CARD, ON COMMAND OF THE FUEL SELECTOR SWITCH, GENERATES THE STEER SIGNALS FOR THE VRCC REVERSIBLE COUNTER. A"1" ON GAS STEERS THE COUNTER UP (OUTPUT ST1="1") AND A "1" ON LIQ STEERS THE COUNTER DOWN (OUTPUT STO="1"). OUTPUT PUL PROVIDES COUNT PULSES TO VRCC. PULSES ARE INHIBITED WHEN ANY OF THE FOLLOWING ARE TRUE.

1. LIQ AND GAS ARE BOTH "0" OR "1".
2. INPUT CARRY IS A "1".
3. DGAS IS "0" AND GAS IS "1".
4. OTD IS "1", NPG IS "0" AND LIQ IS "1".

INPUTS OTD AND DGAS PROVIDE MEANS OF ALLOWING A PURGE TIME AFTER SIGNALS LIQ OR GAS ARE RECEIVED. NPG PROVIDES A WAY TO BYPASS THE PURGE WHEN TRANSFERRING FROM GAS TO LIQUID. INPUT CAP MAY BE USED TO ALTER THE FREQUENCY OF THE COUNT PULSES ON PUL. OUTPUT $\overline{FG}$ IS A "0" WHEN FULLY ON GAS AND $\overline{FL}$ IS A "0" WHEN FULLY ON LIQUID.

THE OUTPUT OF THE VRCC COUNTER IS THE INPUT TO THE D/A CONVERTER ON THE SFKD CARD. THE SIGNAL ON INPUT VCE IS DIVIDED INTO TWO SIGNALS WITH THEIR VALUES DEPENDING ON THE OUTPUT OF THE D/A CONVERTER. A SIMPLIFIED DIAGRAM FOLLOWS. THIS DIAGRAM IS USED ONLY AS AN EXAMPLE AND DOES NOT INCLUDE ALL ASPECTS OF THE ACTUAL CIRCUIT.

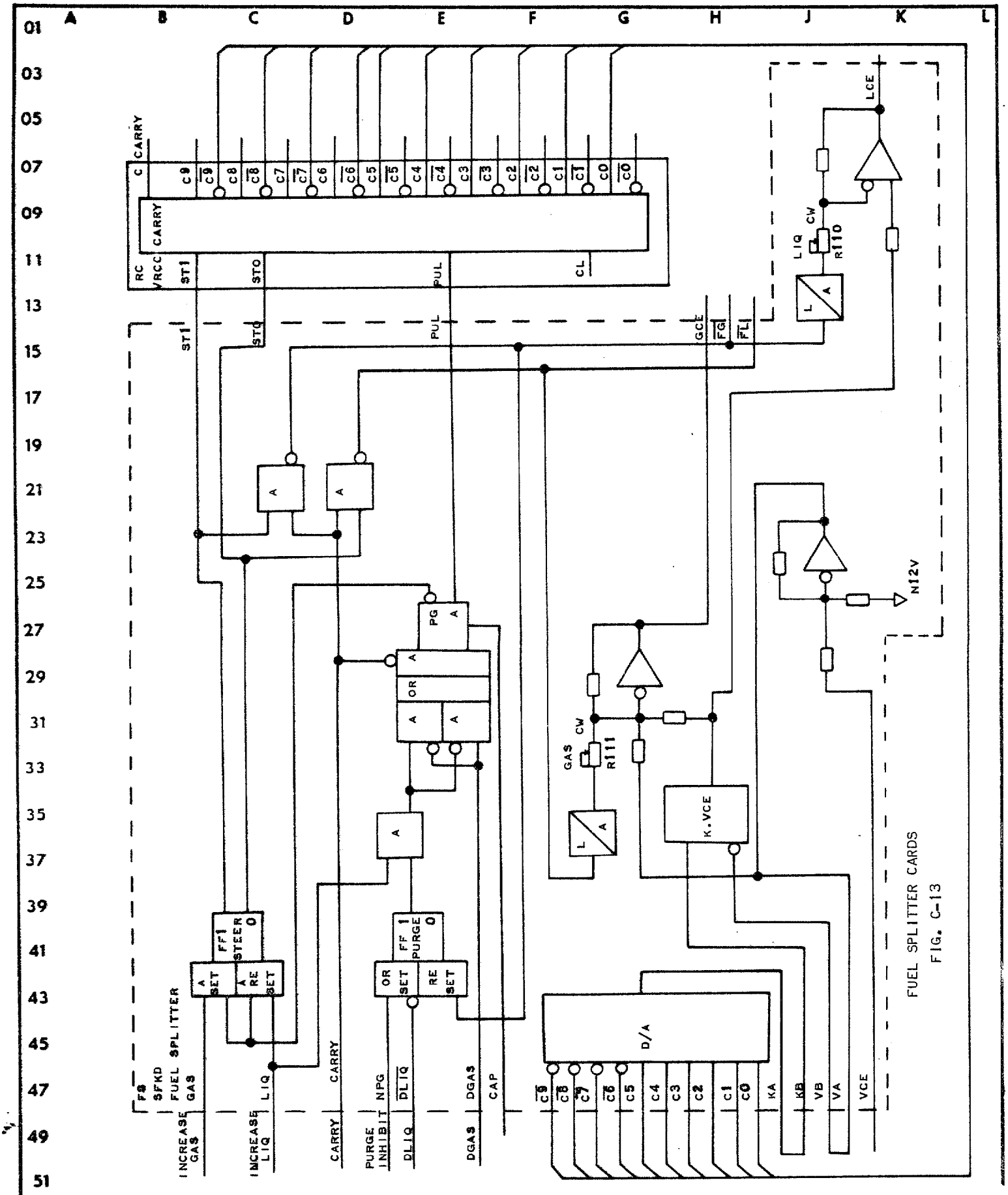
FORM 1C 2390 2-72



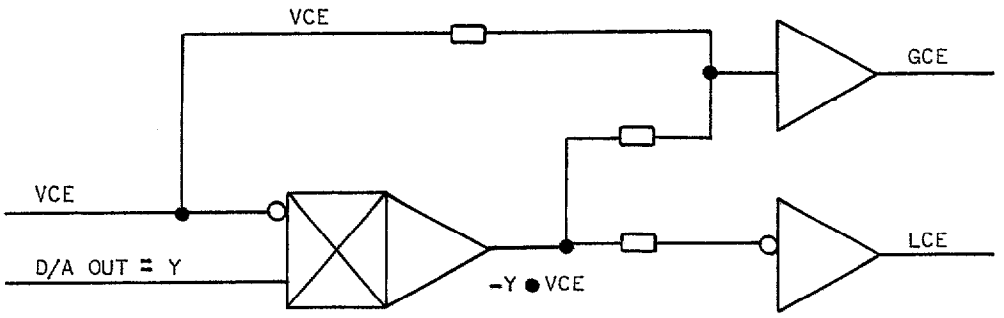
TYPICAL FUEL SPLITTER CIRCUIT CONNECTIONS
FIG. C-12

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-12
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		FUEL SPLITTER
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 59 SH. NO. 58

FORM 1C 2390 2-72



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-13
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	FUEL SPLITTER CARDS
REV. 3	MADE BY L M Worley			I.C. NO.		GEK-41053
						CONT. ON SH. 60 SH. NO. 59



FOR EXAMPLE, THE SIGNALS VCE, GCE AND LCE ARE EACH BROKEN INTO ACTIVE AND INACTIVE SIGNALS WITH THE INACTIVE SIGNAL BEING THAT PORTION REQUIRED TO CRACK THE VALVE. ALL ABOVE THIS IS CONSIDERED TO BE THE ACTIVE PART. THE ACTUAL CIRCUIT IS DESIGNED SUCH THAT THE ACTIVE PORTION OF LCE AND GCE ARE ALWAYS EQUAL TO THE ACTIVE PORTION OF VCE. LOOKING AT THE DIAGRAM AND CONSIDERING THE ACTIVE PORTION ONLY, IT IS SEEN THAT

$$LCE = (Y)(VCE).$$

WHERE Y IS PROPORTIONAL TO THE COUNT IN THE COUNTER. ALSO IT IS SEEN THAT

$$GCE = (1-Y)(VCE).$$

THEREFORE, WHEN Y EQUALS 1, LCE IS EQUAL TO VCE AND GCE EQUALS ZERO. WHEN Y EQUALS ZERO, LCE EQUALS ZERO AND GCE EQUALS VCE. THE COUNTER AND D/A CONVERTER ARE CONNECTED SUCH THAT WHEN THE COUNTER IS FULL, Y IS ZERO, AND WHEN THE COUNTER IS EMPTY, Y EQUALS 1. FOR MIX OPERATION, Y IS SOMEWHERE BETWEEN 0 AND 1. THREE TYPICAL FUEL TRANSFERS ARE GIVEN ON THE FOLLOWING PAGE.

ADJUSTMENTS

GAS - R111 - ADJUSTMENT FOR INACTIVE PORTION OF GCE

LIQ - R110 - ADJUSTMENT FOR INACTIVE PORTION OF LCE

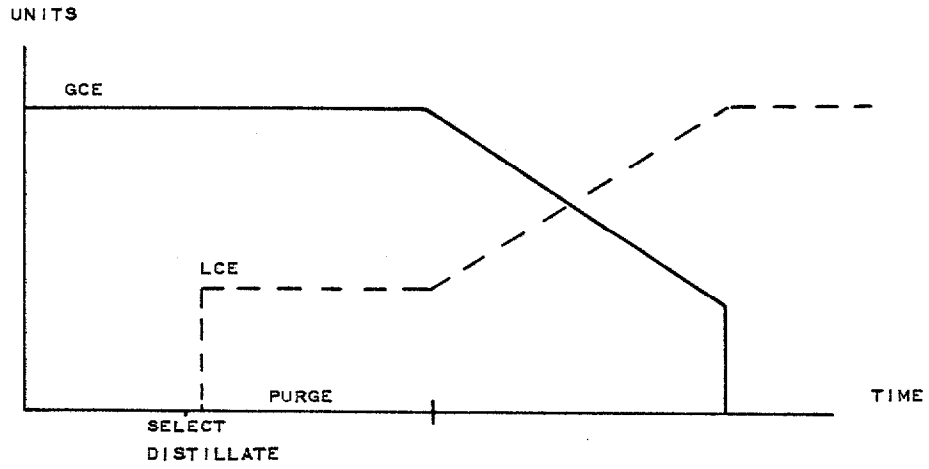
C-7 - UNIVERSAL LIQUID FUEL CONTROL (MODULATED BY-PASS VALVE)

THE UNIVERSAL LIQUID FUEL CONTROL SYSTEM CONSIST OF OUTER AND INNER CONTROL LOOPS. THE OUTER LOOP IS FOR FLOW CONTROL. IT CONSIST OF AN SFUA CARD (PROVIDING FLOW FEEDBACK DERIVED FROM MAG. PICKUPS MONITORING FLOW DIVIDER SPEED) AND AN SFKC CARD WHICH SUMS THE FLOW REFERENCE SIGNAL (FUNCTION OF VCE AND TURBINE SPEED) WITH THE FLOW FEEDBACK TO GENERATE A FLOW ERROR. SIGNAL. THE INNER LOOP IS FOR BYPASS POSITION CONTROL. IT CONSISTS OF AN SSVF CARD (PROVIDING VALVE POSITION FEEDBACK) AND AN SSVG CARD WHICH SUMS THE POSITION REFERENCE AND POSITION FEEDBACK AND PROVIDES FORWARD GAIN TO DRIVE THE LIQUID FUEL SERVO VALVE. FOR A LIQUID FUEL SYSTEM WITH MODULATED PUMP STROKE, ONLY THIS INNER POSITION LOOP IS NECESSARY. INTERFACE (OUTER LOOP FORWARD GAIN) BETWEEN THE LOOPS IS PROVIDED BY TWO CIRCUITS ON AN AOAH CARD. THE AOAH(D) CIRCUIT INTEGRATES THE FLOW ERROR TO GENERATE THE POSITION REFERENCE SIGNAL, THUS FORMING AN INTEGRAL FLOW CONTROL LOOP. THE OTHER CIRCUIT, AOAH(A), PROVIDES A FAST RAMP OR INITILIZATION EFFECT TO A PRE DEFINED VALVE POSITION WHEN VCE GOES TO THE "FIRE" LEVEL OR WHEN CHANGING FROM COMPLETE GAS TO LIQUID, TO MINIMIZE EFFECTS OF INTEGRATOR WINDUP.

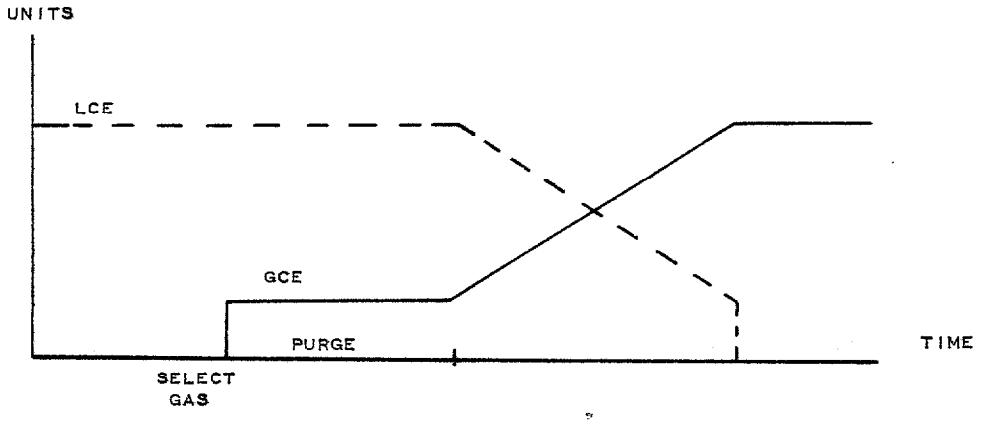
SFUA CARD (REFER TO FIGURE C-15)

THE SFUA CARD CONTAINS TWO PR/A CONVERTERS. EACH OF THESE CONVERTERS RECEIVES A SIGNAL FROM A SEPARATE SENSOR AND PROVIDES AN ANALOG SIGNAL AT OUT 1 AND OUT 2 RESPECTIVELY WHICH IS PROPORTIONAL TO THE FREQUENCY OF THE SIGNAL FROM THE SENSOR. THE LARGEST SIGNAL FROM THE TWO PR/A CONVERTERS IS SELECTED AND AMPLIFIED TO PROVIDE A NEGATIVE ANALOG SIGNAL OUT OF FFN TO BE USED AS THE FUEL FLOW FEEDBACK SIGNAL TO THE SFKC REGULATOR.

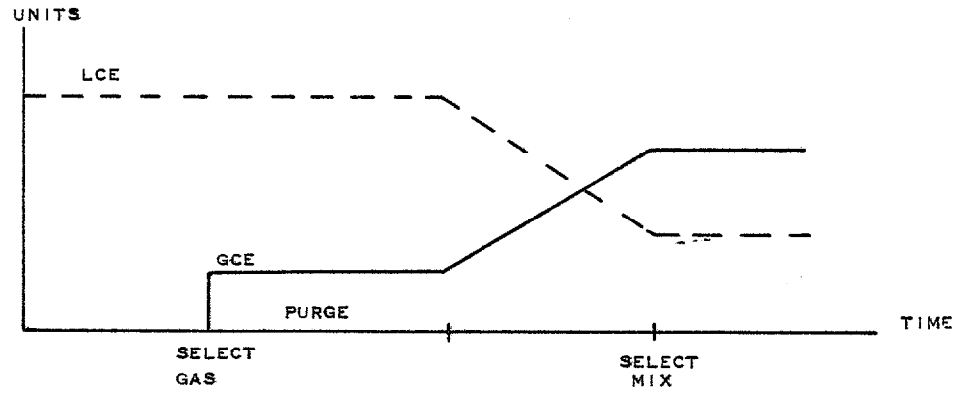
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-7
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LIQUID FUEL CONTROL
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 61 SH. NO. 60



TRANSFER FROM FULL GAS TO FULL DISTILLATE



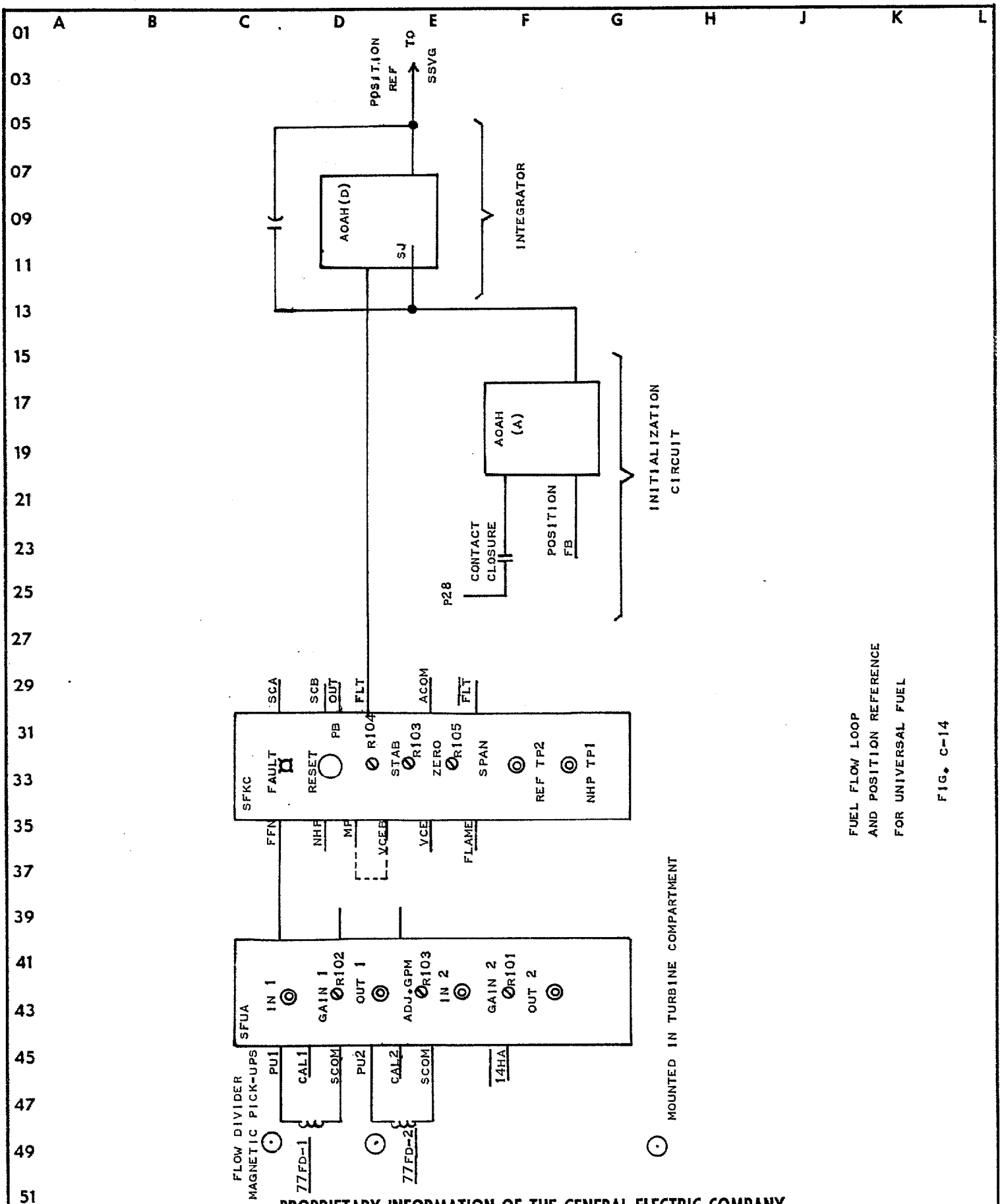
TRANSFER FROM FULL DISTILLATE TO FULL GAS



TRANSFER FROM FULL DISTILLATE TO MIXTURE

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	C-6
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		FUEL TRANSFER	
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 62	SH. NO. 61

FORM 1C 2390 4-74



FUEL FLOW LOOP
AND POSITION REFERENCE
FOR UNIVERSAL FUEL

FIG. C-14

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

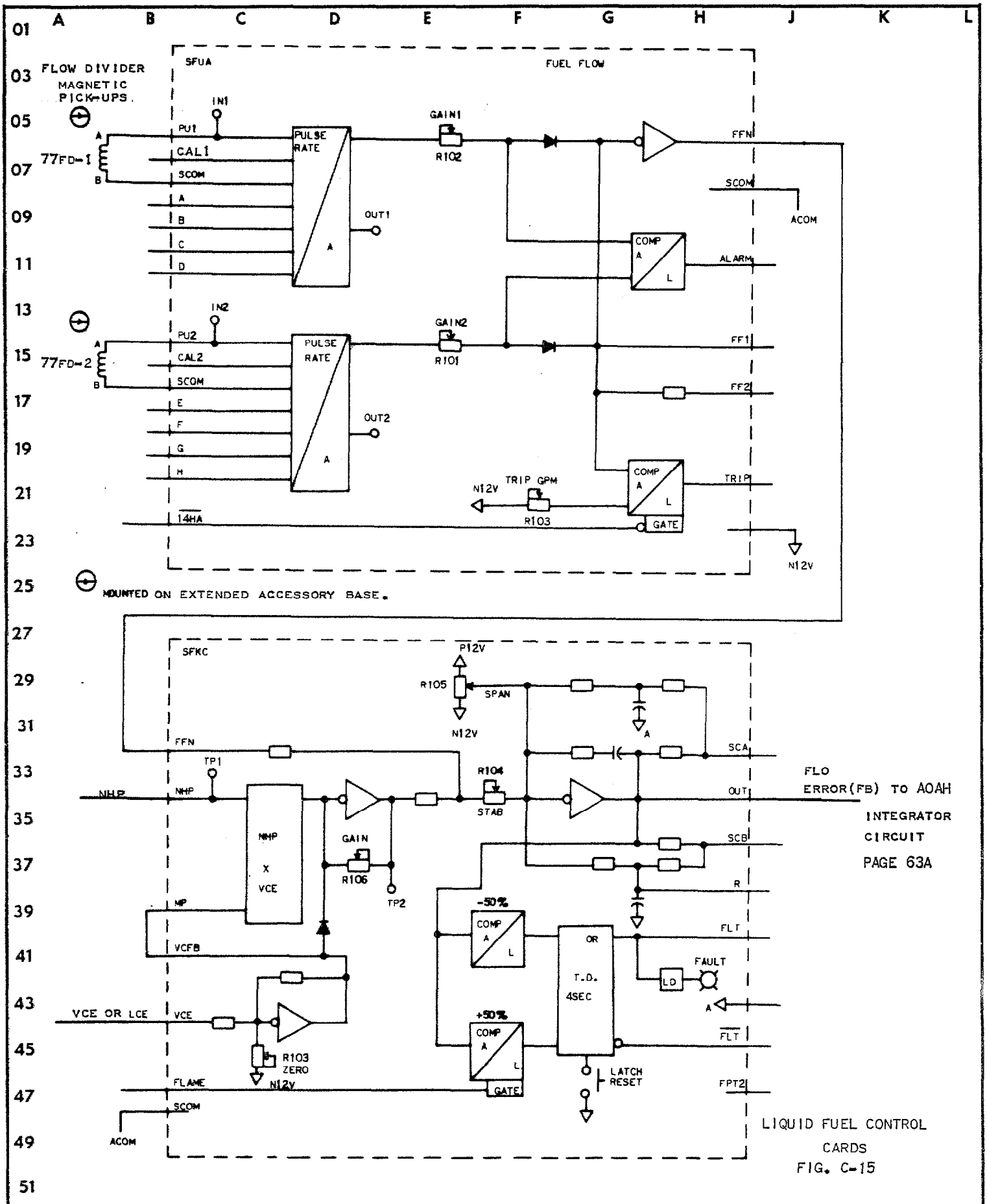
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS
REV. 2	ISSUED			FIRST MADE FOR
REV. 3	MADE BY			I.C. NO.

GENERAL ELECTRIC
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
UNIVERSAL FUEL
FUEL FLOW LOOP
C-14

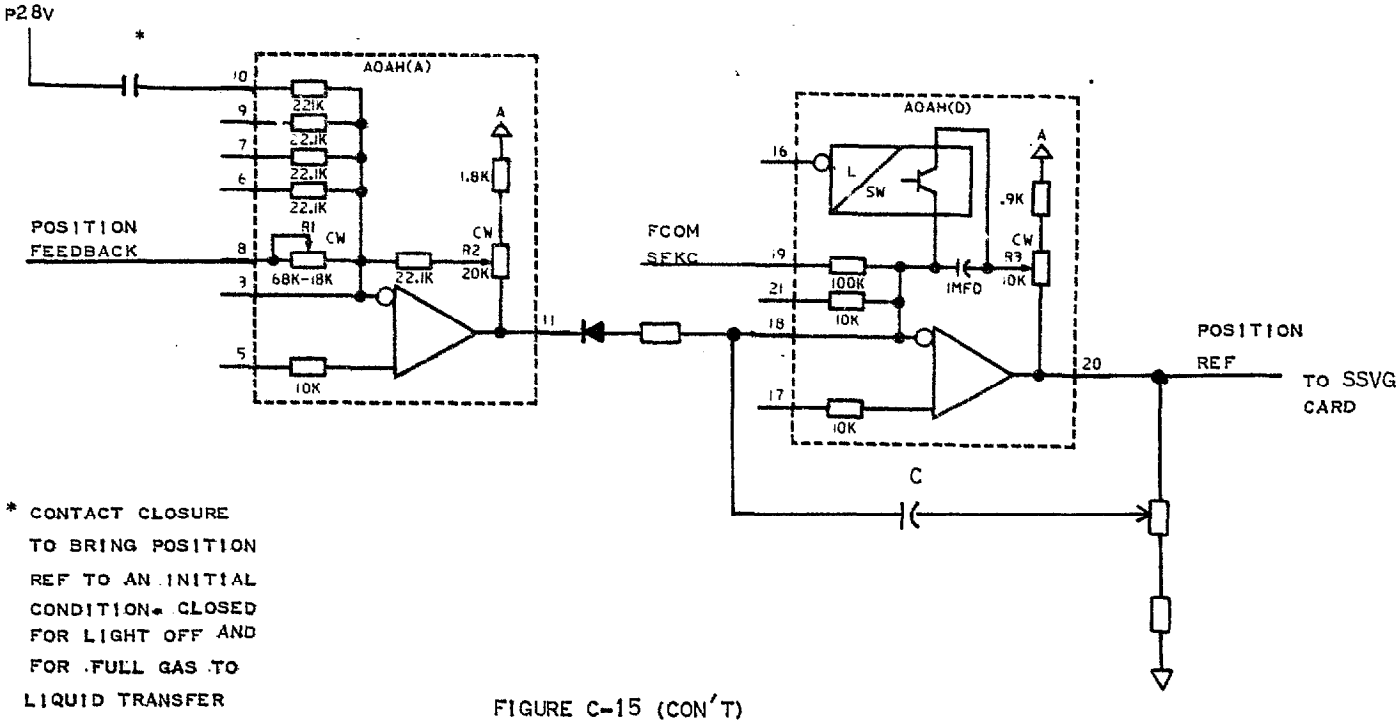
GEK41053

CONT. ON SH. 63 SH. NO. 62



LIQUID FUEL CONTROL
CARDS
FIG. C-15

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-15
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		LIQUID FUEL CONTROL CARDS
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 63A SH. NO. 63



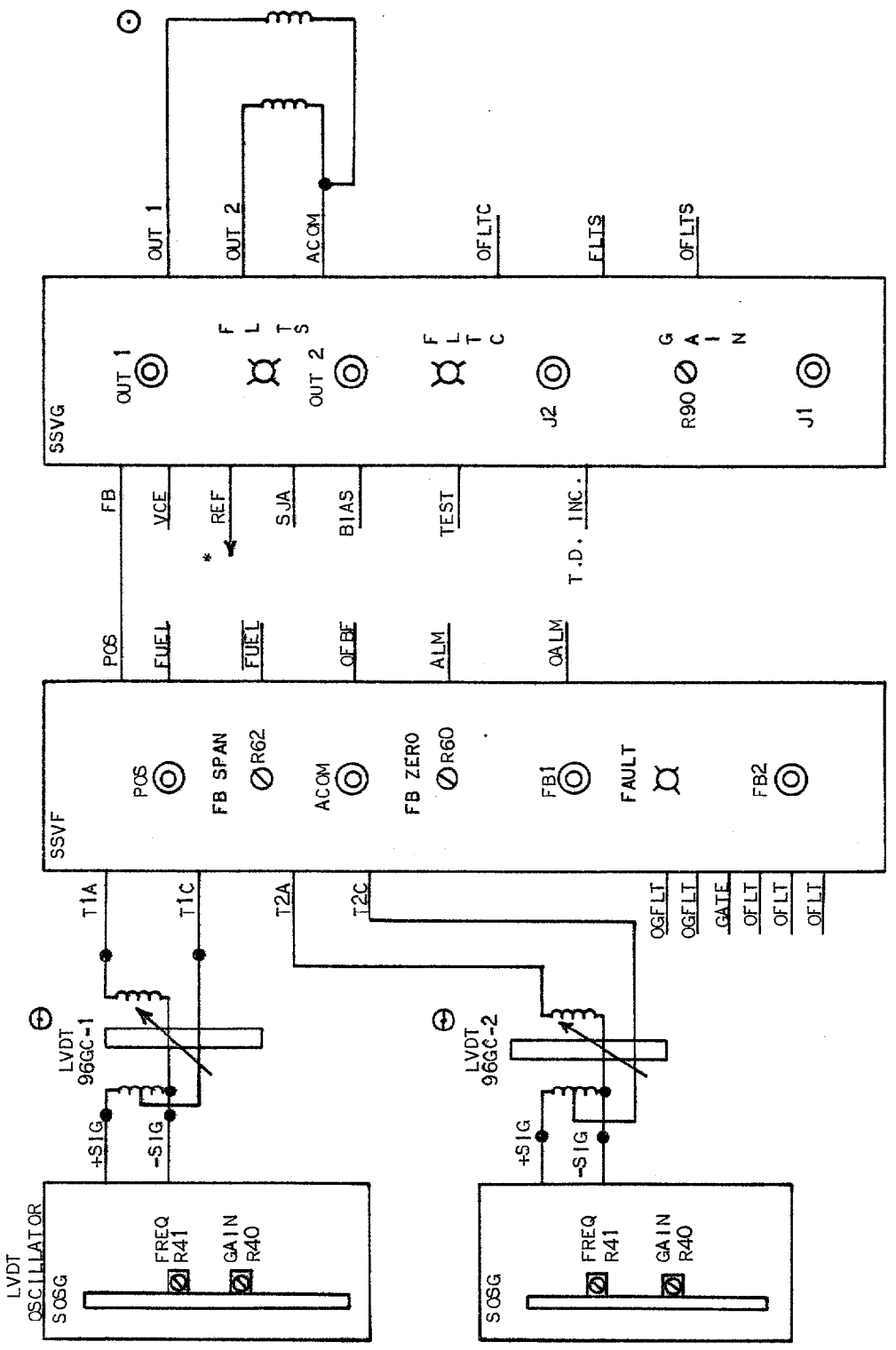
* CONTACT CLOSURE TO BRING POSITION REF TO AN INITIAL CONDITION. CLOSED FOR LIGHT OFF AND FOR FULL GAS TO LIQUID TRANSFER

FIGURE C-15 (CON'T)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM INITIALIZATION AND INTEGRATION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 64 SH. NO. 63A

FORM 1C 2390 4-74



* GCE FOR GAS CONTROL
REF (SEE FIG. C-15) FOR UNIVERSAL LIQUID COUNTER
LCE FOR OPTIONAL LIQUID CONTROL

TYPICAL FUEL VALVE CONTROL
FIG. C-16

⊕ MOUNTED ON EXTENDED ACCESSORY BASE

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-16
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TYPICAL FUEL VALVE CONTROL
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 65 SH. NO. 64

DIFFERENT FREQUENCY RANGES AT THE INPUT TO THE PR/A CONVERTER CAN BE PROCESSED BY MAKING THE APPROPRIATE JUMPERS BETWEEN A,B,C AND D FOR THE FIRST PR/A CONVERTER AND E,F,G AND H FOR THE SECOND PR/A CONVERTER. THE FOLLOWING TABLE GIVES REQUIRED JUMPERS FOR VARIOUS RANGES. A MS7001C GAS TURBINE AS AN INPUT FREQUENCY RANGE OF 0 TO 3340 HZ.

FREQUENCY RANGES, HZ	JUMPERS
0 TO 4290	NO JUMPERS
0 TO 3340	A-D AND E-H
0 TO 2640	A-C AND E-G
0 TO 2180	A-B AND E-F
0 TO 1910	A-B-D AND E-F-H
0 TO 1660	A-B-C AND E-F-G
0 TO 1440	A-B-C AND E-F-G-H

A COMPARATOR LOOKS AT THE TWO ANALOG SIGNALS FROM THE PR/A CONVERTERS AND OUTPUT ALARM BECOMES A "1" IF THEY DIFFER BY MORE THAN A FIXED PERCENTAGE OF THE NOMINAL SIGNAL. THE SELECTED HIGH SIGNAL, OUT OF THE CONVERTER, IS ALSO COMPARED TO THE FUEL FLOW REFERENCE AND GIVES A "1" OUT OF TRIP IF THE FEEDBACK SIGNAL IS GREATER THAN THE REFERENCE AND INPUT 14HA IS A "1". WHEN INPUT 14HA GOES TO A "0" OUTPUT TRIP IS INHIBITED.

ADJUSTMENTS

GAIN 1 - R102
GAIN 2 - R101
TRIP - R103

SFKC CARD - (REFER TO FIGURE C-15)

THE SFKC CARD CONTAINS THE FUEL REGULATOR AMPLIFIER, A MULTIPLIER, AND SERVO VALVE FAULT DETECTOR. THE REFERENCE TO THE REGULATOR AMPLIFIER IS THE PRODUCT OF THE SIGNALS ON INPUTS MP AND NHP. TYPICALLY ONE REFERENCE INPUT IS RECEIVED AT VCE AND AN OFFSET IS ADDED TO REMOVE THE INACTIVE PORTION OF THE SIGNAL. THE OFFSET SIGNAL IS INVERTED AND APPEARS ON PIN VCEB. VCEB IS CONNECTED TO MP TO PROVIDE ONE INPUT TO THE MULTIPLIER. THE REGULATOR SUMS THE REFERENCE SIGNAL FROM THE MULTIPLIER WITH THE FEEDBACK SIGNAL ON FFN AND PROVIDES A REFERENCE TO THE SERVO CARD SSVG. REFER TO PAGE 65 FOR A DESCRIPTION OF THE POSITION SERVO SECTION OF THIS CIRCUIT.

ADJUSTMENTS

ZERO - R103 1.3 TO 2.6 VDC
STAB - R104
SPAN - R105

C-8 FUEL POSITION SERVO

A TYPICAL DIAGRAM OF THE GAS OR LIQUID CONTROL VALVE LOOP IS SHOWN IN FIGURE C-16. THE SSVG REGULATOR CARD RECEIVES A FEEDBACK SIGNAL FROM SSVF AND COMPARES IT TO A FUEL REFERENCE SIGNAL TO MODULATE THE SERVO-VALVE. THE FEEDBACK IS DEVELOPED USING A LINEAR, VARIABLE DISPLACEMENT TRANSFORMER (LVDT) WHOSE COUPLING CO-EFFICIENCY IS A FUNCTION OF THE CONTROL VALVE POSITION. AN OSCILLATOR OF APPROXIMATELY 3 KHz PROVIDES THE INPUT FOR THE LVDT. THE AMPLITUDE OF LVDT SECONDARY VARIES WITH THE POSITION OF THE CONTROL VALVE. TWO LVDT'S AND OSCILLATORS MAY BE PROVIDED FOR INCREASED RELIABILITY. AN ALARM IS GENERATED IF THE FEEDBACK CIRCUIT FAILS.

SOSG CARD

THE SOSG CARD CONTAINS AN ADJUSTABLE OSCILLATOR TO SUPPLY A SIGNAL FOR THE LVDT'S ONE SOSG CARD CAN DRIVE A 200 MA RMS LOAD AT 6 TO 8 VOLTS. THIS WILL DRIVE UP TO 4 LVDT'S.

ADJUSTMENT

GAIN - R40 - 4.5 TO 8.5 VOLTS RMS
FREQ - R41 - 2350 TO 3359 HZ.

SSVF CARD - (REFER TO FIGURE C-17)

THE SSVF CARD RECEIVES ITS INPUT(S) SIGNAL FROM THE LVDT AND DETECTS THE PEAK VALUE OF THE SIGNAL(S) AND AMPLIFIES IT TO PROVIDE A NEGATIVE FEEDBACK SIGNAL ON POS FOR THE SSVG REGULATOR CARD. IF EITHER OF THE SIGNALS OUT OF THE PEAK DETECTORS GOES BELOW A SET THRESHHOLD, OUTPUT OFBF BECOMES A "1". A MINIMUM NO FUEL SIGNAL OF 0.5 VOLTS RMS IS REQUIRED FROM THE LVDT'S TO PREVENT THE ALARM SIGNAL ON OFBF. A FUEL DETECTOR CIRCUIT IS ALSO PROVIDED TO GIVE A "1" ON OUTPUT FUEL WHEN THE FEEDBACK SIGNAL REACHES A SET THRESHHOLD.

ADJUSTMENT

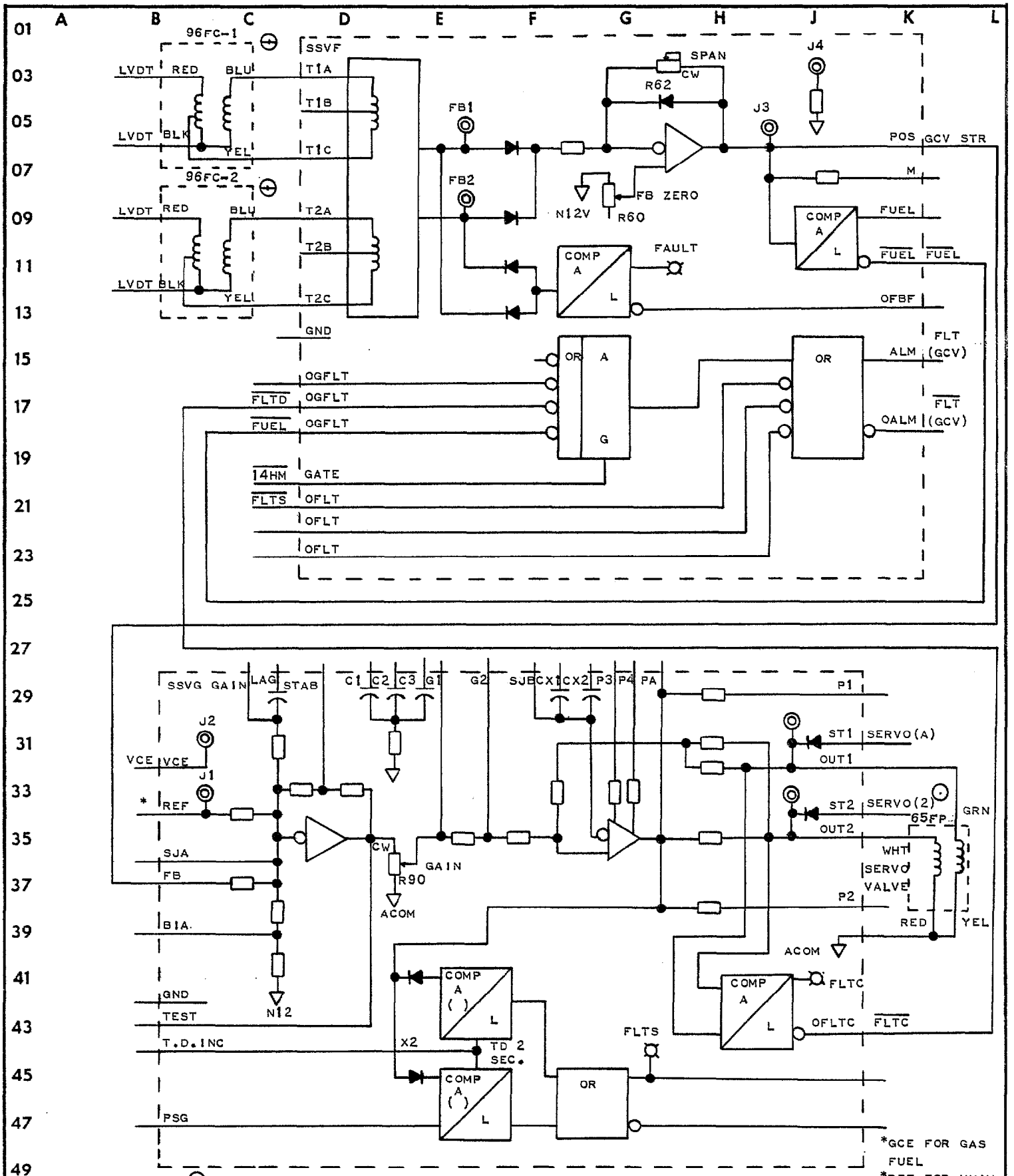
FB ZERO - R60
FB SPAN - R62

SSVG CARD - (REFER TO FIGURE C-17)

THE SSVG CARD SUMS THE FEEDBACK SIGNAL ON FB WITH THE REFERENCE SIGNAL ON REF TO MODULATE THE SERVOVALVE. A SIGNAL ON INPUT VCE CAN BE USED AS THE REFERENCE BY JUMPERING J2 TO J1 AND REMOVING THE SIGNAL ON REF. A NEGATIVE BIAS EQUIVALENT TO 2 VOLTS ON REF IS ALSO SUMMED WITH THE REFERENCE TO HOLD THE GAS CONTROL VALVE CLOSED UNTIL THE REFERENCE IS GREATER THAN 2 VOLTS. A VOLTAGE MAY BE PUT ON ST1 TO INPOSE A BIAS DIRECTLY ON THE SERVO VALVE.

THE CARD CONTAINS A COMPARATOR CIRCUIT WHICH WILL GIVE A "1" ON FLTS AND ILLUMINATE THE LIGHT FLTS IF THE ABSOLUTE VALUE OF THE OUTPUT VOLTAGE STAYS ABOVE 3.5 VOLTS FOR AN EXTENDED PERIOD OF TIME. A COMPARATOR CIRCUIT IS ALSO PROVIDED TO DETECT AN OPEN OR SHORTED SERVOVALVE COIL. OUTPUT OFLTC WILL BECOMES A "0" FOR AN OPEN OR SHORTED COIL.

FORM 1C 2390 4-74



FUEL CONTROL CARDS (POSITION SERVO)
FIG. C-17

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM FUEL CONTROL (POSITION) SERVO C-17
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 68 SH. NO. 67

ADJUSTMENT

GAIN - R90

C-9 SPEED RATIO VALVE (GAS FUEL)

WHEN A GAS FUEL SYSTEM IS USED, A SPEED RATIO VALVE IS REQUIRED TO MAINTAIN THE PRESSURE UP STREAM OF THE GAS CONTROL VALVE AT A VALUE THAT IS A FUNCTION OF TURBINE SPEED. (SEE FIGURE C-18). THE REST OF THE GAS CONTROL IS IDENTICAL TO THE POSITION LOOP (SSVF,SSVG) DESCRIBED UNDER UNIVERSAL FUEL IN THE PREVIOUS SECTION.

SSVD CARD - (REFER TO FIGURE C-19)

THE SSVD CARD RECEIVES A SIGNAL ON P1 THAT IS PROPORTIONAL TO PRESSURE. THIS SIGNAL IS COMPARED WITH THE SPEED SIGNAL ON NHP AND PROVIDES AN OUTPUT ON TRD. AN OFFSET AND GAIN IS PROVIDED ON THE NHP SIGNAL. THE SIGNAL ON TRD IS THE REFERENCE FOR THE SERVOVALVE REGULATOR ON THE CARD. A "1" ON INPUT SR WILL FORCE TRD TO ZERO VOLTS.

THE FEEDBACK SIGNAL FOR THE REGULATOR IS DERIVED USING A LVDT AS WAS DONE IN THE GAS CONTROL VALVE. ONLY ONE LVDT IS USED IN THIS CASE.

ALARM OUTPUT FLT WILL BE A "1" AND FAULT LIGHT WILL BE ON FOR THE FOLLOWING:

(1) IF TPC GOES MORE POSITIVE THAN -1.2 VOLTS INDICATING LOSS OF FEEDBACK.
(2) IF INPUT D IS NOT BETWEEN -1.2 AND -2.8V AND INPUT INH IS A "0".
(3) WHEN TWO SSVD'S ARE USED TO DRIVE THE SERVOVALVE, OUTPUT C OF EACH CARD CAN BE TIED TO OUTPUT SCB ON THE OTHER CARD TO DETECT A FAULTY VALVE OR CIRCUIT. FOR THIS CASE, AN ALARM WILL OCCUR WHEN THE SIGNALS ON SCB AND C ARE OF OPPOSITE SENSE. THE ALARM WILL BE INDICATED ON THE CARD THAT SCB IS MORE POSITIVE THAN C.

ADJUSTMENT

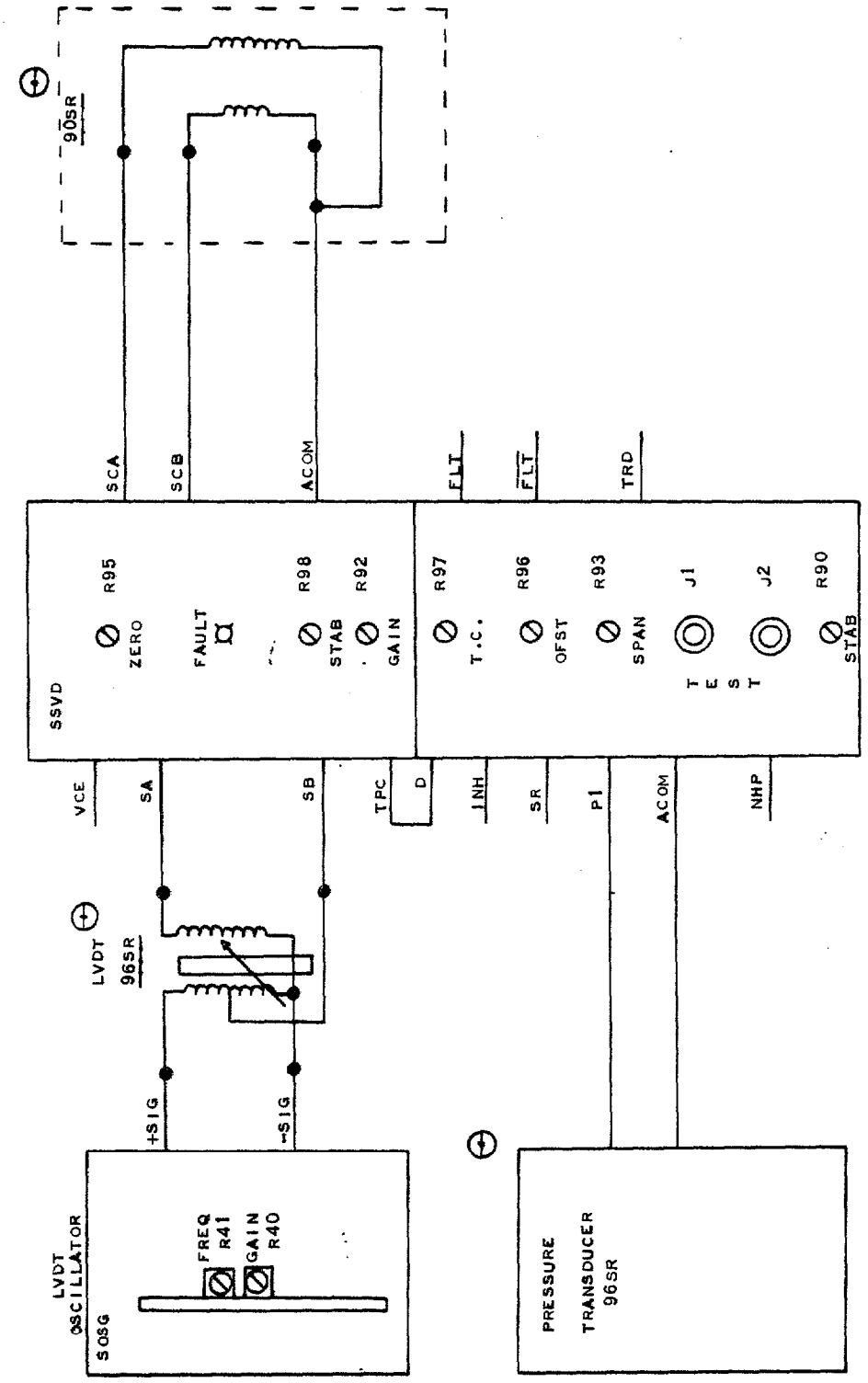
ZERO - R95
GAIN - R92
PRESS SPAN - R93
NHP OFFSET - R96
STAB - R90

C-10 OPTIONAL LIQUID FUEL (MODULATED PUMP STROKE)

AN OPTIONAL LIQUID FUEL CONTROL IS AVAILABLE. THIS SYSTEM, WHICH MODULATES THE LIQUID FUEL PUMP STROKE, USES THE SSVF AND SSVG CARDS ONLY AS DESCRIBED IN THE SECTION ON UNIVERSAL FUEL CONTROL. THE CIRCUIT IS IDENTICAL TO GAS CONTROL EXCEPT THAT WHEN A VCE SPLITTER IS USED, LCE IS THE REFERENCE SIGNAL.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-9, C-10
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GAS SPEED RATIO VALVE
REV. 3	MADE BY L.N. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 69 SH. NO. 68

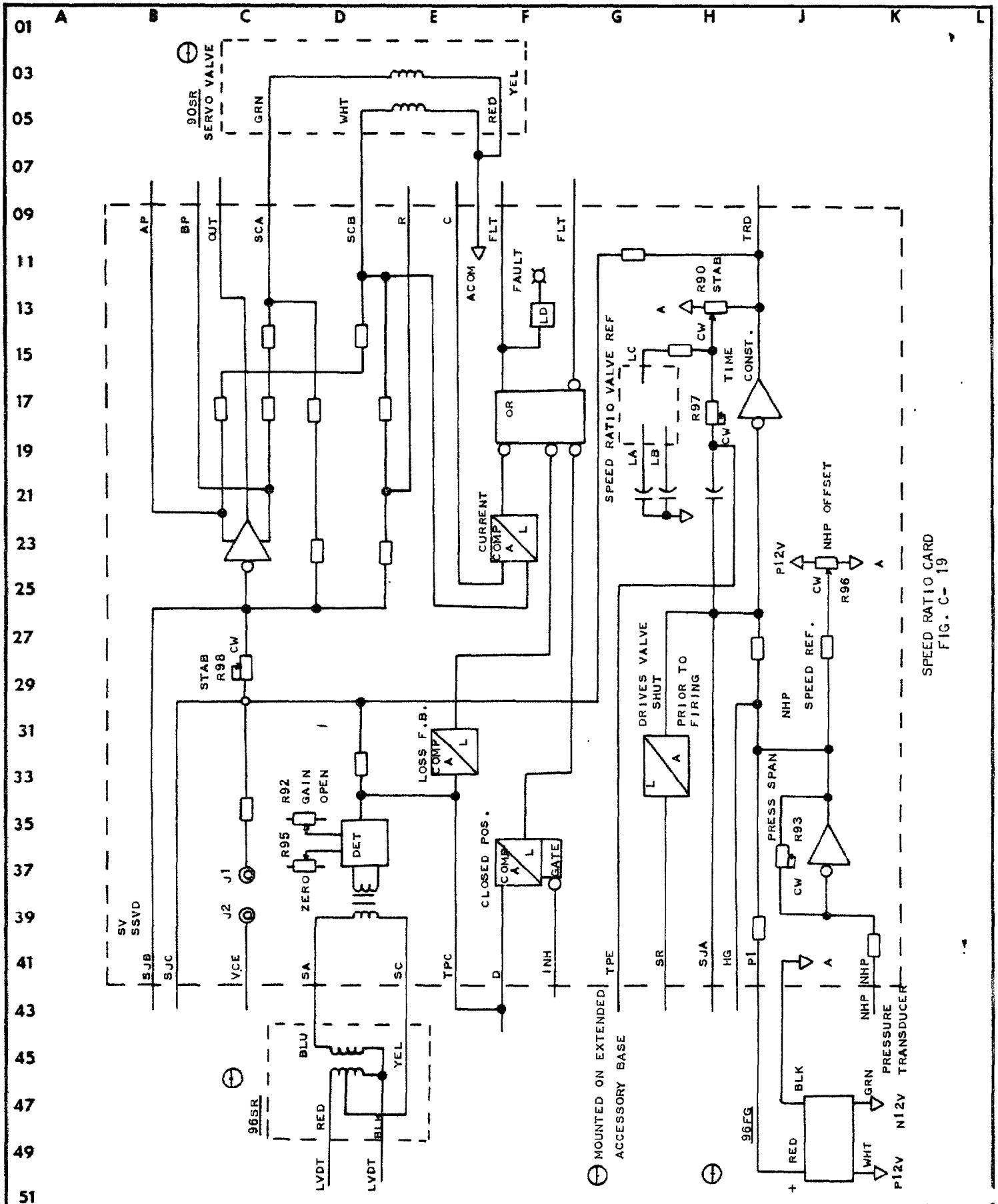
FORM 1C 2390 2-72



TYPICAL SPEED RATIO VALVE SERVO
FIG. C-18

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-18
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SPEED RATIO VALVE
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 70 SH. NO. 69

FORM IC 2390 2-72



REV. 1	REV 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-19
REV 2	ISSUED 4-9-75			FIRST MADE FOR		GAS SPEED RATIO CARD
REV 3	MADE BY L. M. Worley			I C NO.	SALEM, VA. U.S.A.	GEK-41053 CONT ON SH 70A SH. NO 70

C-11 FUEL SEQUENCING

FOR A DUAL FUEL MACHINE INGITION IS ATTEMPTED AT 20 °/o SPEED IF THE TURBINE CONTROLS ARE ON EITHER 100 °/o GAS FUEL OR 100 °/o LIQUID FUEL. IF THE TURBINE IS OPERATING ON GAS FUEL A PURGE TIMER PRESENTS FIRING UNTIL THE SYSTEM IS CLEARED OF GAS FROM ANY OTHER FILLED ATTEMPTS TO FIRE. FROM THE TIME FIRING IS ATTEMPTED UNTIL APPROXIMATELY 95 °/o FULL SPEED CHANGING RATIO OF GAS TO LIQUID IS INHIBITED.

FOR MACHINES WHICH BURN HEAVY FUEL OR DISTILLATE FUEL, THE MACHINE MUST START UP ON DISTILLATE FUEL, EVEN IF HEAVY FUEL WAS SELECTED. WHEN A SHUTDOWN SIGNAL IS RECEIVED THE TURBINE WILL AUTOMATICALLY TRANSFER FROM HEAVY TO DISTILLATE AND BEGIN TO UNLOAD THE TURBINE ON AN ORDERLY SHUTDOWN.

IF HEAVY FUEL WAS SELECTED AT STARTUP THAN THE CONTROLS WOULD ALLOW TRANSFER FROM DISTILLATE TO HEAVY AT 95 °/o TURBINE SPEED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM C-11 FUEL SEQUENCING
REV. 2	ISSUED			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY			I.C. NO.		CONT. ON SH. 71 SH. NO. 70A

C-12. WATER INJECTION CONTROL

WHEN WATER IS INJECTED INTO THE IGNITION CHAMBERS OF THE TURBINE, THE PEAK TEMPERATURE IN THE CHAMBERS IS LOWERED. THIS REDUCES THE QUANTITY OF NITROUS OXIDES (NOX) PRODUCED AS A BY-PRODUCT OF COMBUSTION (SINCE NOX IS PRODUCED AT HIGHER TEMPERATURES). THE RESULT IS LESS NOX IN THE EXHAUST AND THUS A CLEANER BURNING TURBINE.

THE SPEEDTRONIC SYSTEM PROVIDES CONTROL AND PROTECTION FOR THE WATER INJECTION SYSTEM (REFER TO FIGURE C-20). A FLO REF (FLOW REFERENCE) SIGNAL IS GENERATED AND THEN SUMMED AND INTEGRATED WITH A FLOW FEEDBACK (FFB) SIGNAL TO GENERATE A POSITION REFERENCE SIGNAL (POS REF). THE POSITION REFERENCE SIGNAL CONTROLS THE WATER SERVO VALVE IN MUCH THE SAME WAY THAT VCE CONTROLS THE FUEL SERVO VALVES USING THE SSVF AND SSVG CARDS.

THE FLO REF SIGNAL IS GENERATED AUTOMATICALLY OR MANUALLY. THE MANUAL SIGNAL IS ONLY CONNECTED TO THE MINIMUM SELECT BUS WHEN MANUAL OPERATION IS SELECTED. THE AUTOMATICALLY GENERATED SIGNAL IS ALWAYS CONNECTED TO THE MINIMUM SELECT BUS. THEREFORE, THE MANUAL SIGNAL CAN CALL FOR LESS WATER BUT NEVER MORE WATER THAN THE AUTO SIGNAL CALLS FOR. IN ADDITION, THE FLO REF SIGNAL IS LIMITED TO A MAXIMUM VALVE BY AN ADJUSTABLE CIRCUIT.

THE MANUAL CONTROL SIGNAL IS GENERATED EITHER LOCALLY OR REMOTELY (OPTIONAL). THE LOCAL MANUAL CONTROL SIGNAL IS ESTABLISHED BY A POTENTIOMETER ON THE FRONT PANEL AND A BUFFER AMPLIFIER (AOAH). WHEN MANUAL OPERATION IS SELECTED AND THE PANEL IS IN LOCAL CONTROL THE FLO REF SIGNAL FOLLOWS THE POT SETTING. IN REMOTE OPERATION THE AIAD CARD RECEIVES A REMOTELY GENERATED CURRENT SIGNAL WHICH IS TRANSLATED TO THE OUTPUT OF THE AIAD CARD AS A VOLTAGE SIGNAL. THIS VOLTAGE SIGNAL OPERATES IN A MANNER SIMILIAR TO THE LOCAL VOLTAGE SIGNAL. WHEN MANUAL OPERATION IS NOT SELECTED, AN AUTOMATIC SIGNAL DETERMINES FLO REF. THE AMOUNT OF WATER INJECTED IS PROPORTIONAL TO TF (TURBINE FIRING TEMPERATURE) AS COMPUTED BY AN AOAH CIRCUIT FROM PCD (COMPRESSOR DISCHARGE PRESSURE) AND TX (EXHAUST TEMPERATURE).

OPTIONALLY AN AMBIENT TEMPERATURE MODIFIED TF IS USED IN PARALLEL WITH THE TF SIGNAL TO GENERATE FLO REF. IN THIS CASE THE FIRING TEMPERATURE SIGNAL IS CONNECTED TO THE MINIMUM SELECT GATE AS A BACK UP. BUT ANOTHER, SMALLER SIGNAL IS ALSO CONNECTED. THIS IS THE AMBIENT COMPENSATED NOX SIGNAL (NOXC). NOXC IS DIRECTLY PROPORTIONAL TO TF AND INVERSLY PROPORTIONAL TO TA (AMBIENT TEMPERATURE).

THE SFUC CARD PROVIDES A PULSE RATE TO ANALOG CIRCUIT. THE OUTPUT OF THIS CARD IS A FEEDBACK TO WATER FLOW.

THE FLOW FEEDBACK (FFB) SIGNAL IS SUMMED WITH THE FLOW REFERENCE AND THE ERROR IS INTEGRATED (AOAH CARD) TO GENERATE A POSITION REFERENCE (POS REF).

A LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) PROVIDES A VALVE POSITION FEEDBACK. THE SSVF CARD CONVERTS THE RMS VOLTAGE FROM THE LVDT TO A DC POSITION FEEDBACK SIGNAL (PFB).

THE SSVG CARD SUMS THE POS REF AND THE PFB TO CONTROL THE WATER CONTROL VALVE.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM WATER INJECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 71A SH. NO. 71

C-12. WATER INJECTION PROTECTION

THE WATER INJECTION FUNCTION ALSO CONTAINS THREE PROTECTION CIRCUITS WHICH ALL USE AMPLIFIERS ON THE AOAH CARD.

ONE OF THESE CIRCUITS MONITORS THE RATE OF CHANGE OF VCE. IF VCE IS DECREASING BY MORE THAN TWO VOLTS PER SECOND, WATER INJECTION WILL BE CUT OFF.

THE SECOND CIRCUIT MONITORS THE FLO REF SIGNAL. WHEN THAT SIGNAL IS CALLING FOR AT LEAST 10 GALLONS PER MINUTE, A PERMISSIVE LOGIC SIGNAL IS GENERATED TO ENABLE WATER INJECTION.

THE THIRD SIGNAL MONITORS POSITION FEEDBACK AND DISABLES WATER INJECTION IF THE SERVO POSITION IS GREATER THAN A CERTAIN VALUE.

THE AOAH, AIAD, AND STKJ ARE DESCRIBED BELOW. THE SFUC CARD (WHICH GENERATES AN ANALOG SIGNAL BASED ON A PULSE RATE FLOW SIGNAL) IS SIMILAR TO THE SFUA CARD DESCRIBED IN THE UNIVERSAL FUEL SYSTEM. THE SFKC, SSVF AND SSVG ARE ALSO DESCRIBED IN THE SECTION ON UNIVERSAL FUEL.

AOAH CARD

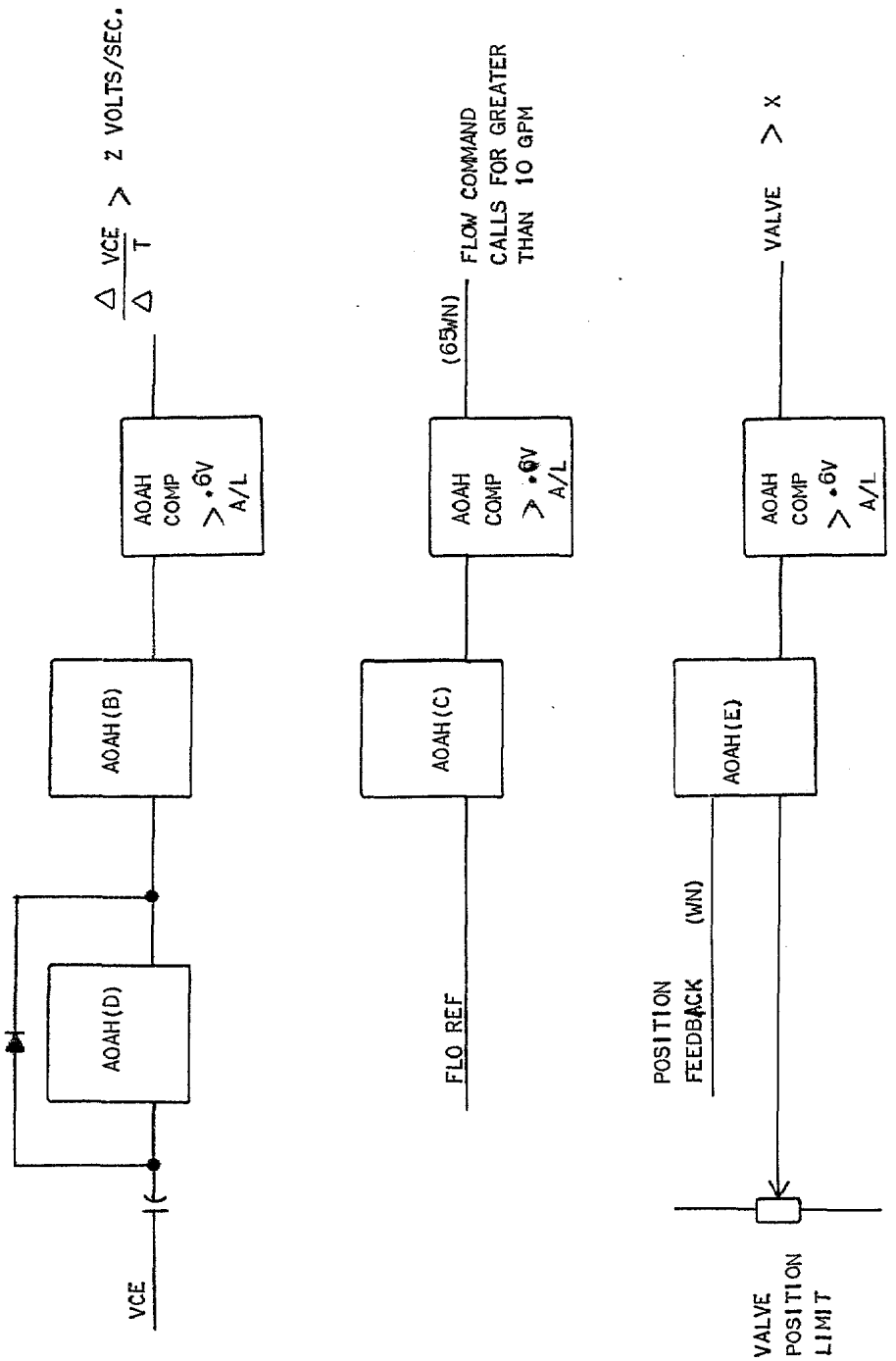
THE AOAH CARD, (REFER TO FIGURE C-22A), IS COMPOSED OF SIX OPERATIONAL AMPLIFIER CIRCUITS AND TWO ANALOG TO LOGIC LEVEL DETECTORS. THESE ARE GENERAL PURPOSE TYPE CIRCUITS WHICH MAY BE COMBINED TOGETHER OR WITH OTHER CARDS TO PERFORM A SPECIFIC FUNDTION.

REFER TO PAGE 28 FOR A DESCRIPTION OF OPERATIONAL AMPLIFIERS. CIRCUITS A, B, AND C ARE INVERTING AMPLIFIERS AS DESCRIBED ON PAGE 29. CIRCUIT E HAS AVAILABLE POINTS TO OPERATE IN THE DIFFERENTIAL MODE. CIRCUIT D MAY BE USED AS AN INTERGRATOR AND CIRCUIT F HAS ADJUSTMENTS TO LIMIT THE RANGE OF THE OUTPUT.

THERE ARE ALSO TWO ANALOG/LOGIC COMPARATORS ON THIS CARD. ONE OF THE COMPARATORS HAS AN OUTPUT OF LOGIC "1" WHEN ITS INPUT IS GREATER THAN +.6 VOLTS, AND THE OTHER HAS AN OUTPUT OF LOGIC 1 WHEN ITS INPUT IS MORE NEGATIVE THAN -.6 VOLTS. BOTH OF THESE COMPARATORS HAVE A PERMISSIVE GATE WHICH MUST BE A LOGIC 1 TO ENABLE THE OUTPUT TO GO TO A LOGIC 1.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FORM 1C 2390 4-74

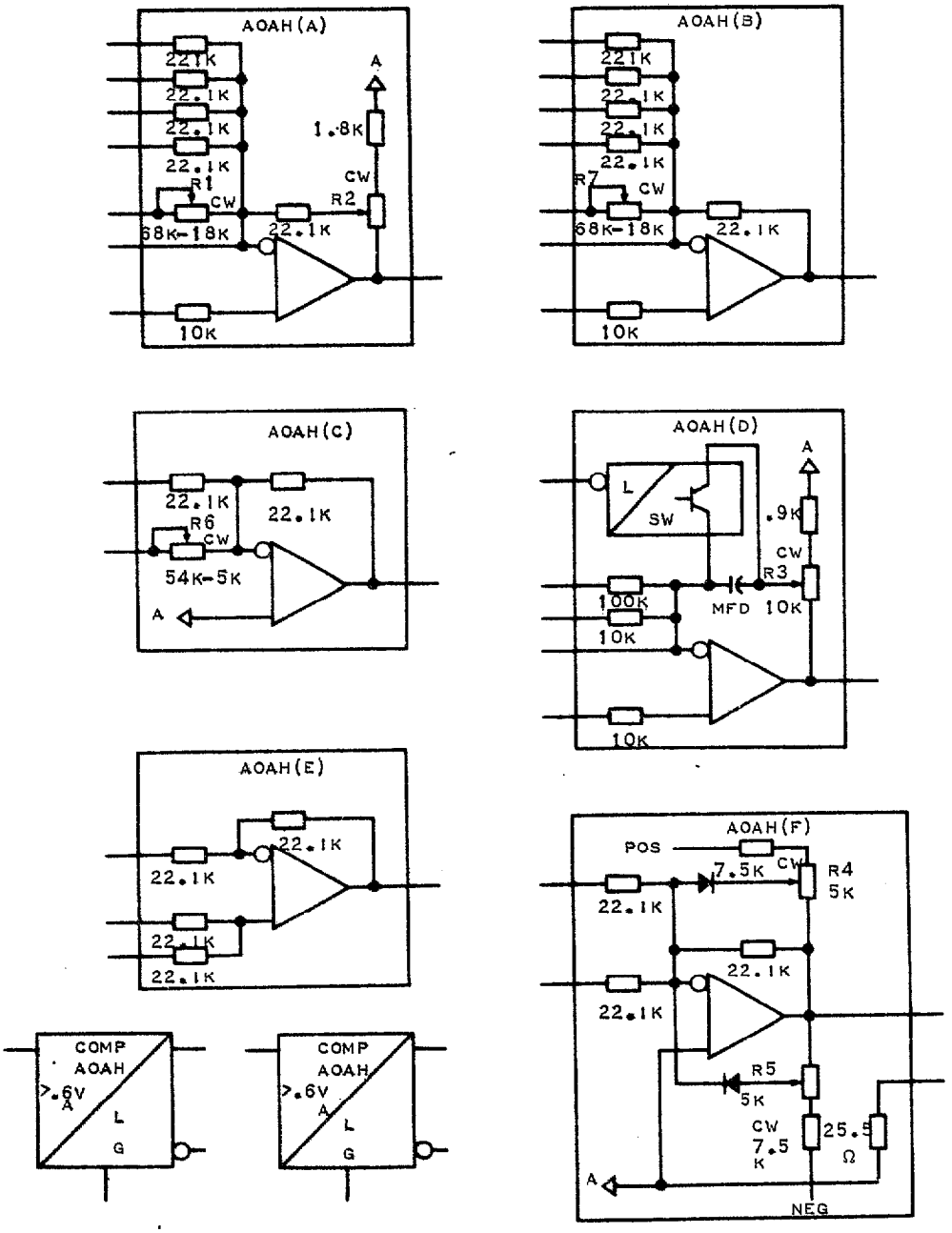


PROTECTION BLOCK DIAGRAM
WATER INJECTION
FIGURE C-21

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION BLOCK DIAG.
REV. 2	ISSUED			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 72 SH. NO. 71B

FORM 1C 2390 4-74



MULTIPURPOSE OPERATIONAL AMPLIFIER CARD
AOAH
FIG. C-22A

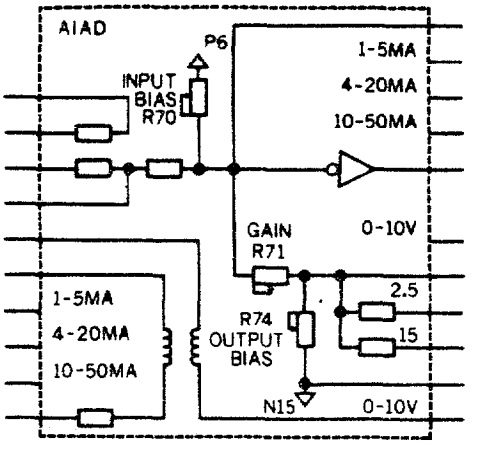
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		AOAH CARD
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 73 SH. NO. 72

AIAD CARD

THE AIAD CARD IS AN ISOLATION AMPLIFIER THAN CAN TRANSFER A CURRENT OR VOLTAGE INPUT SIGNAL TO THE OUTPUT AS A CURRENT OR VOLTAGE SIGNAL.

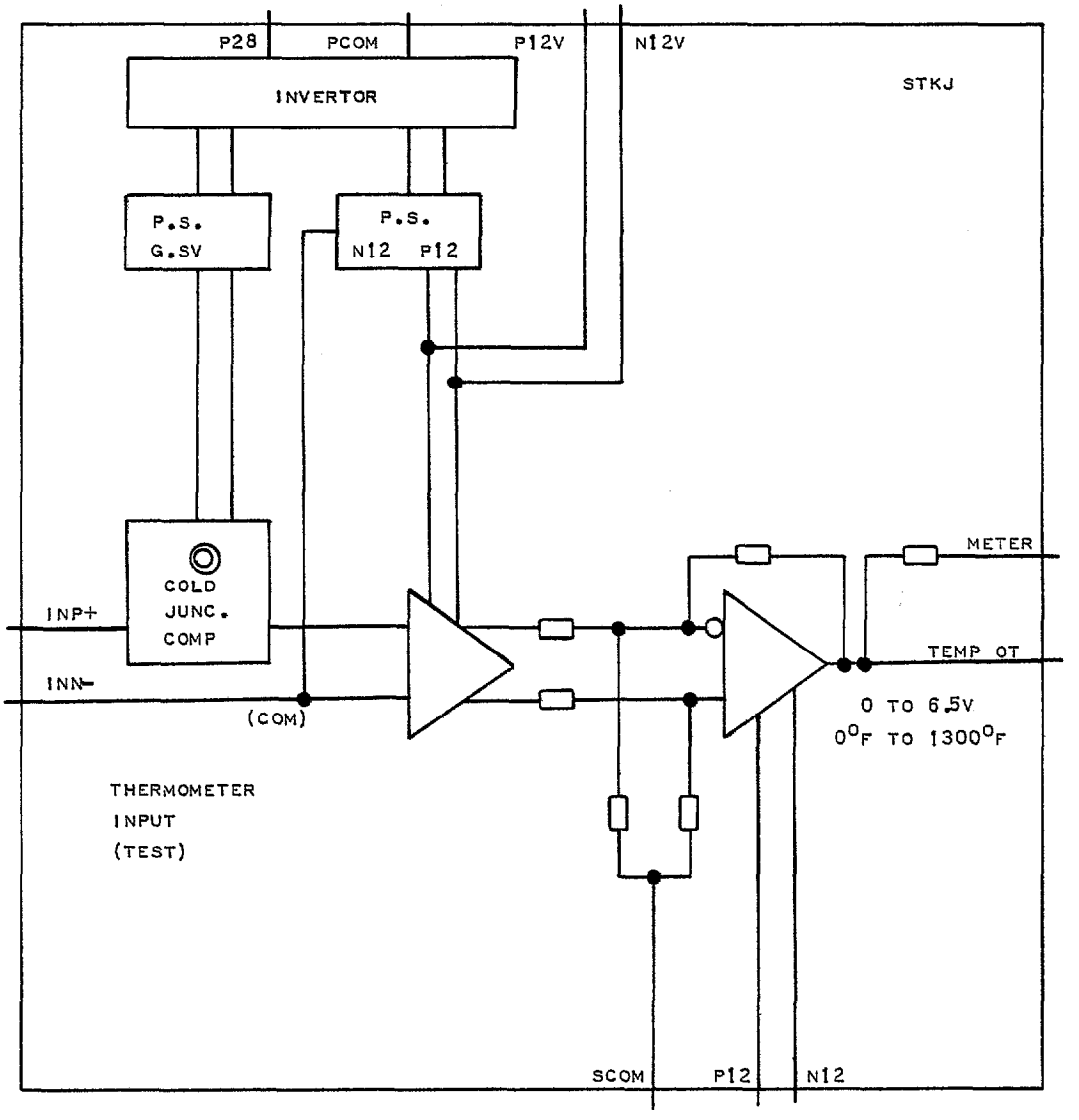
ANY OF THE VOLTAGES OR CURRENTS IN THE INPUT TABLE BELOW CAN BE TRANSLATED TO A CURRENT OR VOLTAGE SIGNAL LISTED IN THE OUTPUT TABLE BY MAKING THE APPROPRIATE JUMPERS.

INPUTS	OUTPUTS
0 TO (3V TO 5V)	0 TO 10V
0 TO +10V	+1 TO +5 MA
0 TO +20V	+4 TO +20 MA
0 TO 25V	+10 TO +50 MA
1 MA TO +5 MA	
+4 MA TO 20 MA	
+10 MA TO +50 MA	



STKJ CARD - (REFER TO FIGURE C-22B

THE STKJ CARD RECEIVES AN ANALOG INPUT FROM A CHROMEL-ALUMEL THERMOCOUPLE. THE JUNCTION OF THE THERMOCOUPLE AND THE CARD IS COLD JUNCTION COMPENSATED WITH A RESISTANCE TEMPERATURE DETECTOR (RTD). THEREFORE, THE VOLTAGE GENERATED BY A CHANGE IN TEMPERATURE AT THE CARD/THERMOCOUPLE JUNCTION IS CANCELLED BY THE CHANGE IN VOLTAGE OF THE RTD.

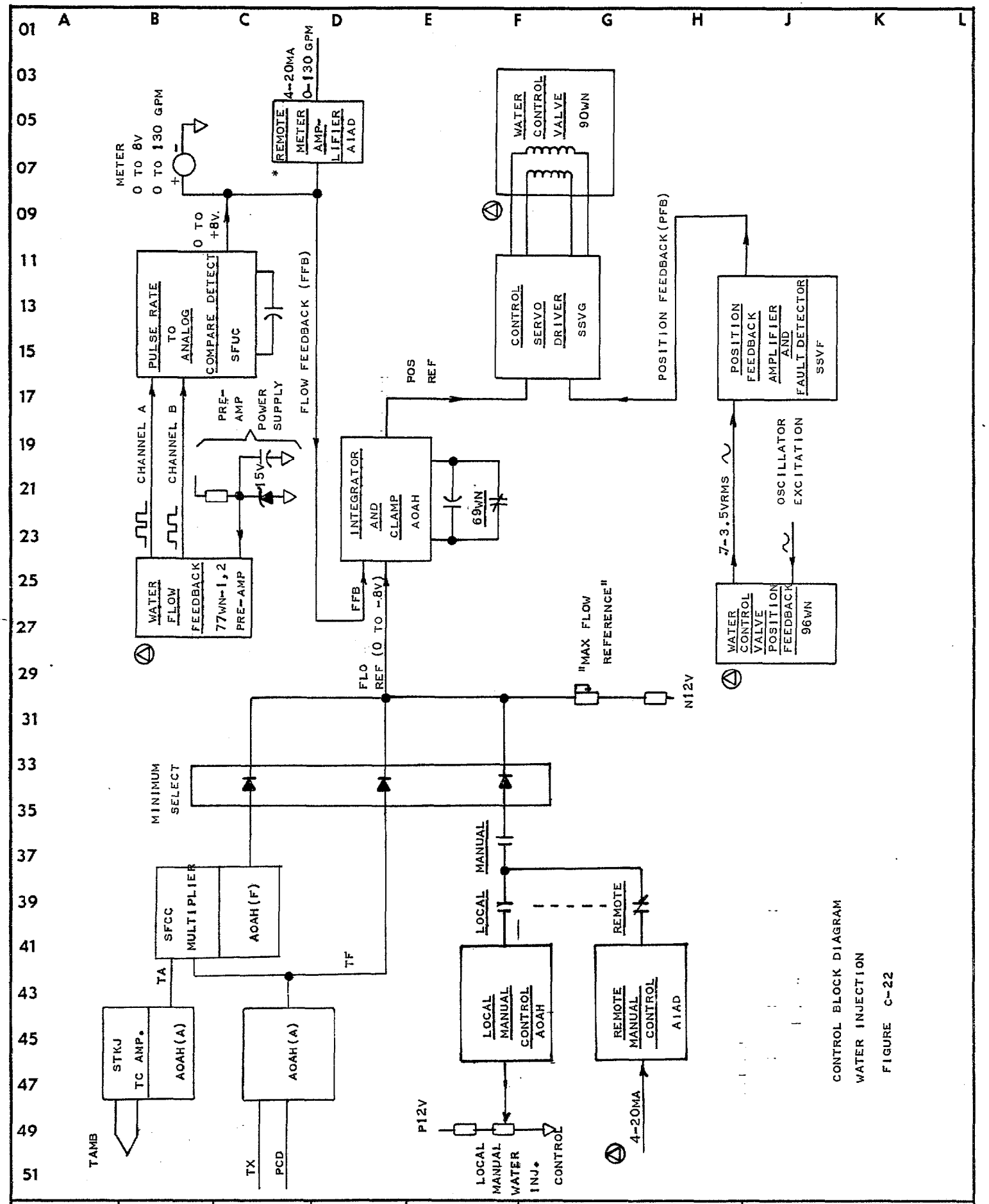


CARD SYMBOL FOR

STKJ THERMOCOUPLE CARD
FIG. C-2'2B

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM STKJ CARD
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 76 SH. NO. 74

FORM 1C 2390 4-74



CONTROL BLOCK DIAGRAM
WATER INJECTION
FIGURE C-22

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.
REV. 3	MADE BY L.M. Worley			I.C. NO.	

ELEMENTARY DIAGRAM C-22
WATER INJECTION BLOCK DIAGRAM
GEK-41053
CONT. ON SH. 76A SH. NO. 76

C-12 WATER INJECTION SEQUENCE

BEFORE WATER MAY BE INJECTED, VARIOUS CONDITONS MUST BE MET WHICH CONSIST, IN GENERAL OF THE FOLLOWING.

NON-LATCHING

WATER PRESSURE (TIME DELAY)
WATER INJECTION SELECTED
AT LEAST 10 GALLONS PER MINUTE (GPM) OF WATER PRODUCED
GENERATOR BREAKER CLOSED
VCE CHANGING BY LESS THAN-2 VOLTS PER SECOND.
"4'S" ARE UP. (MASTER PROTECTIVE CIRCUIT)

LATCHING

NO POSITION LOOP FEEDVACK FAULT
NO COMBUSTION MONITOR ALARM
NO MORE THAN APPROSIMATELY 80 GALLONS PER MINUTE BASED ON POSITION FEEDBACK
NO MORE THAN 10 GPM COMMANDED BELOW 95 °/₀ SPEED
POSITION FEEDBACK AND FLOW FEEDBACK AGREE

POWER AUGMENTATION

PEAK OR PEAK RESERVE TEMPERATURE CONTROL
NOT OFF OF TEMPERATURE CONTROL FOR MORE THAN 10 SECONDS

IF THE ABOVE CONDITIONS ARE MET WATER MAY BE INJECTED BASED ON A MANUAL OR AUTOMATIC REFERENCE.

IF ONE OF THE CONDITIONS LISTED UNDER "LATCHING" OCCURS TO HALT WATER INJECTION, THEN THE SYSTEM MUST BE RESET VIA A CARD MOUNTED RESET PUSHBUTTON. BEFORE WATER INJECTION WILL RESUME (ASSUMING THAT THE FAULT CONDITION HAS CLEARED). IF A COMBUSTION MONITOR ALARM OCCURS, THE COMBUSTION MONITOR MUST BE RESET AND THEN THE WATER INJECTION SYSTEM MUST BE RESET BEFORE WATER FLOW WILL RESUME.

WHEN ONE OF THE ABOVE CONDITIONS IS NOT MET (EXCEPT VCE LIMITS, WHICH ARE DESCRIBED LATER) WATER FLOW IS STOPPED IN SEVERAL WAYS, BOTH ELECTRICALLY AND MECHANICALLY.

FIRST OF ALL THE RELAYS CONTROLLING THE STOP VALVE AND WATER PUMP DROP OUT. WHEN WATER INJECTION IS STOPPED BY THE WATER INJECTION SELECTOR SWITCH, THE PUMP TURNS OFF 5 SECONDS BEFORE THE STOP VALVE CLOSES.

SECOND, WHEN THE STOP VALVES CLOSE, A POSITIVE BIAS IS PLACED ON THE WATER CONTROL VALVE BY A CONTACT CLOSURE THROUGH A RESISTOR TO P5.3 VOLTS. THE POSITIVE BIAS CLOSES THE CONTROL VALVE. THIS IS SIMILAR TO THE BIAS PLACED ON THE FUEL CONTROL VALVES BY THE VMPA CARDS WHEN THE "4'S" (MASTER PROTECTIVE CIRCUIT) TRIP.

THIRD, A CONTACT CLOSURE ACROSS THE FEEDBACK OF THE COMMAND SIGNAL AMPLIFIER EFFECTIVELY SHORTS THE POSITION COMMAND SIGNAL TO ZERO VOLTS. THIS INDEPENDENTLY CLOSES THE CONTROL VALVE. WHEN VCE CHANGES BY MORE THAN -2 VOLTS PER SECOND THE POSITION COMMAND SIGNAL IS SHORTED FOR TWO MINUTES AFTER VCE RETURNS TO NORMAL. IN THIS CASE, THE STOP VALVE STAYS OPEN AND THE PUMP(S) CONTUNUES TO RUN.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM C-12 WATER INJECTION SEQ.
REV. 2	ISSUED			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY			I.C. NO.		CONT. ON SH. 77 SH. NO. 76A

C-13 MODULATED INLET GUIDE VANE CONTROL - (REFER TO FIGURE C-23)

THE INLET GUIDE VANE (IGV) FUNCTION IS SIMILAR TO WATER INJECTION IN THAT BOTH USE THE MULTI-PURPOSE AOAH CARD AND ISOLATOR AMPLIFIER AIAD AS DESCRIBED IN THE WATER INJECTION DESCRIPTION. THE INLET GUIDE VANE FUNCTION USES, FOR SERVO CONTROL, THE SSVF AND SSVG CARDS AS DESCRIBED IN THE FUEL SYSTEM.

THE IGV FUNCTION HAS THREE CONTROL SIGNAL SOURCES - AUTO, LOCAL MANUAL, AND REMOTE MANUAL. ONE OF THESE THREE SIGNALS IS SWITCHED TO THE FINAL CONTROL SIGNAL AMPLIFIER BASED ON LOGIC SIGNALS.

THE FINAL CONTROL AMPLIFIER OUTPUT GOES THROUGH A MINIMUM SELECT DIODE ARRANGEMENT TO THE SERVO CONTROL. ANOTHER SIGNAL, THE IGV CLOSE COMMAND SIGNAL, ALSO GOES THROUGH THE MINIMUM SELECT CIRCUIT TO THE SERVO CONTROL. THE LEAST OF THESE TWO SIGNALS ACTUALLY CONTROLS THE IGV.

NOW, LET'S LOOK AT EACH OF THESE FUNCTIONS WITHIN THE IGV CONTROL SEPARATELY. A TEMPERATURE ERROR SIGNAL IS GENERATED FROM THE EXHAUST TEMPERATURE (TX), COMPRESSOR DISCHARGE PRESSURE (PCD), AND THE BASE REFERENCE (BASE); THIS ERROR GOES THROUGH DOUBLE DEAD BAND AMPLIFIERS. THAT IS, AN ERROR DEAD BAND IS ESTABLISHED BY A POTENTIOMETER SETTING. THE POTENTIOMETER VOLTAGE AND THE TEMPERATURE ERROR SIGNAL ARE SUMMED ON THE AOAH (F) CIRCUIT, AND THE INVERTED POTENTIOMETER VOLTAGE AND THE TEMPERATURE ERROR ARE SUMMED ON ANOTHER AOAH (F) CIRCUIT. THE AOAH (F) CIRCUIT HAS DIODES WHICH CLAMP THE OUTPUT OF THE AMPLIFIER IN THE POSITIVE AND/OR NEGATIVE DIRECTION. IN THIS CASE, THE UPPER CIRCUIT IS CLAMPED IN THE POSITIVE DIRECTION AND THE LOWER IN THE NEGATIVE DIRECTION. IF THE TEMPERATURE ERROR IS NEGATIVE AND HAS AN ABSOLUTE VALUE GREATER THAN THE DEAD BAND REFERENCE, THEN THE LOWER AOAH (F) CIRCUIT WILL HAVE A POSITIVE OUTPUT. IF THE TEMPERATURE ERROR IS POSITIVE AND HAS AN ABSOLUTE VALUE GREATER THAN THE INVERTED DEAD BAND VOLTAGE, THE UPPER AOAH (F) CIRCUIT WILL HAVE A NEGATIVE OUTPUT. THE OUTPUT SIGNAL OF EITHER CIRCUIT IS AMPLIFIED THROUGH A PROPORTIONAL PLUS INTEGRATOR (REFER TO PAGE 32) TO BECOME THE AUTO SIGNAL.

THE LOCAL MANUAL CONTROL SIGNAL IS GENERATED FROM A MANUALLY CONTROLLED POTENTIOMETER. THE POTENTIOMETER WIPE VOLTAGE IS INVERTED THROUGH AN AOAH OPERATION AMPLIFIER TO GENERATE THE LOCAL MANUAL CONTROL SIGNAL.

A REMOTE CURRENT SIGNAL IS RECEIVED INTO THE AIAD CARD AND IS TRANSLATED TO THE OUTPUT AS A VOLTAGE SIGNAL. THIS VOLTAGE SIGNAL IS THE REMOTE MANUAL CONTROL.

ONE OF THE THREE SIGNALS (AUTO, LOCAL MANUAL, REMOTE MANUAL) IS CONNECTED TO THE FINAL CONTROL AMPLIFIER THROUGH RELAY CONTACTS WHICH ARE CONTROLLED BY LOGIC. THE CONTROL AMPLIFIER HAS DIODE CLAMPS WHICH LIMIT ITS OUTPUT SIGNAL TO CONTROL THE INLET GUIDE VANES FROM A MINIMUM OF 50° TO A MAXIMUM OF 80°. SEE THE CONTROL SPECIFICATION FOR EXACT SETTINGS FOR A SPECIFIC TURBINE. THIS OUTPUT SIGNAL IS CONNECTED THROUGH A MINIMUM SELECT DIODE TO THE GVE SIGNAL AS PREVIOUSLY MENTIONED.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-13 MODULATED INLET GUIDE VANE CONTROL
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 78 SH. NO. 77

THE AUTO/MANUAL CIRCUIT IS ONE OF TWO LEGS IN THE CONTROL CIRCUIT. THERE IS ANOTHER LEG WHICH DOESN'T ALLOW THE INLET GUIDE VANES TO MODULATE UNTIL CERTAIN CONDITIONS ARE MET. THIS IS THE IGV CLOSE LEG.

THE INLET GUIDE VANES RECEIVE TWO LOGIC SIGNALS -4 (PROTECTION) AND 14HS (OPERATING SPEED). WHEN BOTH 4 AND 14HS ARE TRUE (LOGIC 1), THE INVERSE OUTPUT OF THE SLEH GOES TO A LOGIC 0. THIS OUTPUT IS CONNECTED, THROUGH A DIODE, TO THE SUMMING JUNCTION OF AN OP. AMP. IT IS ALSO CONNECTED THROUGH THE SAME DIODE AND A SERIES RESISTOR TO P12V. WHEN THE OUTPUT IS A LOGIC 1 (EITHER 4 AND/OR 14HS NOT TRUE) A POSITIVE VOLTAGE (CURRENT) IS PLACED ON THE SUMMING JUNCTION.

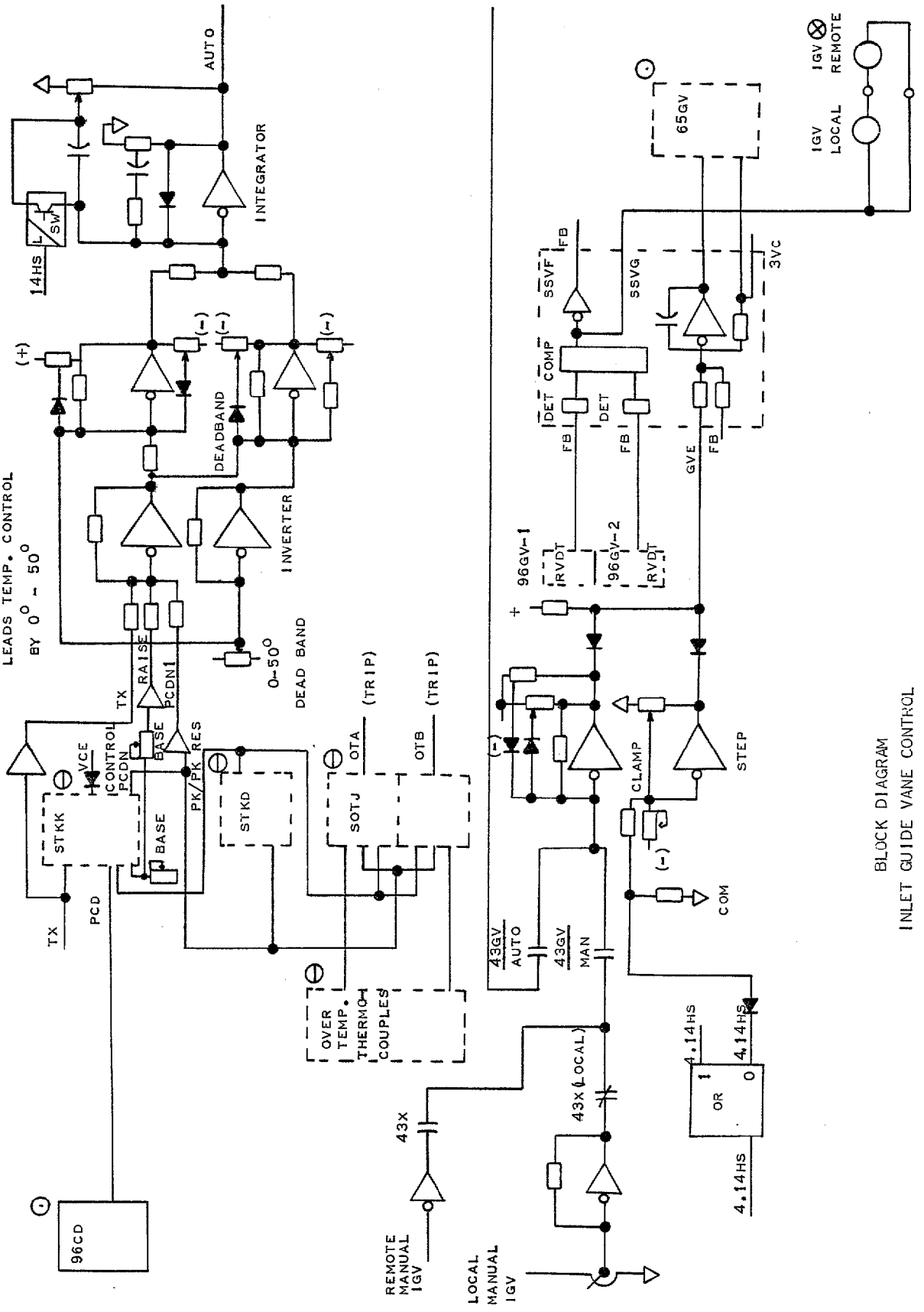
SINCE THE SUMMING JUNCTION HAS N12V PERMANENTLY CONNECTED, THE RESULTANT OUTPUT IS A LOW POSITIVE VOLTAGE. THIS OUTPUT IS CONNECTED TO GVE THROUGH A MINIMUM SELECT DIODE. SINCE THIS SIGNAL IS SMALLER THAN THE OUTPUT OF THE OTHER CONTROL LEG, THIS "CLOSE" SIGNAL WILL CONTROL GCE AND CONSEQUENTLY KEEP THE INLET GUIDE VANES CLAMPED CLOSED.

WHEN THE OUTPUT OF THE SLEH GOES TO A LOGIC 0 (4 AND 14HS ARE TRUE) THEN THE NET INPUT ON THE SUMMING JUNCTION IS VERY NEGATIVE AND ITS OUTPUT GOES MORE POSITIVE THAN THE MAXIMUM CONTROL LEG OUTPUT. THEREFORE, THE CONTROL LEG WILL MODULATE GVE.

THE GVE SIGNAL AND THE FEEDBACK SIGNAL CONTROL THE 90TV VALVE FOR INLET GUIDE VANE MODULATION USING THE SSVF AND SSVG CARDS. AS DESCRIBED IN THE GAS FUEL CONTROL SECTION (C-8).

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM C-13 MODULATED INLET GUIDE VANE CONTROL
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH 79 SH. NO. 78

FORM 1C 2390 4-74



BLOCK DIAGRAM
INLET GUIDE VANE CONTROL
FIG. C-23

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.
REV. 3	MADE BY L. M. Worley			I.C. NO.	

ELEMENTARY DIAGRAM C-23
BLOCK DIAGRAM IGV CONTROL
GEK-41053
CONT. ON SH. 80 SH. NO. 79

SECTION D
PROTECTION

D-1 "4" CIRCUIT

THE FOCAL POINT OF THE PROTECTION SYSTEM IS THE VMPA CARD WHICH GENERATES THE "4" SIGNAL AND THE SHUTDOWN BIAS TO THE SERVOVALVES. (DUE TO THE MAGNITUDE OF ITS IMPORTANCE, A DETAILED DESCRIPTION WILL BE PRESENTED). THE SYMBOL FOR THE VMPA CARD IS SHOWN IN FIGURE D-1. TO HELP ELIMINATE SOME CONFUSION, A "*" IS PLACED NEAR THOSE PINS NOT NORMALLY USED IN UTILITY APPLICATIONS. A TYPICAL PROTECTION SYSTEM ILLUSTRATING THE "4" BUS IS SHOWN IN FIGURE D-2.

THE THREE PRIMARY OUTPUTS OF THE VMPA CARD : (1), THE START CHECK SIGNAL ON PIN OSTCK WHICH MUST BE A LOGIC "0" TO START THE TURBINE, (2), THE SERVO TRIP SIGNAL ON PIN SRT WHICH PROVIDES A SHUTDOWN BIAS TO THE FUEL SERVO-VALVES, AND (3), THE "4" SIGNAL ON PINS 4RD, 4 AND $\overline{4}$. THE "4" CIRCUIT IS CONSIDERED A "PERMISSIVE TO OPERATE" SIGNAL. OPERATING CONDITIONS THAT MAY BE HARMFUL TO THE TURBINE ARE SENSED BY THE VMPA CARD WHICH STOPS THE TURBINE BY DROPPING THE "4" CIRCUIT.

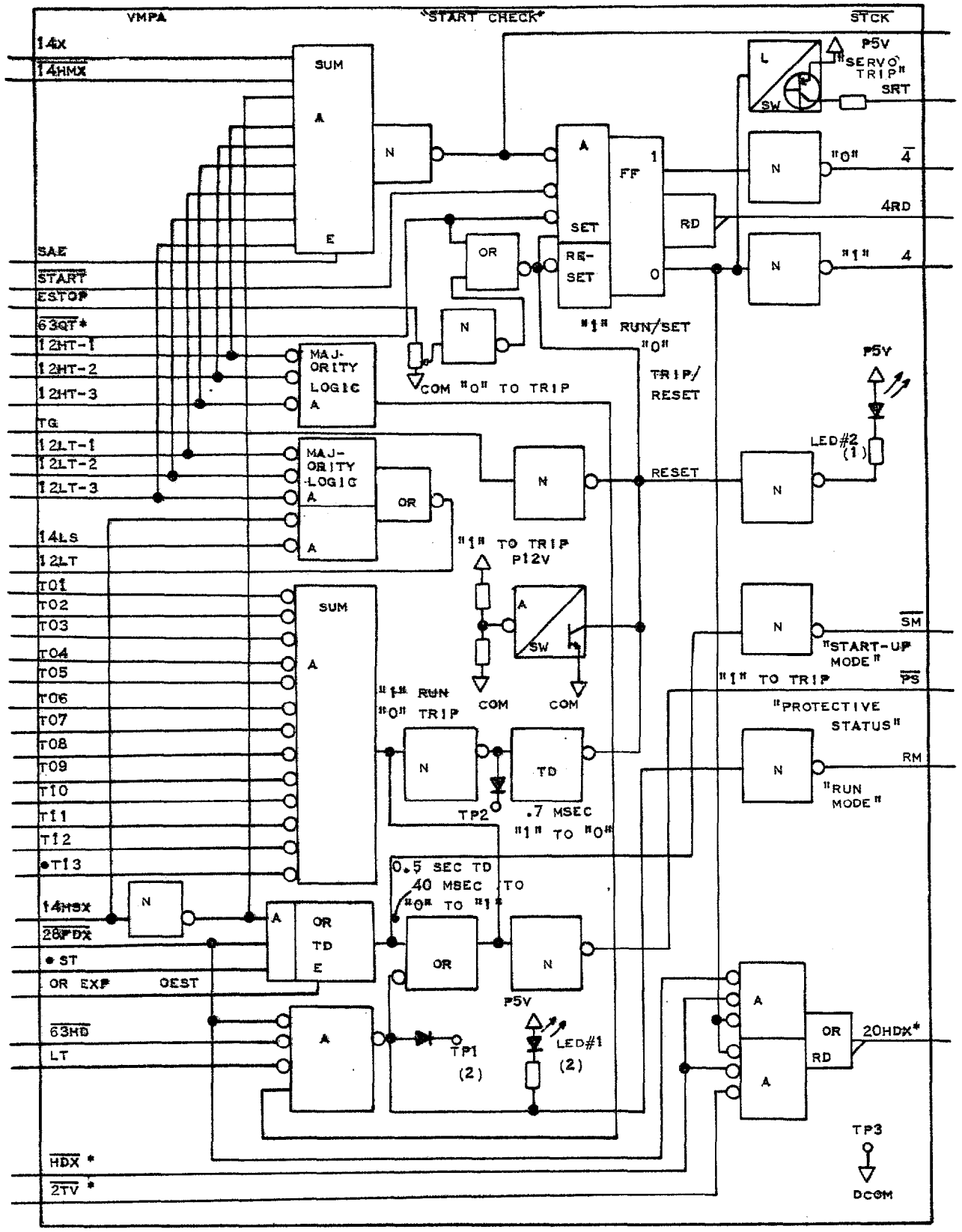
THE VMPA CARD INTERROGATES SELECTED INTERLOCKS, AND GENERATES A START CHECK SIGNAL AND A "4" SIGNAL IF APPROPRIATE INTERLOCKS ARE TRUE. THE START CHECK CIRCUIT CONSISTS OF THE "SUM 1" GATE IN THE TOP LEFT CORNER OF THE SYMBOL SHOWN IN FIGURE D-1. IF ALL OF THE INPUTS TO THE "SUM 1" GATE ARE "1", THE OUTPUT WILL BE A LOGIC "0" WHICH IS THE NECESSARY LEVEL TO START THE TURBINE. ANY LOGIC "0" INPUT TO THE "SUM 1" GATE WILL INHIBIT STARTING THE TURBINE. TO BRING THE "4" CIRCUIT UP, THE FLIP-FLOP IN THE TOP CENTER OF THE SYMBOL MUST BE SET. TO SET THE FLIP-FLOP THE $\overline{\text{START}}$ CHECK SIGNAL MUST BE "0", THE START SIGNAL ON PIN $\overline{\text{START}}$ MUST BE "0", PIN $\overline{\text{63QT}}$ MUST BE "0" (TIED TO DCOM), AND THE TRIP BUS MUST BE A LOGIC "1". ONCE THE FLIP-FLOP IS SET, THE TOP TWO SIGNALS CAN BE REMOVED AND IT WILL REMAIN IN THE SET POSITION. HOWEVER, IF THE TRIP BUS GOES TO A LOGIC "0" THE FLIP-FLOP WILL BE RESET AND THE "4" CIRCUIT WILL DROP OUT. THERE ARE MANY PATHS TO PLACE A LOGIC "0" ON THE TRIP BUS. THE DISCUSSION WILL BEGIN AT THE TOP MOST PATH ON THE SYMBOL AND PROCEED DOWNWARD.

THE FIRST PATH BEGINS AT PIN ESTOP WITH THE EMERGENCY STOP SIGNAL. THIS PIN IS CONNECTED TO P28 THROUGH NORMALLY CLOSED RELAY CONTACTS. OPENING THE CONTACTS WILL CAUSE THE FLIP-FLOP TO RESET. THE NEXT PATH IS THROUGH THE MAJORITY LOGIC GATE. A LOGIC "0" ON ANY TWO INPUTS WILL GIVE A LOGIC "1" OUT AND KEEP THE TRIP BUS A "1".

IF TWO INPUTS ARE "1", THE BUS WILL GO TO A "0" AND RESET THE FLIP-FLOP. IN UTILITY APPLICATION, EXTERNAL CIRCUITRY IS ADDED TO MAKE THE MAJORITY LOGIC GATE ACT AS A NEGATIVE "AND" GATE. A DIAGRAM OF THE ACTUAL CIRCUIT AND AN EQUIVALENT CIRCUIT IS SHOWN ON PAGE 83.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		PROTECTION -"4" CIRCUIT
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 81 SH. NO. 80

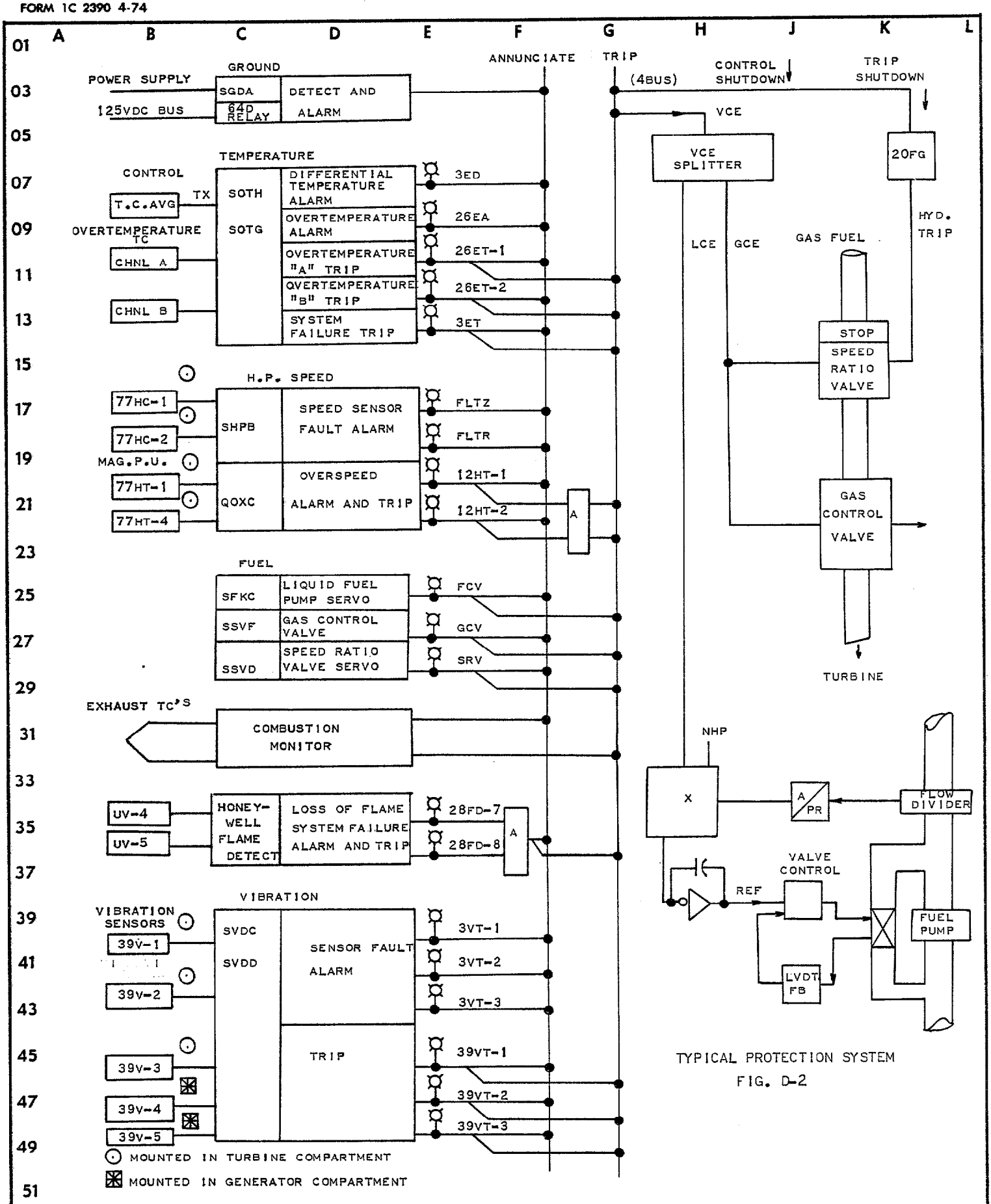
FORM 1C 2390 4-74



VMPA CARD
FIG. D-1

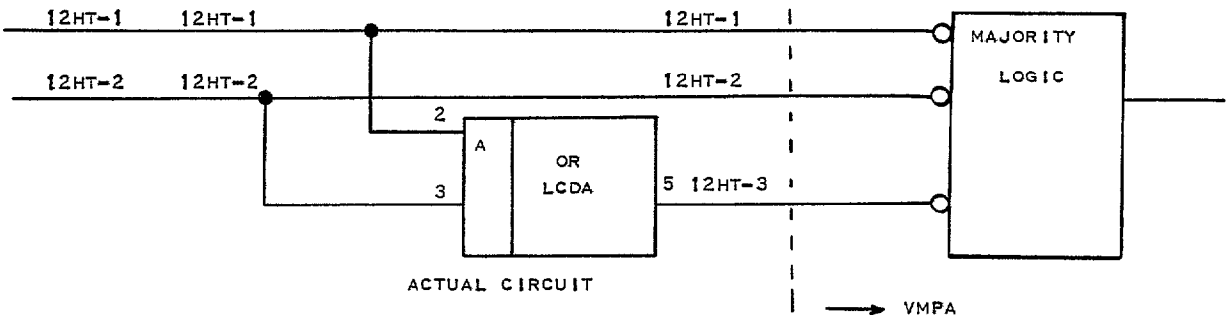
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		VMPA CARD
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 82 SH. NO. 81

FORM 1C 2390 4-74



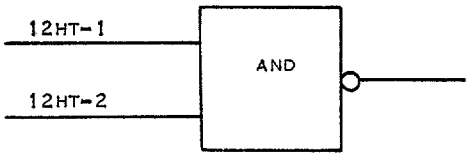
TYPICAL PROTECTION SYSTEM
FIG. D-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS LEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-2
REV. 2	ISSUED			FIRST MADE FOR		TYPICAL PROTECTION SYSTEM
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 83 SH. NO. 82



ACTUAL CIRCUIT

ACTUAL CIRCUIT



EQUIVALENT CIRCUIT

THE NEXT TRIP PATH IS THROUGH THE "SUM 0" GATE. IF ANY INPUT IS A LOGIC "1" THE TRIP BUS WILL GO TO A LOGIC "0" AND RESET THE FLIP-FLOP. (NOTE: THE OUTPUT OF THE "SUM 0" GATE IS ALSO TIED TO THE OUTPUT "OR" GATE. RECALLING FROM THE SEQUENCING SECTION THAT A LOGIC "1" CAN ALWAYS BE FORCED TO A LOGIC "0", IT IS SEEN THAT A LOGIC "0" OUT OF THE "SUM 0" GATE OR THE "OR" GATE WILL FORCE THE TIE POINT TO A LOGIC "0" AND TRIP THE CIRCUIT). CONTINUING ON, IT IS SEEN THAT THE REMAINING INPUTS ARE TO A PAIR OF "AND" GATES WHICH FEED A COMMON "OR" GATE. BOTH OF THE "AND" GATES HAVE TO BE IN THEIR FALSE STATE TO TRIP THE CIRCUIT. THE FLAME DETECTOR INVERSE SIGNAL 028FDX IS COMMON TO BOTH "AND" GATES; THEREFORE, THE GATE WITH INPUTS ON PINS 14HSX, 028FDX, ST, AND OR EXP HAS CONTROL OF THIS PATH BEFORE FLAME IS DETECTED, AND THE GATE WITH INPUTS ON PINS 063HD, LT, 028FDX AND THE MAJORITY LOGIC OUTPUT IS IN CONTROL AFTER FLAME HAS BEEN DETECTED.

D-2 OVERSPEED

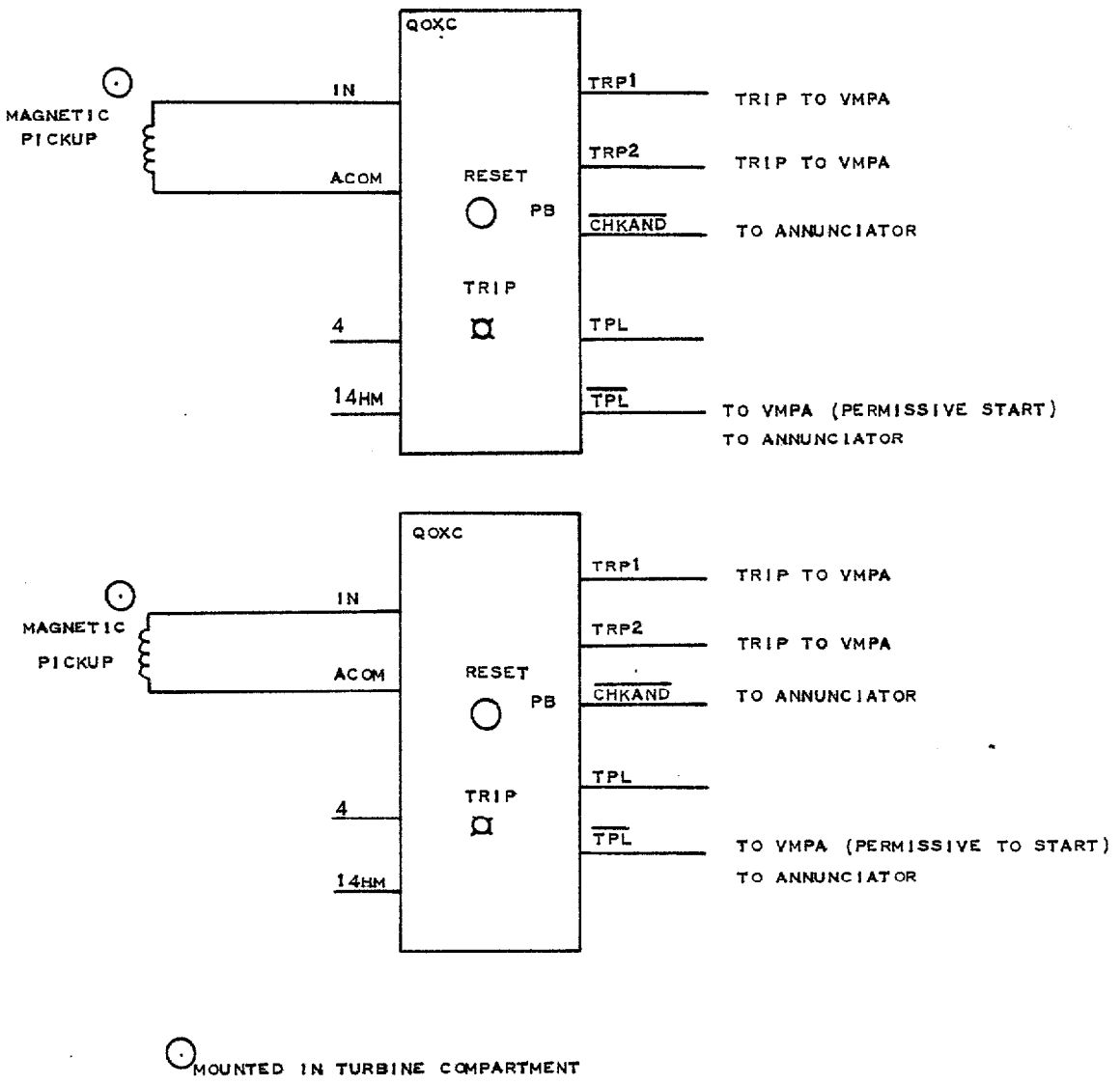
THE OVERSPEED TRIP SYSTEM CONSISTS OF TWO SEPARATE SENSING CIRCUITS TO PROVIDE INCREASED RELIABILITY. A DIAGRAM IS SHOWN IN FIGURE D-3.

QOXC CARD - (REFER TO FIGURE D-4)

THE QOXC CARD RECEIVES A FREQUENCY PROPORTIONAL TO THE SPEED OF TURBINE BETWEEN PINS IN AND ACOM. OUTPUTS TRP1 AND TRP2 WILL BE A "1" WHEN THE FREQUENCY OF THE INPUT IS BELOW APPROXIMATELY 9% ON ACCELERATION AND BELOW APPROXIMATELY 4% ON DECELERATION. HOWEVER, THE PRIMARY PURPOSE OF THE CIRCUIT IS TO GENERATE A "1" ON OUTPUTS TRP1 AND TRP2 WHEN THE TURBINE IS OPERATING OVER A PREDETERMINED SPEED. A LATCHING CIRCUIT IS PROVIDED TO HOLD THE TRIP INDICATION SIGNAL ON TPL. THE LATCHING CIRCUIT MAY BE INHIBITED BY A "0" ON INPUTS 4 AND 14HM. THE PUSH-BUTTON ON THE CARD FRONT MUST BE DEPRESSED TO CLEAR THE LATCH.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OVERSPEED
REV. 3	MADE BY L. M. Dooley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 84 SH. NO. 83

FORM 1C 2390 4-74

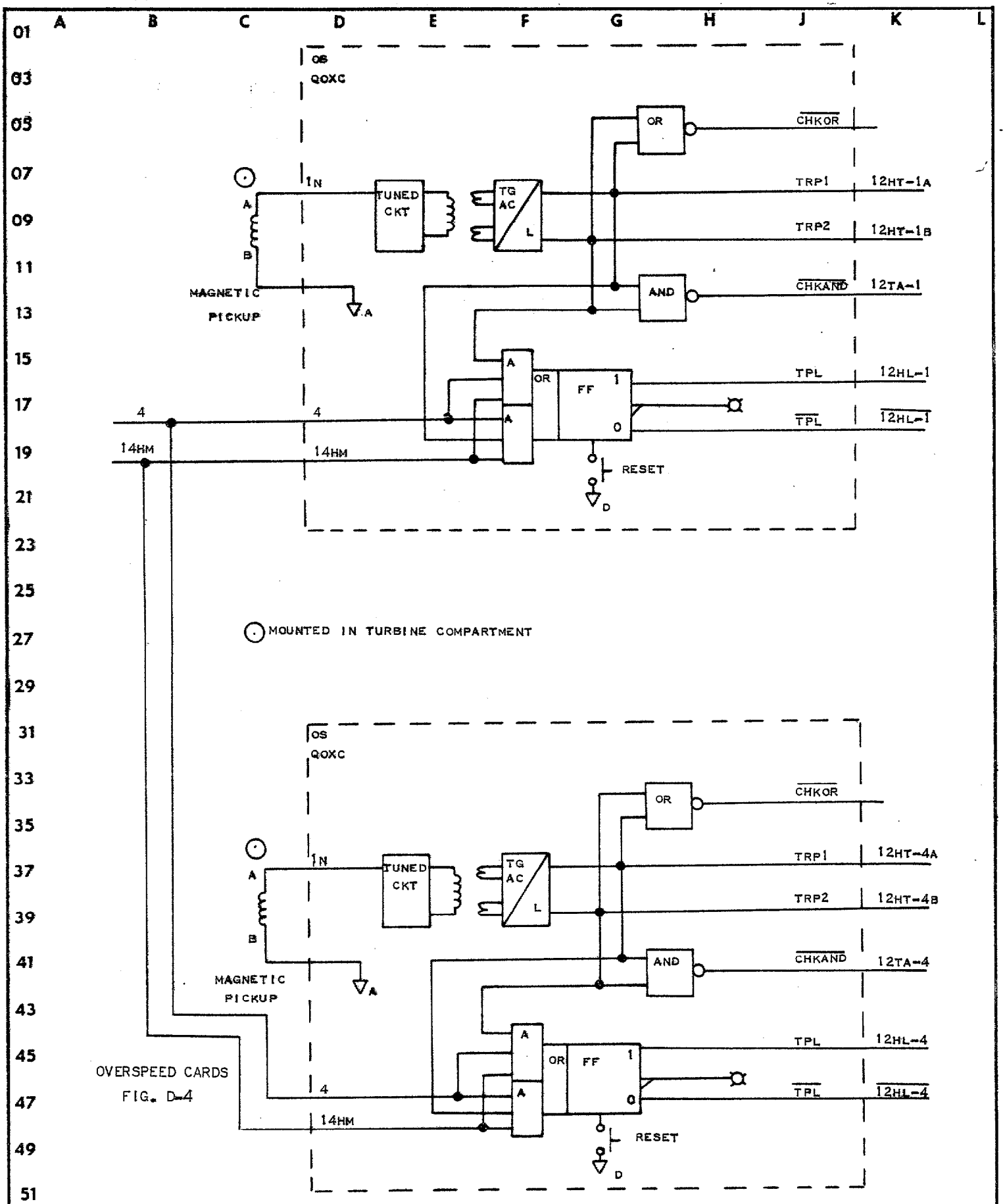


TYPICAL OVERSPEED TRIP

FIG. D-3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		TYPICAL OVERSPEED TRIP
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 85 SH. NO. 84

FORM 1C 2390 4-74



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-4
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	QOXC CARDS
REV. 3	MADE BY L.M. Worley			I.C. NO.		GEK-41053
						CONT. ON SH. 86 SH. NO. 85

D-3 OVERTEMPERATURE TRIP AND ALARM

THE OVERTEMPERATURE PROTECTION SYSTEM CONSISTS OF TWO SEPARATE CHANNELS. EACH CHANNEL LOOKS AT THE AVERAGE VALUE FROM THREE THERMOCOUPLES PLACED IN THE TURBINE EXHAUST DUCT. THE OVERTEMPERATURE SYSTEM INCLUDES TWO SOTJ CARDS, SPOA CARD AND THE TEMPERATURE AVERAGING CABINET. THE AVERAGING CABINET IS THE SAME ONE USED IN THE TEMPERATURE CONTROL LOOP AND WILL NOT BE DISCUSSED HERE. (SEE FIGURE D-5 FOR DIAGRAM OF PROTECTION SYSTEM). EACH CHANNEL HAS ITS OWN STKJ AMPLIFIER CARD IN THE TEMPERATURE AVERAGING CABINET.

SOTJ CARD (REFER TO FIGURE D-6)

THERE ARE TWO SOTJ CARDS USED FOR THE OVERTEMPERATURE TRIP AND ALARM FUNCTION. ONE CARD IS USED FOR EACH OVERTEMPERATURE CHANNEL, SINCE THE FUNCTION OF BOTH IS IDENTICAL, ONLY ONE IS DESCRIBED HERE.

THE SOTJ CARD PROVIDES THE FOLLOWING FIVE FUNCTIONS.

1. OVERTEMPERATURE TRIP FUNCTION: GENERATES A LATCHED SIGNAL WHEN THE EXHAUST TEMPERATURE EXCEEDS A PRESET VALUE. THE CIRCUIT SUMS THE EXHAUST TEMPERATURE WITH AMBIENT COMPENSATION (PCDN), PRESSURE SENSOR FAILED (PCDF) AND PEAK RESERVE BIAS. WHEN THE TOTAL EXCEEDS THE ALLOWABLE VALUE, THE OUTPUT WILL TRIGGER AND LATCH. IT CAN ONLY BE RESET WITH PB1.

2. OVERTEMPERATURE ALARM: INDICATES THAT THE TURBINE IS APPROACHING THE OVER-TEMPERATURE TRIP POINT. ADJUSTABLE BY R2.

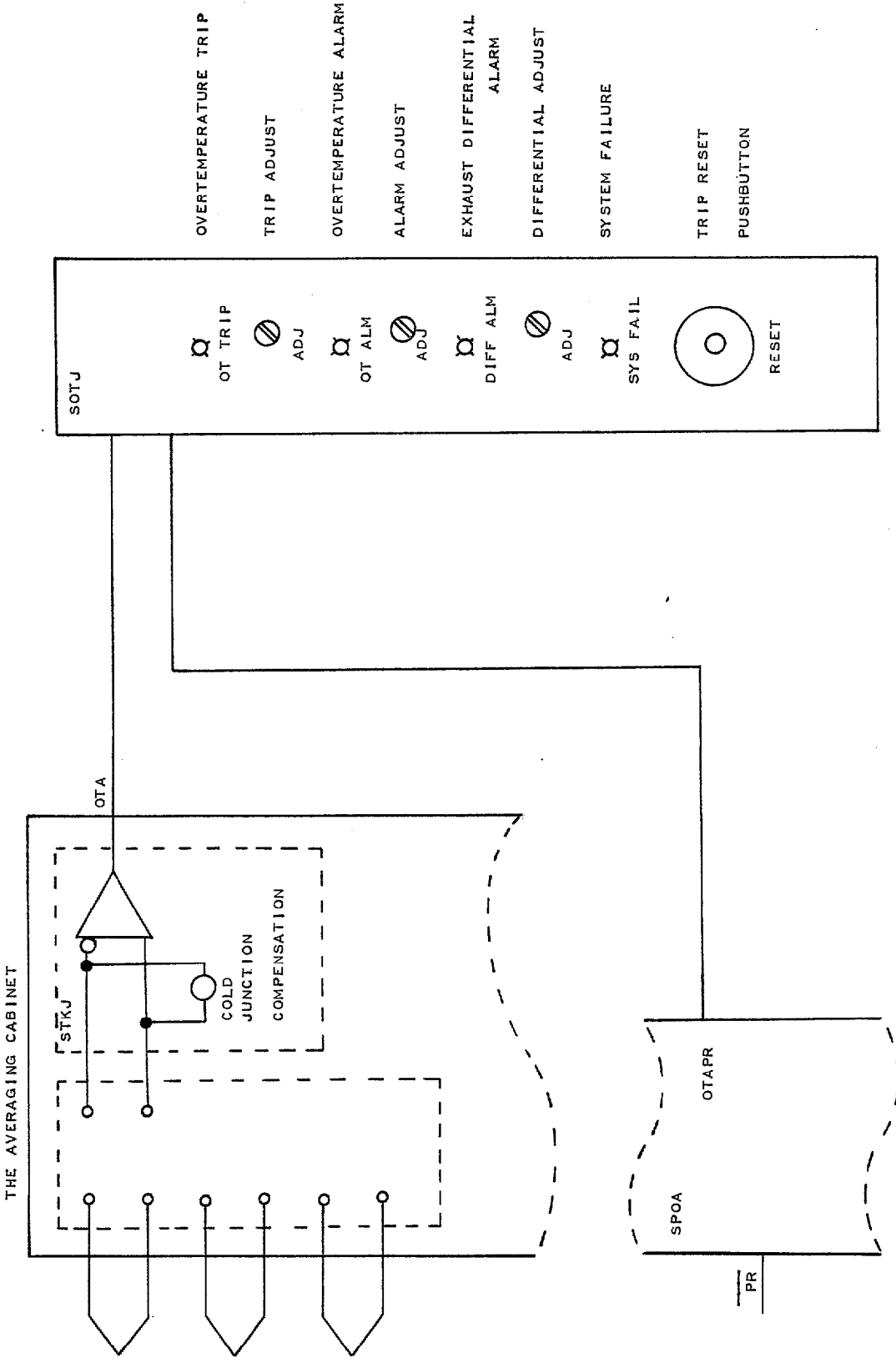
3. A DIFFERENTIAL ALARM: COMPARES THE TWO CHANNELS. IF AN ADJUSTABLE DIFFERENCE OF BETWEEN 20°F. AND 100°F. EXIST, AN ALARM SIGNAL IS GENERATED. ADJUSTABLE BY R3.

4. SYSTEM FAILURE TRIP: PROVIDES A LATCHED OUTPUT WHEN ALL OF THE OVERTEMPERATURE THERMOCOUPLES READ "COLD".

5. A TRANSIENT LOAD CAPABILITY WHICH ALLOWS THE TURBINE TO TAKE ON EXTRA TRANSIENT LOADS BELOW THE 100 SPEED CONDITION. THIS FUNCTION IS NOT USED ON MS7001C'S.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS RBM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-3 OVERTEMPERATURE
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 87 SH. NO. 86

FORM 1C 2390 4-74

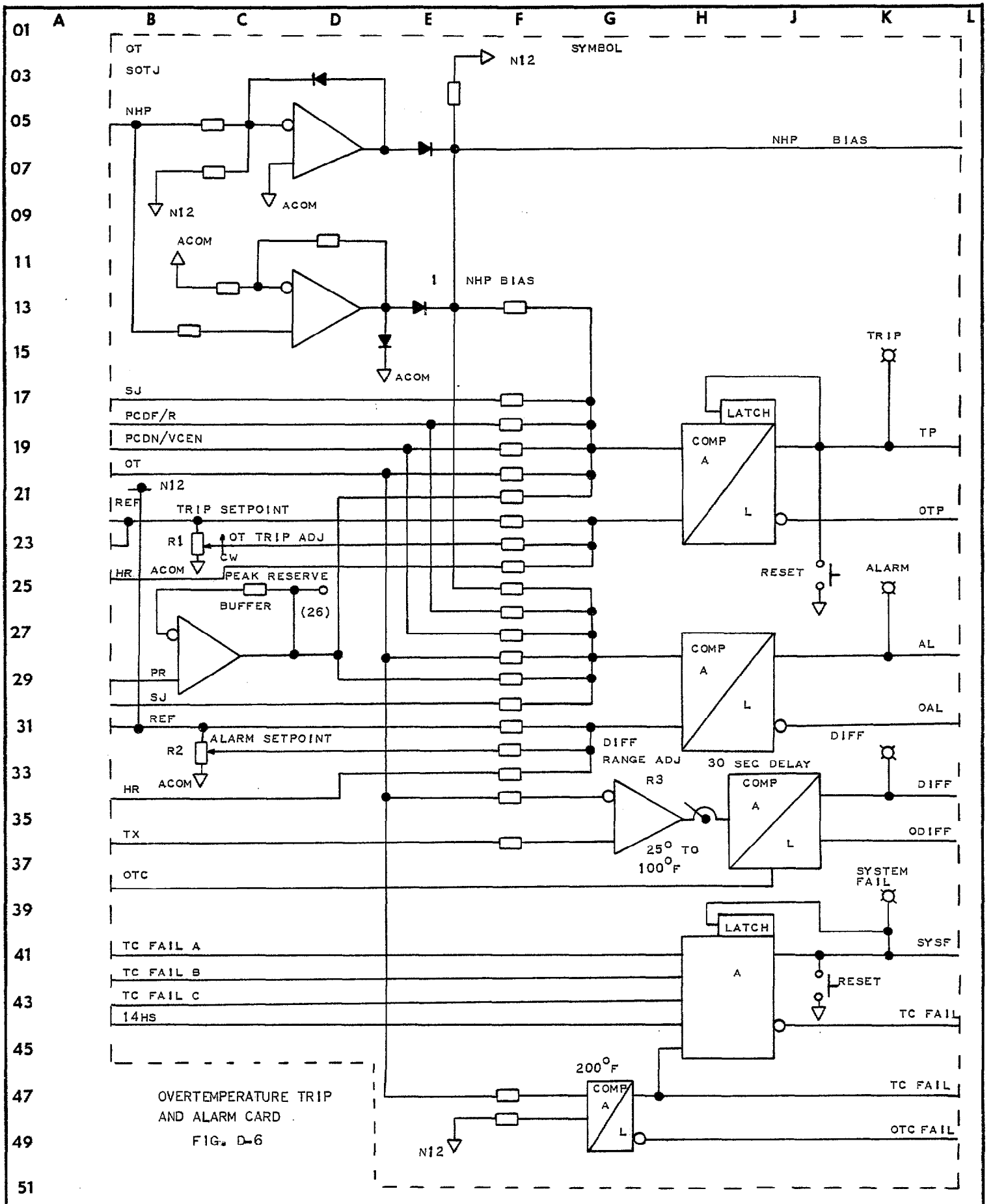


OVERTEMPERATURE ALARM AND TRIP (ONE CHANNEL)

FIG. D-5

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-5
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		OVERTEMP. PROTECTION
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 88 SH. NO. 87

FORM 1C 2390 4-74



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	D-6
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SOTJ CARD	
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053	
						CONT. ON SH. 89	SH. NO. 88

D-4 VIBRATION PROTECTION

A TYPICAL VIBRATION PROTECTION SYSTEM HAS FIVE TRIP CHANNELS. A TRIP ON ANY CHANNEL WILL TRIP THE TURBINE. A TRIP IS LATCHING AND MUST BE RESET BY PUSHBUTTON OR REMOTE SIGNAL. THREE VELOCITY VIBRATION SENSORS PROVIDE THE INPUT SIGNALS TO THE THREE CHANNELS. A TYPICAL DIAGRAM IS SHOWN IN FIGURE D-7. EACH CHANNEL HAS CIRCUITRY TO TEST FOR AN OPEN OR SHORTED SENSOR.

SVDC CARD - (REFER TO FIGURE D-8)

THE SVDC CARD RECEIVES THE SIGNAL FROM THE SENSOR BETWEEN PINS VIB AND ACOM. THE PEAK VALUE OF THIS SIGNAL IS DETECTED AND COMPARED TO A REFERENCE SET BY R1. IF THE PEAK VALUE OF THE SIGNAL EXCEEDS THE REFERENCE, A "1" IS LATCHED UP ON OUTPUT T AND TROUBLE LIGHT IS ILLUMINATED. THE LATCHING CIRCUIT MAY BE RESET VIA THE LOCAL RESET PUSHBUTTON OR A "1" ON INPUT RST. THE SETPOINT REFERENCE MAY BE MONITORED BETWEEN TP3 AND TP4. THE PEAK VALUE (VIB LEVEL) OF THE INPUT SIGNAL MAY BE MONITORED BETWEEN TP1 AND TP2. IF THE SENSITIVITY OF THE SENSOR USED IS 0.2 VOLTS PEAK FOR 3"/SEC., PIN GAIN IS CONNECTED TO ACOM TO DOUBLE THE GAIN OF THE DETECTOR.

A SENSOR FAULT DETECTOR IS PROVIDED TO DETECT A SHORTED OR OPEN SENSOR. THIS IS A LATCHING CIRCUIT AND MUST BE RESET BY LOCAL PUSHBUTTON OR REMOTE SIGNAL. THE TROUBLE LIGHT IS ILLUMINATED AND OUTPUT "A" BECOMES A "1" FOR A FAULTY SENSOR. THE FOLLOWING TABLE LISTS THE IMPEDANCE REQUIRED FOR NO FAULT CONDITIONS:

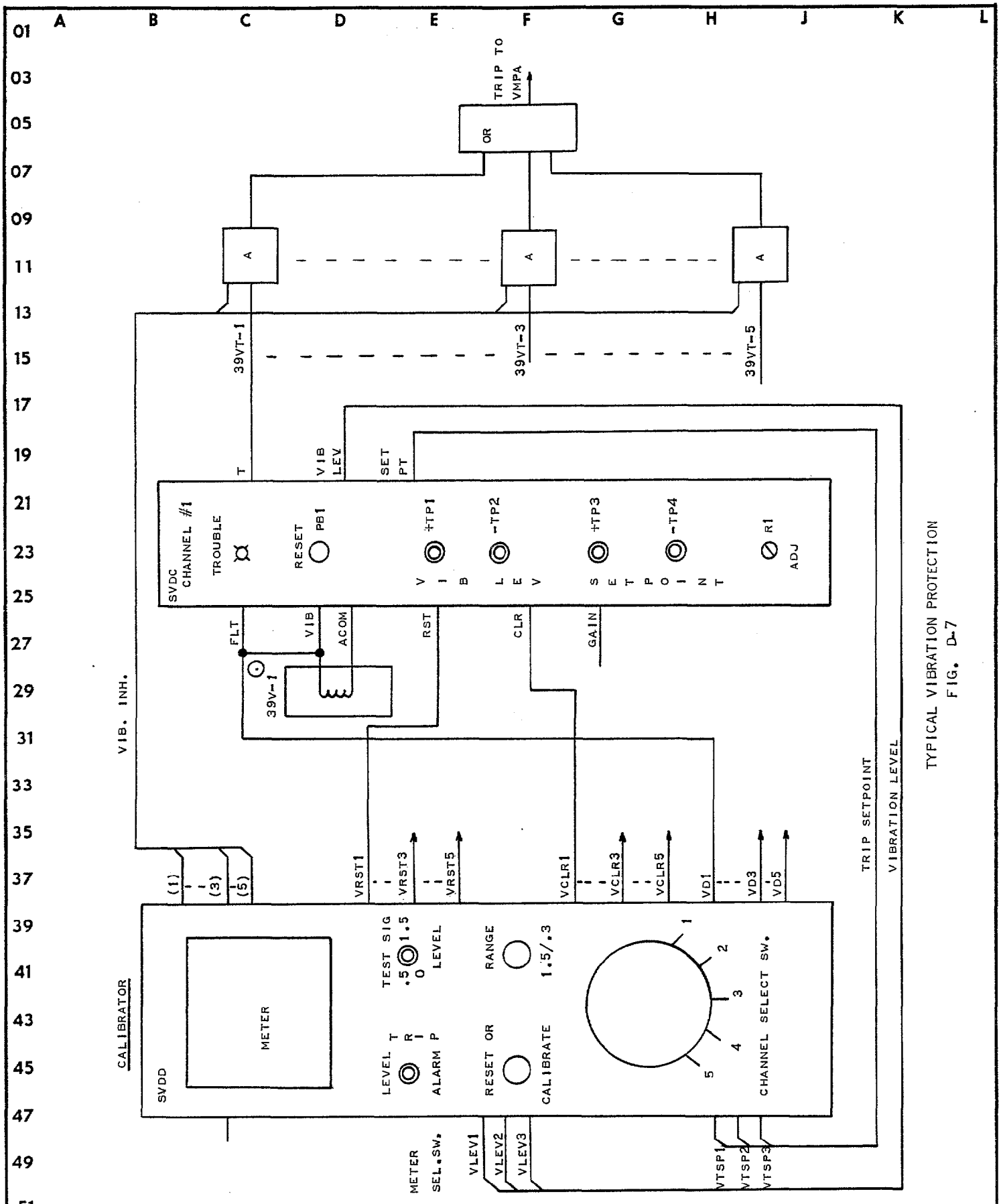
JUMPERS	NO FAULT IMPEDANCE
NONE	2.9K TO 50K OHMS
PIN 9 - P12V	80 TO 800 OHMS
PIN 6 - P12V	160 TO 1.5K OHMS

ADJUSTMENT

SET PT - R1 0.25 TO 1.6 IN/SEC.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-4 VIBRATION PROTECTION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY L.M. Darley			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 90 SH. NO. 89

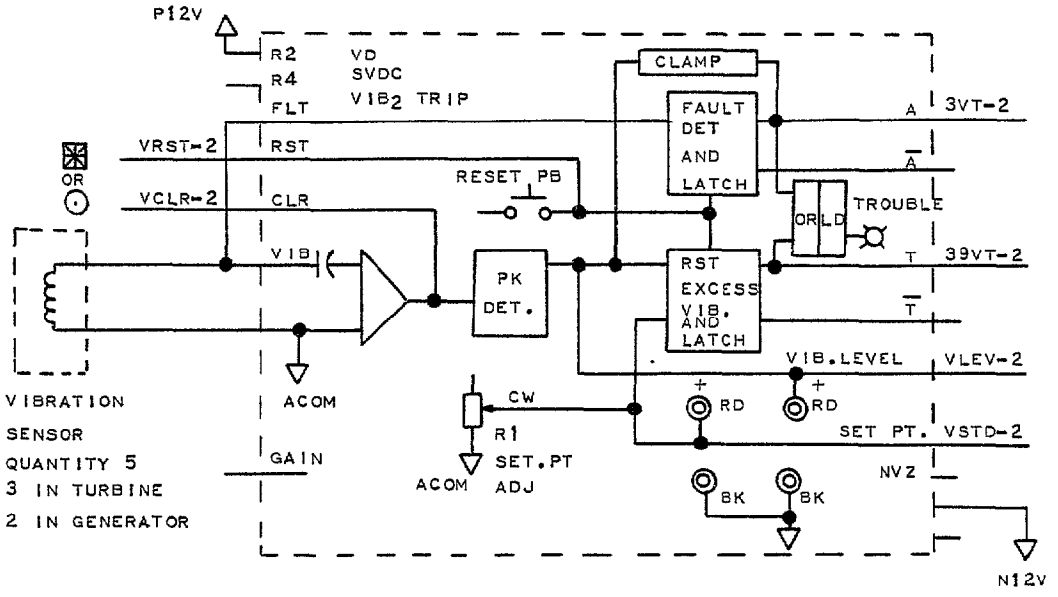
FORM 1C 2390 4-74



TYPICAL VIBRATION PROTECTION
FIG. D-7

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC SALEM, VA. U.S.A.
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	
REV. 3	MADE BY L. M. Worley			I.C. NO.	

ELEMENTARY DIAGRAM D-7
TYPICAL VIBRATION PROTECTION
GEK-41053
CONT. ON SH. 91 SH. NO. 90



ONE CARD FOR EACH VIBRATION CHANNEL (5 TOTAL)

SVDC CARD
FIG. D-8

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	D-8
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SVDC CARD	
REV 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 92	SH NO. 91

SVDD CARD - (OPTIONAL) (REFER TO FIGURE D-9)

THE SSVD CARD PROVIDES A MEANS OF CHECKING AND ADJUSTING THE SVDC CARDS. THE "CHANNEL SELECTOR" SWITCH ON SVDD SELECTS THE SVDC CARD TO BE CHECKED.

THE SVDD CALIBRATOR CARD HAS FOUR OUTPUT SIGNALS; (1) AN INHIBIT SIGNAL TO PREVENT TRIPPING THE SYSTEM WHEN THE CALIBRATE PUSHBUTTON IS DEPRESSED, (2) A CLEAR SIGNAL TO CLEAR TRIP AT COMPLETION OF TEST, (3) A RESET SIGNAL TO RESET LATCHED TRIPS, (4) A 100 HERTZ SIGNAL WITH ADJUSTABLE MAGNITUDE TO SIMULATE THE SENSOR SIGNAL.

THE INPUTS TO THE CALIBRATOR ARE THE VIBRATION LEVEL AND SETPOINT LEVEL FROM THE SVDC CARD. THE "METER SELECTOR SWITCH" DETERMINES WHICH OF THESE WILL BE DISPLAYED ON THE SVDD METER. THE "CALIBRATE" PUSHBUTTON MUST BE DEPRESSED TO CONNECT THE SVDD OUTPUTS TO THE SVDC CARD. ANY TRIPS IN A SVDC CARD WILL BE RESET UPON DEPRESSING OR RELEASING THE CALIBRATE PUSHBUTTON.

ADJUSTMENT

LEVEL - R56 0 TO 0.7 VOLTS RMS

D-5 FLAME DETECTOR

A TYPICAL HONEYWELL FLAME DETECTION SYSTEM IS SHOWN IN FIGURE D-10. EACH CHANNEL OF THE SYSTEM RECEIVES AN INPUT FROM A SEPARATE ULTRA-VIOLET SCANNER. THE SEQUENTIAL LOGIC ASSOCIATED WITH THE DETECTOR MAY BE SUCH TO PREVENT A TRIP FOR A FLAME FLICKER CONDITION PRIOR TO THE TURBINE REACHING A PARTICULAR SPEED. REFER TO GEK-28158 FOR MORE INFORMATION ON THE HONEYWELL FLAME DETECTOR.

D-6 COMBUSTION MONITOR

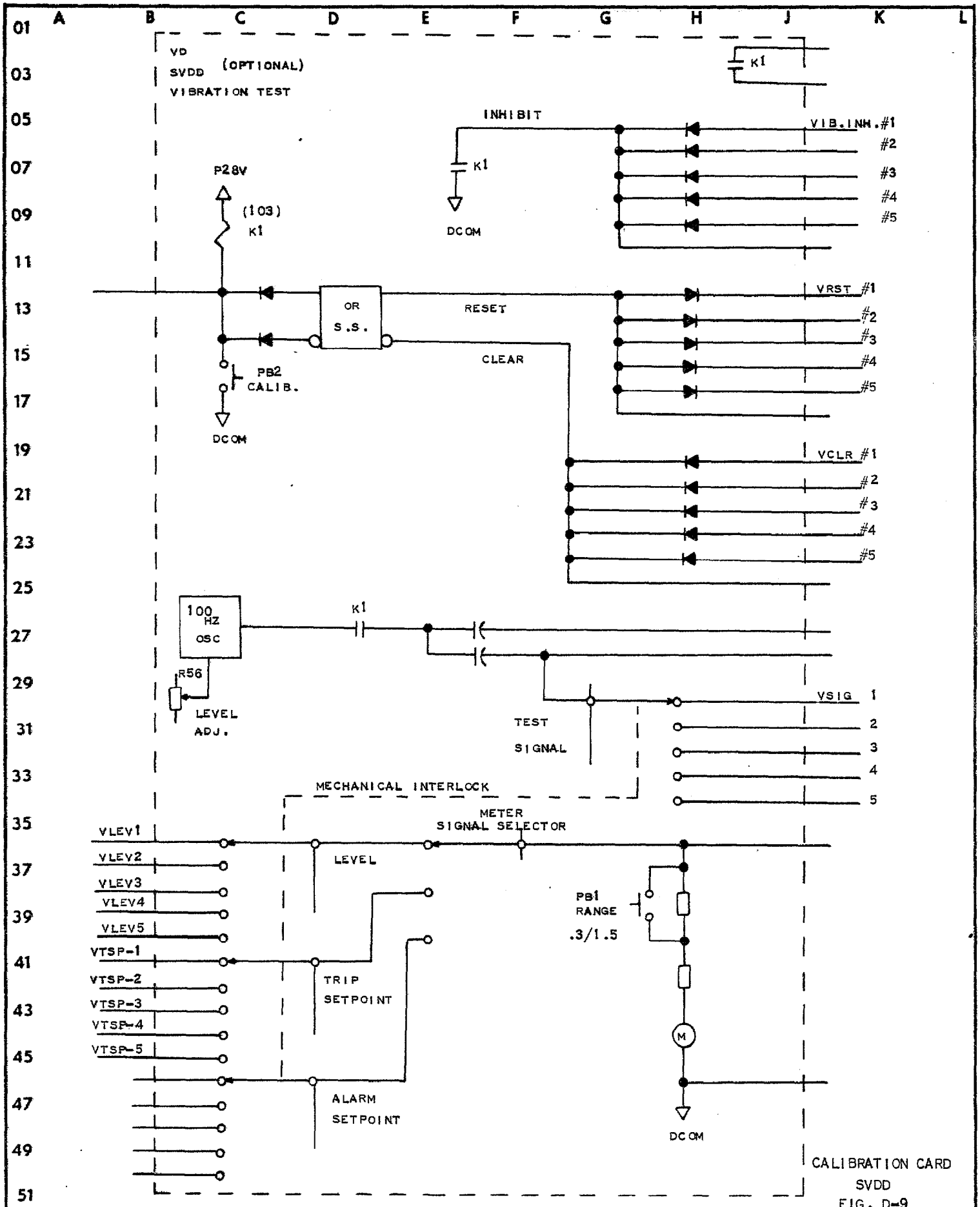
THE COMBUSTION MONITOR (REFER TO FIGURE D-11) PERIODICALLY EXAMINES THE TWELVE EXHAUST TEMPERATURE THERMOCOUPLE AND THE COMPRESSOR DISCHARGE TEMPERATURE THERMOCOUPLE. THE MONITOR IS PROGRAMMED TO ANTICIPATE COMBUSTION PROBLEMS AND EITHER ALARM, SHUTDOWN, OR TRIP THE TURBINE WHEN POTENTIAL TROUBLE IS INDICATED. REFER TO GEK-28165. FOR A MORE DETAILED DESCRIPTION OF THE COMBUSTION MONITOR .

D-7 GROUND DETECTORS

GROUND DETECTOR CIRCUITS ARE PROVIDED FOR INDICATION OF A GROUND ON THE P12, P28, P5.3, N12, XCOM AND P125 VOLT BUSES. A GROUND ON ANY OF THESE BUSES WILL BE ANNUNCIATED. THE FOLLOWING TABLE GIVES THE IMPEDANCE TO GROUND REQUIRED FOR DETECTION. A TYPICAL GROUND DETECTOR SYSTEM IS SHOWN IN FIGURE D-12 WHICH CONSISTS OF A SGDE CARD AND 64D RELAY.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		FLAME GROUND DETECTORS
REV. 3	MADE BY L.M. Dailey			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 93 SH. NO. 92

FORM IC 2390 4-74



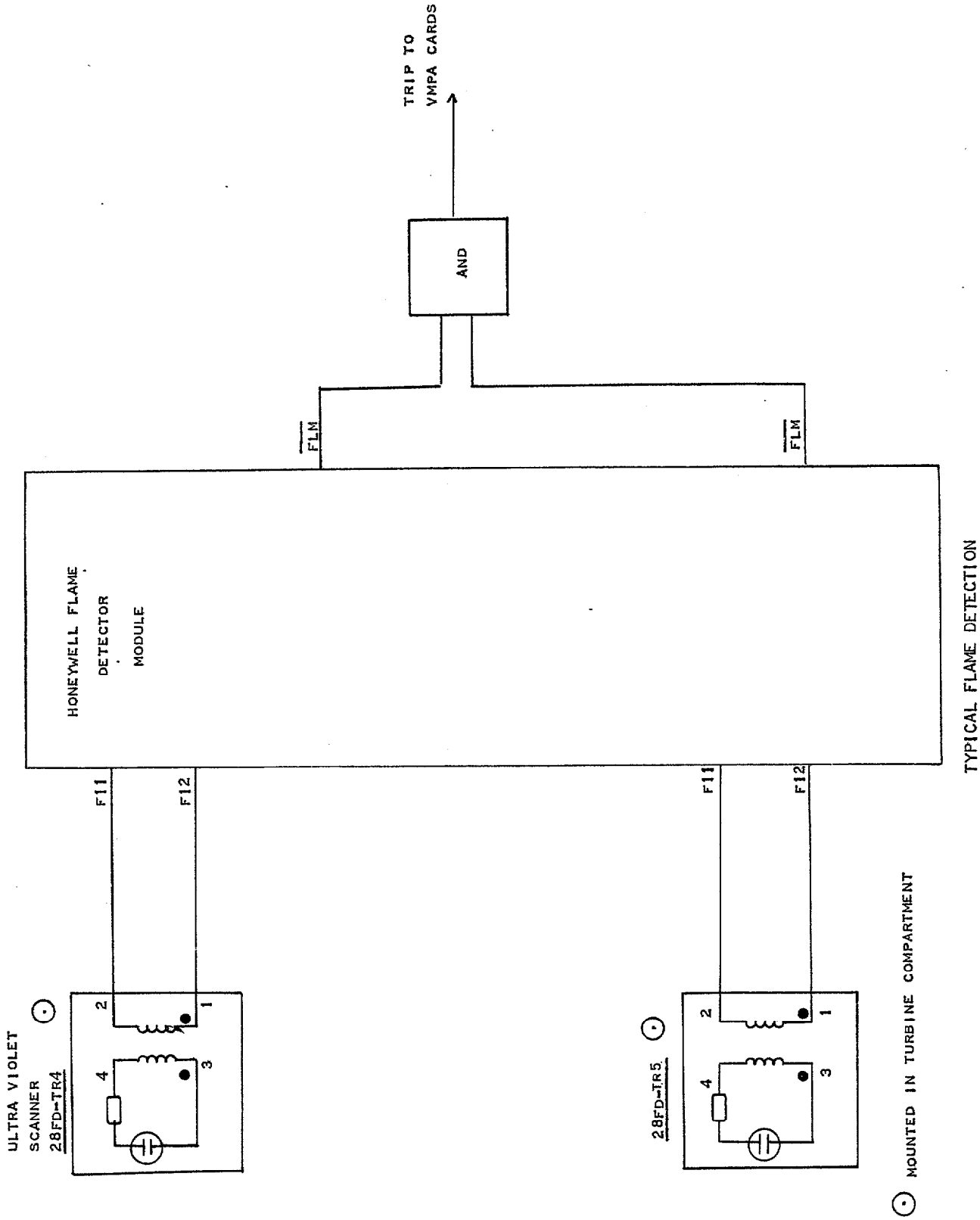
CALIBRATION CARD
SVDD
FIG. D-9

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-9
REV. 2	ISSUED 4-9-75			FIRST MADE FOR	SALEM, VA. U.S.A.	CALIBRATION CARD SVDD
REV. 3	MADE BY L. M. Worley			I.C. NO.		GEK-41053 CONT. ON SH. 94 SH. NO. 93

FORM 1C 2390 4-74

BUS	IMPEDANCE FOR FAULT
P28	2.0 – 3.3 MEG OHMS
P12	0.82– 1.5 MEG OHMS
P5.3	390 – 680 K OHMS
N12	7.5 – 15 K OHMS
ACOM	7.5 – 15 K OHMS

REV. 1	REV. 4	REV. 5	PRINTS TO:	APPROVALS:	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-7
REV. 2	ISSUED: 4-9-75			FIRST MADE FOR:	SALEM, VA. U.S.A.	GROUND DETECTORS
REV. 3	MADE BY: L.M. Dailey			I.C. NOS.		GEK-41053 CONT. ON SHT. 95 SHT. NO. 94



TYPICAL FLAME DETECTION
FIG. D-10

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-10 TYPICAL FLAME DETECTORS
REV. 2	ISSUED			FIRST MADE FOR		GEK-41053
REV. 3	MADE BY B. ALDERMAN			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 96 SH. NO. 95

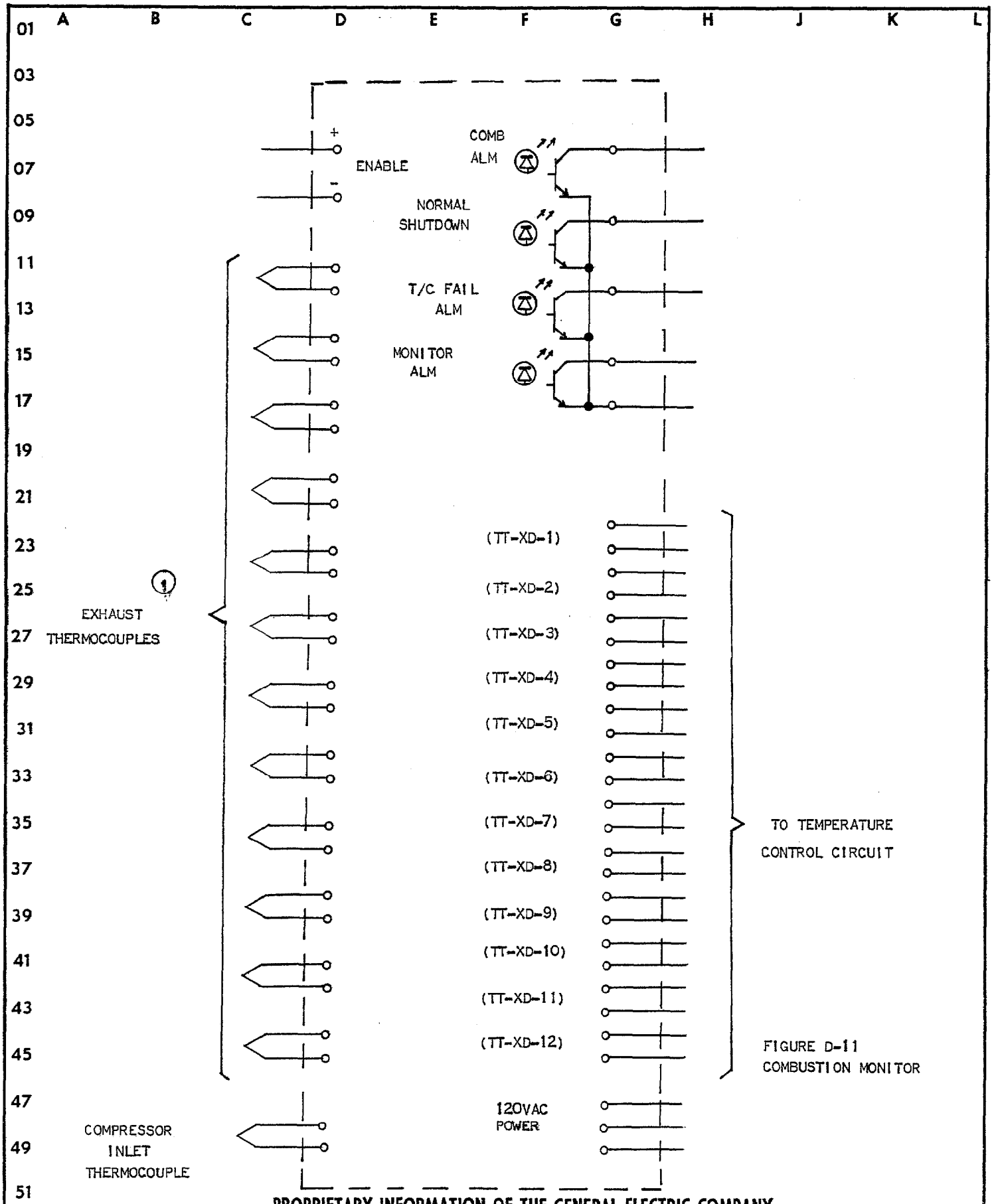
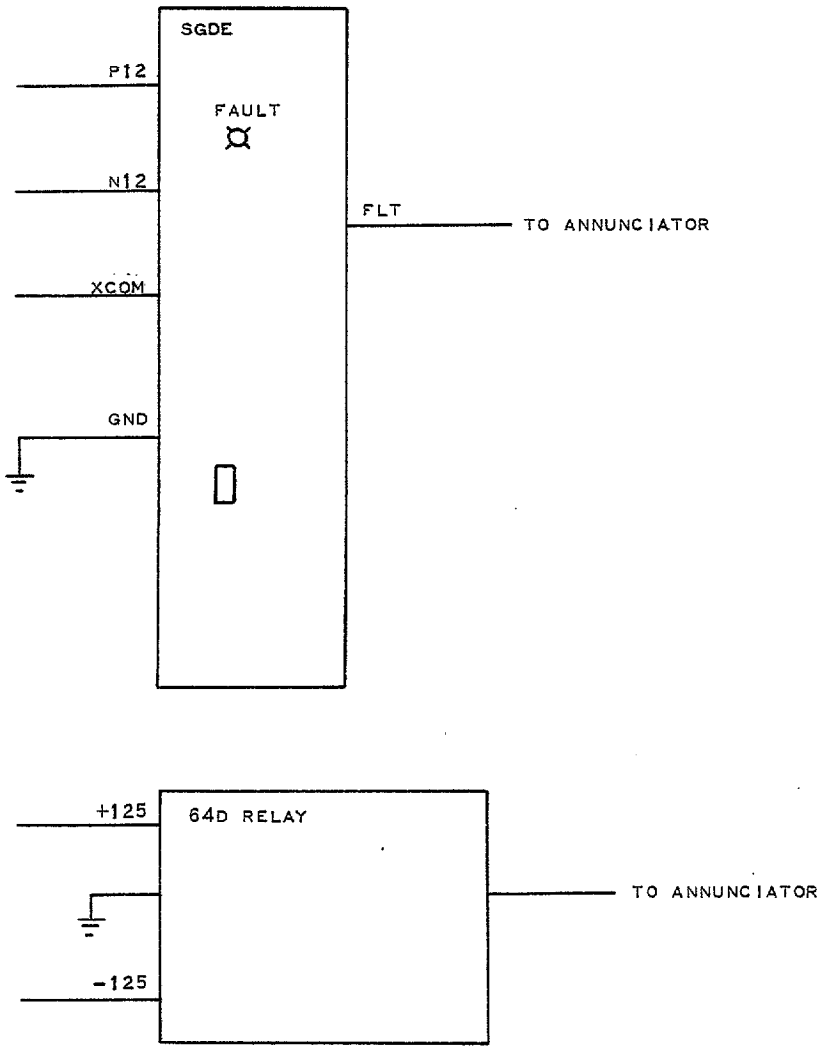


FIGURE D-11
COMBUSTION MONITOR

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM COMBUSTION MONITOR
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 97 SH. NO. 96

FORM 1C 2390 4-74



GROUND DETECTORS

FIG. D-12

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM GROUND DETECTORS D-12
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 98 SH. NO. 97

D-.8 THERMOCOUPLE GROUND DETECTION (FIGURE D-13)

WHEN A THERMOCOUPLE IS GROUNDED, CURRENT FLOWS FROM THE FLOATING ACOM BUS TO THE GROUND. THIS CURRENT WOULD PRODUCE AN ERROR IN THE THERMOCOUPLE SIGNAL. HOWEVER, IN THE FRAME SIZE 7 MK.II SPEEDTRONIC CONTROL, THIS ERROR CURRENT IS EFFECTIVELY CANCELLED BY A CIRCUIT ON THE SGDD GROUND DETECTOR CARD. THE NET ERROR FROM A GROUNDED THERMOCOUPLE IS TWO DEGREES, FAHRENHEIT, OR LESS. AT THE SAME TIME, A THERMOCOUPLE GROUND IS DETECTED BY THE SGDE CARD IN THE THERMOCOUPLE PROCESSING MODULE AND ANNUNCIATED.

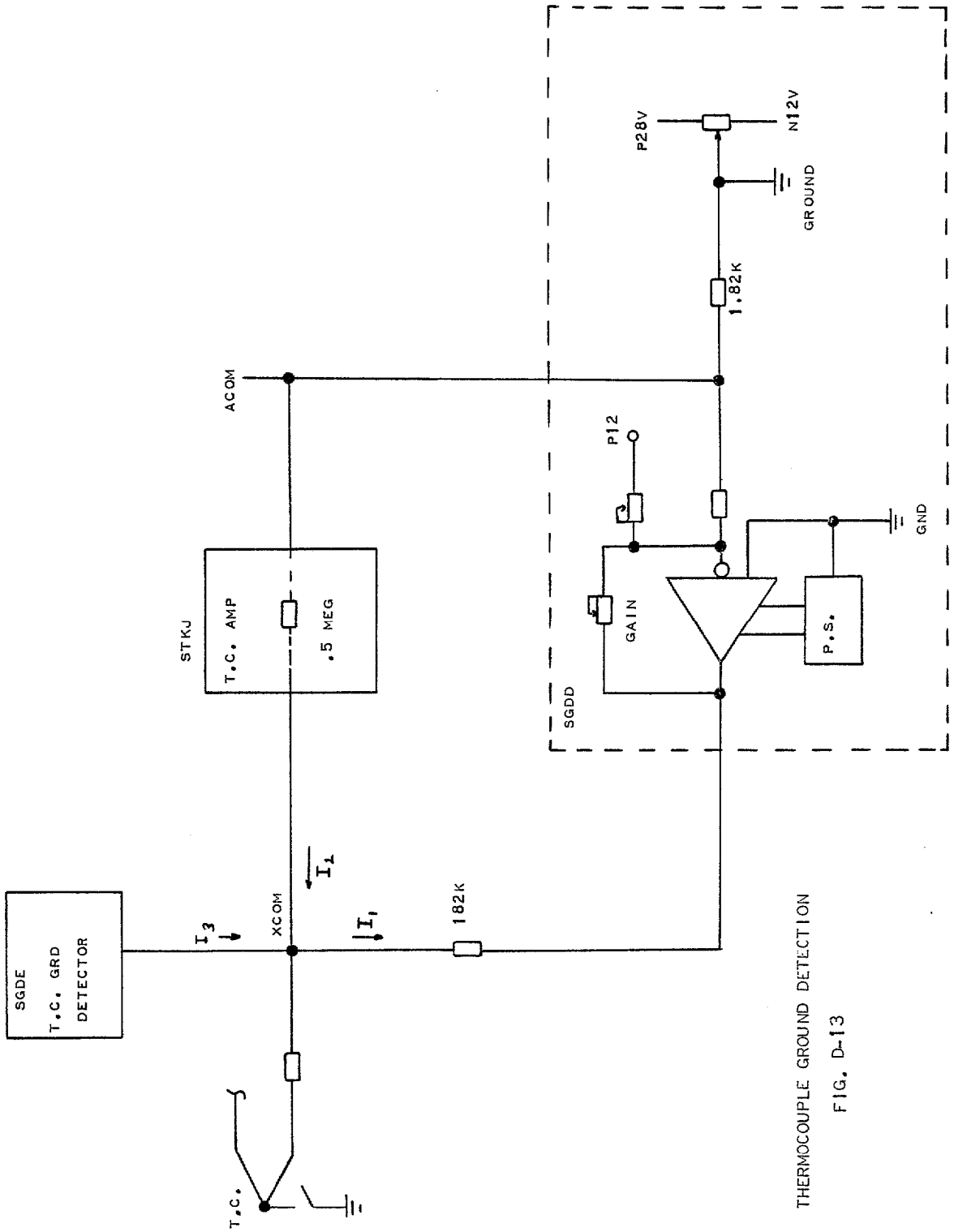
REFER TO FIGURE D-13. THE SGDE CARD HAS THREE GROUND DETECTOR CIRCUITS TO DETECT (1), A GROUND ON ONE OF THE EXHAUST TEMPERATURE TC'S OR, (2) AND (3), TO DETECT A GROUND ON TC'S OF EITHER OVER TEMPERATURE CHANNEL. THE GROUND DETECTOR CONTRIBUTES A SMALL CURRENT (I_3) TO THE XCOM BUS WHICH IS AT APPROXIMATELY +.2 VOLTS ABOVE GROUND. THIS CURRENT IS VERY SMALL AND PRODUCES AN INSIGNIFICANT ERROR.

THE STKJ THERMOCOUPLE AMPLIFIER CARD PROVIDES A HIGH RESISTANCE PATH (.5 MEGAOHMS) FROM ACOM (WHICH CAN VARY FROM +28 TO -12 VOLTS ABOVE GROUND) TO XCOM. WHEN THE THERMOCOUPLE IS GROUNDED, THIS PATH PROVIDES AN ERROR CURRENT I_2 WHICH, UNLESS CANCELLED BY ANOTHER CIRCUIT, WILL PRODUCE SIGNIFICANT ERROR ON THE AMPLIFIED THERMOCOUPLE SIGNAL.

THE SGDD CARD MONITORS THE ACOM BUS, WITH RESPECT TO GROUND AND PROVIDES A CURRENT (I_1) EQUAL AND OPPOSITE TO I_2. CONSEQUENTLY, WHEN A GROUND OCCURS ON A THERMOCOUPLE, ANY ERROR PRODUCED THROUGH THE .5 MEGAOHM LEAKAGE ON THE STKJ CARD IS CANCELLED BY THE SGDD CARD.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM THERMOCOUPLE GROUND DETECTOR D-8
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 99 SH. NO. 98

FORM 1C 2390 4-74



THERMOCOUPLE GROUND DETECTION

FIG. D-13

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS KEM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM THERMOCOUPLES GROUND DETECTION D-13
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 100 SH. NO. 99

D-9 TURNING GEAR SPEED

THE TURNING GEAR SPEED SENSING CIRCUIT IS PART OF THE PROTECTION SYSTEM. IT IS NOT LINKED WITH THE CONTROL CIRCUIT. THE TURNING GEAR SPEED SENSOR (77TG) IS THE INPUT TO AN SSZB CARD. THIS CARD IS DESCRIBED ON PAGE 49 AND IS THE SAME CARD USED FOR SPEED CONTROL OF THE TURBINE.

THE SSZB CARD MONITORS THE TURNING GEAR SPEED AND OUTPUTS A DC SIGNAL PROPORTIONAL TO THE SPEED. THIS ANALOG SIGNAL IS AN INPUT TO THE SVSE CARD AS DESCRIBED ON PAGE 24.

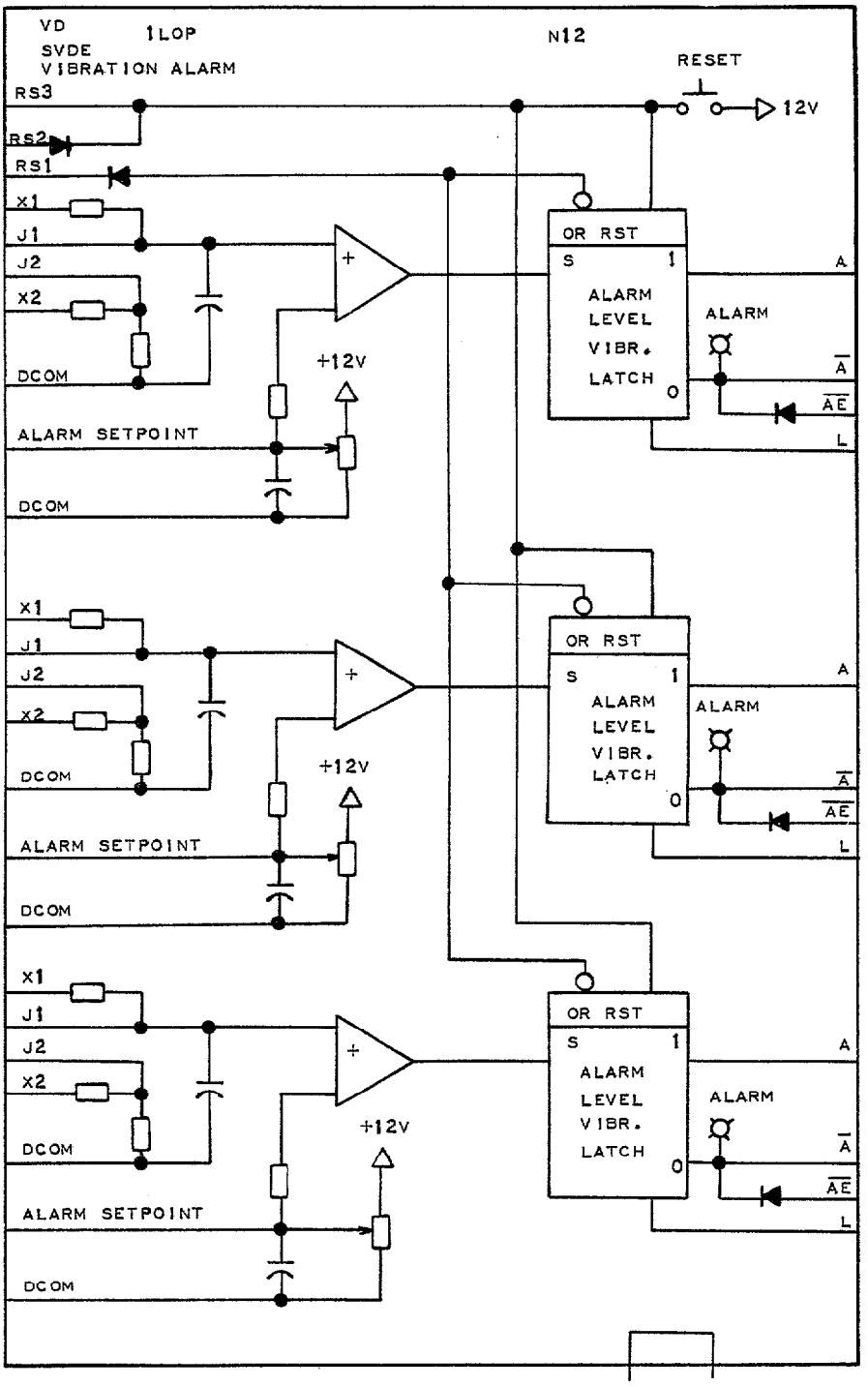
THE SVSE CARD MONITORS THE ANALOG TURNING GEAR SPEED SIGNAL AND PROVIDES A LOGIC OUTPUT ON MR AND TR WHEN THE SIGNAL REACHES A PRESET LEVEL.

D-10 HYDROGEN PURITY DETECTION

THE PURITY OF THE HYDROGEN IN THE GENERATOR IS MONITORED BY TWO GAS ANALYZERS AND DETECTORS. THE OUTPUT CURRENT SIGNAL OF EACH DETECTOR IS CONVERTED TO A VOLTAGE SIGNAL BY AN AIAD CARD IN THE TURBINE PANEL. THESE TWO SIGNALS ARE MONITORED BY AN SVDE CARD. THE AIAD CARD IS DESCRIBED ON PAGE 73. THE SVDE CARD IS DESCRIBED BELOW.

SVDE CARD (REFER TO FIGURE D-15)

THE SVDE CARD HAS THREE LEVEL DETECTOR AND ALARM CHANNELS. TWO ARE USED FOR HYDROGEN PURITY DETECTION. FOR EACH CHANNEL, THE POSITIVE INPUT VOLTAGE IS COMPARED TO AN ADJUSTABLE REFERENCE VOLTAGE. WHEN THE INPUT EXCEEDS ITS REFERENCE, THE LOGIC OUTPUT WILL INDICATE AN ALARM CONDITION, AND A LIGHT ON THE CARD WILL GO ON. IF THE ALARM OUTPUT IS TIED TO THE LATCH SIGNAL FOR THAT CHANNEL, THEN THE ALARM SIGNAL WILL LATCH AND CAN ONLY BE RESET BY THE PUSHBUTTON ON THE CARD OR BY AN EXTERNAL PUSH-BUTTON WHEN THE INPUT GOES BELOW THE ALARM LEVEL. (THERE IS ONE RESET PUSHBUTTON FOR ALL THREE CHANNELS). IF THE LATCH INPUT IS NOT CONNECTED, THE ALARM OUTPUT WILL RETURN TO THE UNALARMED STATE WHEN THE INPUT IS REDUCED TO BELOW THE ALARM LEVEL.



SVDE LEVEL DETECTOR CARD

FIG. D-14

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM D-14
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		SVDE LEVEL DETECTOR CARD
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 102 SH. NO. 101

SECTION E

POWER SUPPLY SYSTEM

A BLOCK DIAGRAM OF THE POWER SUPPLY SYSTEM IS GIVEN IN FIGURE E-1. THE SYSTEM IS BROKEN DOWN INTO THE FOLLOWING THREE FUNCTIONS (PARTS):

(A) SOURCE SELECTION
(B) DC BULK SUPPLY
(C) FINAL REGULATOR

E-1 GENERAL DESCRIPTION

(A) SOURCE SELECTION

THE SOURCE SELECTORS ALLOW THE DC BULK SUPPLY TO UTILIZE POWER FROM ONE OF TWO SOURCES. AS ILLUSTRATED IN FIGURE E-2, THESE SOURCES ARE:

(I) DC BATTERY
(II) COMMERCIAL AC

THE STANDARD CONFIGURATION IS TO USE ONE SOURCE, NAMELY THE DC BATTERY. IN THIS CASE, NO SOURCE SELECTION MODULES ARE NEEDED; THE DC BULK SOURCE IS CONNECTED DIRECTLY TO THE BATTERY. THE SOURCE SELECTORS ARE AVAILABLE AS OPTIONS TO SATISFY SPECIAL CASE REQUIREMENTS. THE OPERATION OF THE SOURCE SELECTORS IS ALMOST OBVIOUS WHEN REFERENCE IS MADE TO FIGURE E-2. SOURCE SELECTOR IC3603A166A HAS TWO POSSIBLE INPUTS, DC BATTERY AND/OR AC COMMERCIAL POWER. THE BATTERY INPUT HAS:

(I) FUSING, 20 AMPS
(II) FILTERING FOR TRANSCIENT VOLTAGE SUPPRESSION
(III) RECTIFICATION WITH FILTERING
(IV) LAMP (DSZ) FOR "POWER ON" INDICATION
(V) DIODE TO ISOLATE THE SOURCE FROM OTHER SOURCES

SOURCE SELECTOR IC3603A167A IS USED ONLY FOR AN INDUCTOR ALTERNATOR SOURCE. THIS MODULE PERFORMS THE FOLLOWING:

(I) FUSING, 20 AMPS
(II) AUTOTRANSFORMER WITH TAPS
(III) RECTIFICATION AND FILTERING
(IV) PRE-REGULATION TO 120V DC
(V) CURRENT LIMIT
(VI) FUSING, 15 AMPS
(VII) LAMP FOR "POWER ON" INDICATION
(VIII) CROWBAR, TO SHORT CIRCUIT OUTPUT IF THE VOLTAGE EXCEEDS 135V DC

THE SOURCE WITH THE HIGHEST VOLTAGE AUTOMATICALLY TAKES OVER AND SUPPLIES THE CURRENT.

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS REM	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		POWER SUPPLY-GEN. DESCPT.
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 103 SH. NO. 102

(B) DC BULK SUPPLY

BASICALLY, THE DC BULK SUPPLY PERFORMS THE FOLLOWING:

(I) ACCEPTS A NOMINAL 125 V DC VOLTAGE
(II) CONVERTS TO AC (350 TO 450 HERTZ)
(III) TRANSFORMS VOLTAGE TO APPROPRIATE LEVELS TO FEED THE P28V, P12V, N12V, AND P5V POWER SUPPLY CIRCUITS
(IV) PROVIDES ISOLATION FROM THE PRIMARY (I.E. THE 125V DC CIRCUITRY)
(V) RECITIFIES AND FILTERS THE ABOVE VOLTAGES
(VI) FEEDS THE P28V BUS DIRECTLY FROM THE DC BULK SUPPLY
(VII) FEEDS THE P12V, N12V, P5V BUSES VIA THE FINAL REGULATOR AS ILLUSTRATED IN FIGURE E-4.

THE MAIN FEATURES OF THE DC BULK SUPPLY ARE ILLUSTRATED IN FIGURE E-3. WORKING FROM LEFT TO RIGHT IN FIGURE E-3:

(I) ON-OFF SWITCH
(II) FUSING, 20 AMP
(III) THYRECTOR, FOR SPIKE SUPPRESSION
(IV) INDUCTOR, TO ISOLATE SWITCHING SPIKES (GENERATED IN BULK SUPPLY) FROM GETTING BACK ONTO THE SOURCE BUS
(V) 11,800 MICROFARADS OF CAPACITANCE TO ALLOW THE POWER SUPPLIES TO RIDE THROUGH A 65 MSEC SOURCE VOLTAGE DIP
(VI) ISOLATING DIODE, TO PREVENT A VOLTAGE DIP ON THE SOURCE FROM DISCHARGING THE CAPACITANCE OF STEP (V)
(VII) 100 MICROFARAD CAPACITOR, TO PROVIDE AC COUPLING TO GROUND
(VIII) FUSING, 15 AMPS, CURRENT LIMIT (FAST) TYPE
(IX) TRANSFORMER; THE 125 VDC (NOMINAL) IS APPLIED TO THE CENTER OF THE PRIMARY, WITH EACH END ALTERNATELY SWITCHED TO THE RETURN 125 VDC, VIA SWITCHING TRANSISTORS. THE SECONDARIES ARE RECTIFIED AND FILTERED TO PROVIDE DC VOLTAGES TO THE FINAL REGULATOR AND THE P28V BUS (REFER TO FIGURE E-4)
(X) THE EPSU CARD PROVIDES CONTROL OF THE SWITCHING TRANSISTORS OF STEP (IX), AND PROVIDES CONTROL POWER TO EPSV. THE EPSU IS IN TURN CONTROLLED VIA PULSES FROM THE EPSV CARD
(XI) THE EPSV CARD IS A COARSE P28V VOLTAGE REGULATOR; IT SENSES THE OUTPUT VOLTAGE ON THE P28V SECONDARY WINDING, COMPARES IT TO A REFERENCE AND SENDS OUT CONTROL PULSES TO EPSU (STEP (X)). EPSV ALSO PROVIDES A SHUNT LOAD CONTROL (POWER TRANSISTOR) TO ENSURE CONTINUOUS CURRENT IN THE FILTER INDUCTOR (MINIMUM OF THREE AMPERES)

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	E-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		POWER SUPPLY	
REV. 3	MADE BY J.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 104	SH. NO 103

(C) FINAL REGULATOR

THE FUNCTIONS OF THE FINAL REGULATOR ARE ILLUSTRATED IN FIGURE E-4, AND ARE AS FOLLOWS:

(I) MODULE IC3601A250B (INCLUDING EPSW):

-STEPS DOWN A COARSELY REGULATED VOLTAGE, P12X, TO A FINELY REGULATED VOLTAGE, P12V;
-PROVIDES CURRENT LIMIT;
-PROVIDES CURRENT AND VOLTAGE JACKS FOR MEASURING PURPOSES;
-PROVIDES OVERVOLTAGE PROTECTION BY MEANS OF A CROWBAR CIRCUIT (I.E. BY FIRING A SCR)

(II) MODULE IC3601A250A: SAME AS STEP (I) EXCEPT P5V BUS IS CONTROLLED INSTEAD OF THE P12V BUS.

(III) CARD EPSX: SAME AS STEP (I), EXCEPT N12V BUS IS CONTROLLED INSTEAD OF THE P12V BUS AND OVERVOLTAGE PROTECTION IS VIA A DIODE INSTEAD OF AN SCR (PROVIDES PROTECTION AGAINST BEING PULLED POSITIVE)

(IV) CARD EPSY:

-PROVIDES AN OUTSIDE, FINE VOLTAGE, CONTROL LOOP ON THE P28V BUS; I.E. IT TRIMS THE COARSE VOLTAGE REGULATOR EPSV.
-PROVIDES CURRENT LIMIT CONTROL VIA A "STOP" SIGNAL TO EPSV
-PROVIDES CURRENT AND VOLTAGE JACKS FOR MEASURING PUSPOSES

(V) SRMB CARD:

THIS CARD MONITORS THE OUTPUT VOLTAGES ON THE FOUR BUSES AND PROVIDES A LATCHING ALARM SIGNAL (GENERALLY TAKEN TO THE ANNUNCIATOR) IF ANY OF THE VOLTAGES EXCEEDS CERTAIN LIMITS (LOW/HIGH). THE ALARM RESET PUSHBUTTONS (FOUR) ARE ON THE CARD FRONT, ALONG WITH THE LED (FOUR) ALARM INDICATORS.

E-2 APPLICATION INFORMATION

(A) SOURCE SELECTOR (NOT USED ON MS7001)

(I) THE DC BATTERY INPUT CHARACTERISTICS ARE AS FOLLOWS:

VOLTAGE RANGE - 99-134V DC
THE BATTERY USUALLY CONSISTS OF 56 CELLS, FLOATED AT 2.4 V/C;
NO LOAD - 2.0 V/C, DISCHARGE = 1.75 V/C

(II) THE COMMERCIAL AC INPUT CHARACTERISTICS ARE AS FOLLOWS:

NOMINAL VOLTS +5%, - 10%; SITE TO SITE VARIATION OF NOMINAL IS 130-105V RMS.
FREQUENCY - 50/60 HZ;
AC IS SOMETIMES GROUNDED.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-2 POWER SUPPLY APPLICATION INFORMATION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 105 SH. NO. 104

THE COMMERCIAL AC INPUT IS TAKEN TO A TRANSFORMER WHICH HAS FIVE PRIMARY TAPS OF APPROXIMATELY 6V VOLTAGE STEPS, THE PRIMARY NOMINAL VOLTAGES ARE:

WINDING	VOLTS IN	VOLTS OUT (X1-X2 WINDING)
H1-H2	111	100
H1-H3	116	100
H1-H4	122	100
H1-H5	128	100
H1-H6	135	100

AT A PARTICULAR INSTALLATION THE NOMINAL VOLTAGE LEVEL SHOULD BE DETERMINED. THE NEXT HIGHER VOLTAGE WINDINGS SHOULD BE USED AS THE INPUT. THIS IS DONE TO AVOID THE POSSIBILITY OF DEVELOPING OVER 140 VOLTS OUTPUT. VOLTAGES IN EXCESS OF 140 VOLTS CAN DAMAGE RELAYS AND AUXILIARY EQUIPMENT WHICH ARE SUPPLIED BY THE OUTPUT.

OUTPUT VOLTAGE - DETERMINED BY INPUT VOLTAGE AND TRANSFORMER WINDING USED. NOMINAL PEAK NO LOAD OUTPUT VOLTAGE WILL BE APPRIXIMATELY 140V DC. THE OUTPUT REGULATION AT AN OUTPUT CURRENT OF 15 AMPS IS APPROXIMATELY 15 VOLTS.

OUTPUT CURRENT - MAX. IS 17 AMPERES.

(B) BULK SUPPLY

(I) INPUT VOLTAGE RANGE IS 95 TO 140V DC

(II) INPUT CURRENT IS 2.1 TO 7.3 AMPS AT 95V DC AND 1.75 TO 6.1 AMPS AT 140V DC

(III) OUTPUTS ARE AS FOLLOWS:

TERMINAL PT.	VOLTAGE (8%-100% LOAD)	NO LOAD VOLTAGE	RATED CURRENT
P30	29.5-31	31	15A
P12X	18-20	34.8	6A
N12X	18-21	32.7	1A
P5X	10-13	20.8	6A
P11	11-11.5		30A

(IV) THE CURRENT SIGNAL ACROSS P30V TO P28X (0.1 OHM SHUNT), IS 0.3V DC TO 1.5V DC FOR A ZERO TO 15 AMP LOAD ON THE P28V BUS. A SHUNT REGULATOR MAINTAINS A MINIMUM TO GIVE CONTINUOUS CURRENT EVEN WHEN THE LOAD ON THE P28V IS ZERO.

(V) THE SUPPLIES WILL NOT DIP DURING A DIESEL START (OR EQUIVALENT), DEFINED AS FOLLOWS:

DIP TO 60V, BACK TO 80V IN 65 MSEC.

(VI) EPSU CARD (PRIMARY SUPPLY CONTROL)

FUNCTIONS: -PRIMARY TRANSISTOR BASE DRIVE,
-PRIMARY CURRENT LIMIT,
-CONTROL POWER SUPPLY,
-PROTECTION - IF BASE AND COLLECTOR HIGH, BASE DRIVE REMOVED AND LED INDICATION

INPUTS: -COLLECTOR VOLTS (CA,CB)
-EMITTER CURRENTS EA1-3, EB1-3, SCA, SCB
-PULSES (CONTROL(FROM COARSE REG. CARD EPSV. (TIME RATIO CONTROL)
-+51V ZENER INPUT FOR CONTROL POWER (FEEDS INVERTER)

OUTPUTS: -BASE DRIVE
-CONTROL POWER SUPPLY TO EPSV, 13V, 5KHz
-LED INDICATION FOR FAILURE

(VII) EPSV CARD (P28V COARSE REGULATOR)

FUNCTIONS: -COARSE VOLTAGE REGULATOR ON 28V SUPPLY OUTPUT VOLTAGE AND PROVIDING CONTROL PULSES TO EPSU (ABOVE)
-CONTROL TO SHUNT CURRENT REG TO ENSURE +12, -12, +5.3 STAY UP
-PROVIDES CONTROL POWER TO OTHER CARDS VIA P11 (11V DC) WHICH DOES NOT DISAPPEAR WHEN INVERTER SHUTS DOWN

INPUTS: -CONTROL POWER ACC1, ACC2, FROM EPSU
-VOLTAGE FEEDBACK, AC1, AC2, COM.
-STOP-LOGIC, "1" SHUTDOWN INVERTER "0" STARTS INVERTER
-REG - INPUT FROM FINE VOLTAGE REGULATOR EPSY
-SIGNAL BIAS OPERATION OF COURSE REGULATOR
-CURRENT BIAS OPERATION OF COURSE REGULATOR
-CURRENT SIGNAL ACROSS SHUNT P30, P28X

OUTPUTS: -PULSES TO EPSU (TIME RATIO CONTROL)
-CONTROL SUPPLY P11, 11V
-S COM - RETURN FOR P11
-SH - BASE DRIVE TO SHUNT CURRENT REG.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-2 POWER SUPPLY APPLICATION INFORMATION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 107 SH. NO. 106

(C) FINAL REGULATOR

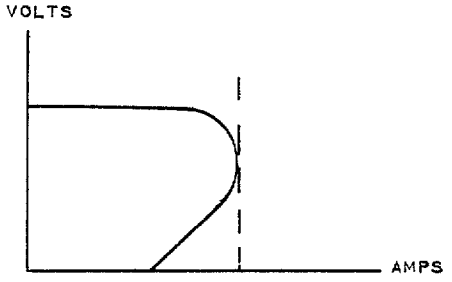
(I) THE FINAL REGULATOR HAS THE FOLLOWING OUTPUT CHARACTERISTICS;

VOLTS	BUS	RETURN	RATED CURRENT	CURRENT LIMIT	RIPPLE UP-P	REG. (LD)	DRIFT (TEMP)	100 CURRENT SHUNT-SIGNAL
+28V*	P28V	P COM	15A	17A	250 MV	300 MV	-	1.5V
+12V	P12V	A COM	6A	7A	10 MV	30 MV	60 MV	1 V
-12V	N12V	A COM	1A	1.2A	10 MV	30 MV	60 MV	2 V
+5.3V	P5V	D COM	6A	7A	100 MV	150 MV	40 MV	1 V

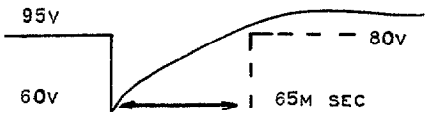
*FOR INPUT OF 105 TO 140V DC; FOR INPUT DOWN TO 95V, REDUCE 28V LOAD TO 7A OR ACCEPT P28V DROP TO 26V.

(II) AN SCR IS USED ON EACH OF THE P12V AND P5V BUSES AND IS CALLED A "CROWBAR" CIRCUIT. ITS PURPOSE IS TO MONITOR THE BUSES AND PROTECT THE LOAD EQUIPMENT FROM OVERVOLTAGE. THE P12V BUS CROWBAR OPERATES AT 15.5 TO 19.5V; THE P5V BUS CROWBAR OPERATES AT 8-10.5V; THE N12V BUS HAS A DIODE TO PROTECT AGAINST POSITIVE GOING VOLTAGES; THE P28V HAS NO CROWBAR, SINCE ITS LOAD IS AS VOLTAGE SENSITIVE AS THE OTHER BUSES.

(III) THE SUPPLIES HAVE A CURRENT LIMIT FOLDBACK AS FOLLOWS:



(IV) THE SUPPLIES WILL NOT DIP DURING A DIESEL START (OR EQUIVALENT), DEFINED AS FOLLOWS:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-2 POWER SUPPLY APPLICATION INFORMATION
REV 2	ISSUED 4-9-75			FIRST MADE FOR		
REV 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 106 SH. NO 107

(VI) AC VOLTAGE, 350-450 Hz, 60V RMS IS AVAILABLE FROM THE BULK SUPPLY; FOR AC LOADING, REDUCE P28V VOLT-AMPS BY AN EQUIVALENT AMOUNT; NON-STANDARD

(VII) MONITORS - OVER/UNDER VOLTAGE ON ALL SUPPLIES, AND +28V SUPPLY OVERCURRENT, MONITORING WITH LED AND LOGIC ANNUNCIATION

(VIII) EPSY (P28 FINE REGULATOR)

FUNCTIONS: -FINE VOLTAGE REGULATION
-CURRENT LIMIT

INPUTS: -P11 - CONTROL SUPPLY
-P28 - 28V BUS, FOR RING BUS MONITOR
-P30 -P28X - FOR CURRENT LIMIT SIGNAL
-P30-P COM-RAW DC SUPPLY
-SENSE - REMOTE VOLTAGE SENSING

OUTPUTS: -REG. - FINE VOLTAGE REG. BIAS TO EPSV
-OCL - CURRENT LIMIT LOGIC SIGNAL "0" = CL; TO BE USED WITH MONITOR
-STOP - TO SHUT OFF INVERTER ON CL "1" = INVERTER OFF

(IX) EPSW (P12V, P5V REGULATOR)

FUNCTIONS: -REGULATE +5.3V OR +12 BUS
-CURRENT LIMIT ON +5.3V OR +12 BUS
-CROWBAR +5.3V OR +12 BUS

INPUTS: -GATE TO CROWBAR SCR
-P12X OR P5X - POWER
-V SENSING - REMOTE SENSING

OUTPUTS: -GATE TO CROWBAR SCR
-VM - TO MONITOR
-BASE DRIVE TO SERIES TRANSISTOR REGULATOR

(X) EPSX (12V REGULATOR

FUNCTIONS: -REGULATE -12V
-CURRENT LIMIT

INPUTS: -N12X - POWER
-VM (+12V SUPPLY) CONTROL POWER
-VS - REMOTE SENSING

OUTPUTS: -VM - MONITOR VOLTAGE
-N12V - OUTPUT POWER

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-2 POWER SUPPLY APPLICATION INFORMATION
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT ON SH 109 SH. NO. 108

(XI) SBMB (BUS MONITOR)

FUNCTIONS: -MONITOR UNDER/OVER VOLTAGE ON FOUR SUPPLIES DETECTION AT 10-15% ERROR
-ANNUNCIATE WITH LED (1ST FAULT INDICATION)
-SIGNAL LOGIC; 1 = ALARM (4)
-MONITOR CURRENT LIMIT ON EPSY (28V REGULATOR), OCL = 0 = CURRENT LIMIT CONDITION
-LOCAL RESET PB AND REMOTE PB

INPUTS: -VM EACH SUPPLY (MONITOR VOLTAGE)
-P11; CONTROL POWER FROM EPSV
-OCL - CL FROM +28 REG.

OUTPUTS: -STOP - "1" FOR OUT OF NORMAL CONDITION; MAY BE USED TO SHUT DOWN INVERTER
-LOGIC SIGNALS - ALARM (4); 1 = ALARM

E-3 TROUBLE SHOOTING GUIDE

THESE INSTRUCTIONS ARE WRITTEN TO AID THE TROUBLESHOOTER IN ISOLATING POWER SUPPLY TROUBLES. THIS IS A GUIDE ONLY, AND CAN NOT BE APPLIED BLINDLY WITHOUT SOME KNOWLEDGE OF TROUBLESHOOTING PROCEDURES AND OF THE POWER SUPPLIES, THEMSELVES. IT IS RECOMMENDED THAT THE TROUBLESHOOTER, IF NOT ALREADY FAMILIAR, BRIEFLY REVIEW THE GENERAL DESCRIPTION OF THE POWER SUPPLY SYSTEM BEFORE PROCEEDING WITH THESE INSTRUCTIONS.

(A) SYMPTOM: P5V OR P12V BUS OUT OF LIMITS, BUT ALL OTHER BUSES CORRECT.

ACTION:

(I) IT IS POSSIBLE THAT THE "CROWBAR" PROTECTION CIRCUIT HAS BEEN ENERGIZED DUE TO EXTERNAL VOLTAGE SPIKING. VERIFY THAT EXTERNAL INSTRUMENTATION, ETC., IS NOT DISTURBING THE BUS. REMOVE POWER FOR SEVERAL SECONDS (TO ALLOW SCR TO RESET) AND REAPPLY THE POWER.

(II) IF THE FAULT IS STILL PRESENT, REMOVE POWER, REPLACE THE EPSW, AND REAPPLY POWER

(III) IF THE FAULT IS STILL PRESENT, VERIFY THAT THE INPUT TO THE FINAL REGULATOR MODULE (P12X OR P5X) IS AS DEFINED IN APPLICATION SECTION E-2B, (III); IF NOT, THE BULK SUPPLY IS SUSPECT. (I.E. THE DIODE-FILTER SECTION OF THE BULK SUPPLY).

(IV) IF STEP (III) WAS CORRECT, REMOVE POWER, ISOLATE ALL LOADING ON THE POWER SUPPLY IN TROUBLE, REAPPLY POWER. IF THE POWER SUPPLY IS CORRECT NOW, THE TROUBLE IS PROBABLY EXTERNAL TO THE SUPPLY (EITHER EXCESSIVE LOAD OR OVERVOLTAGE FROM OTHER SOURCES).

(V) IF THE POWER SUPPLY IS STILL NOT CORRECT, MODULE IC3601A250A (P5V) OR IC3601A250A (P12V) IS SUSPECT.

(B) SYMPTOM: N12V BUS IS OUT OF LIMITS, BUT OTHER BUSES ARE CORRECT.

ACTION:

(I) SUSPECT THE EPSX CARD; REMOVE POWER, REPLACE EPSX

(II) IF THE BUS IS STILL NOT CORRECT, ISOLATE THE LOAD AND CHECK THE N12V BUS. IF THE BUS RETURNS TO NORMAL, THE TROUBLE IS PROBABLY EXTERNAL TO THE POWER SUPPLY SYSTEM.

(III) IF THE BUS IS STILL NOT CORRECT, MEASURE THE VOLTAGE ON THE N12V BUS (INPUT TO EPSX) AND VERIFY THAT IT IS AS DEFINED IN APPLICATION SECTION E-2B, (III). IF NOT, THE BULK SUPPLY IS SUSPECT (I.E. DIODE-FILTER SECTION).

(C) SYMPTOM: P28V BUS OUT OF LIMITS, BUT THE OTHER BUSES ARE CORRECT.

ACTION:

(I) THE EPSY CARD IS SUSPECT; REMOVE POWER, REPLACE EPSY.

(II) IF THE BUS IS STILL NOT CORRECT, ISOLATE THE LOAD AND CHECK THE P28V BUS. IF THE BUS RETURNS TO NORMAL, THE TROUBLE IS PROBABLY EXTERNAL TO THE POWER SUPPLY.

(III) IF THE BUS IS STILL NOT CORRECT, MEASURE THE VOLTAGE ON P28V, AND VERIFY THAT IT IS AS DEFINED IN APPLICATION SECTION E-2B, (III); IF NOT, THE BULK SUPPLY IS SUSPECT; I.E. DIODE-RECTIFIER OUTPUT IS SUSPECT.

(D) SYMPTOM: ALL POWER SUPPLY BUSES ARE LOW, BUT EPSU LIGHT EMITTING DIODE (LED) IS OFF, AND BULK SUPPLY IS MAKING A NOISE (HUM).

ACTION:

(I) CHECK THAT THE BULK SUPPLY INPUT IS 95V DC OR GREATER

(II) CHECK THAT THE "STOP" SIGNAL (WITH RESPECT TO P-COM), LABELED INTERCONNECTIONS BETWEEN BULK SUPPLY AND REGULATOR, IS A LOGIC "ZERO".

USE AN OSCILLOSCOPE IF AVAILABLE. THIS SIGNAL SHOULD STAY AT ZERO VOLTS. IF THE P28V SUPPLY IS RUNNING INTO CURRENT LIMIT, IT WILL BOUNCE BETWEEN "ZERO" AND APPROXIMATELY 2 VDC. ALSO CHECK THE VOLTAGE SIGNAL ACROSS THE CURRENT JACKS OF THE EPSY CARD AND COMPARE IT TO THE NOMINAL CURRENT LIMIT SIGNAL OF 1.7 VDC.

(III) IF THE TROUBLE IS STILL NOT DETERMINED, REMOVE POWER, ISOLATE THE P28V LOAD, REAPPLY POWER AND CHECK FOR THE CORRECT BUS VOLTAGES. IF THE BUSES ARE STILL LOW, JUMPER "STOP" TO "PCOM". IF THE BUS VOLTAGES ARE NOW CORRECT, EPSY IS SUSPECT. IF THE BUS VOLTAGES ARE STILL LOW, BULK SUPPLY CARDS EPSU, EPSV ARE SUSPECT.

(IV) IF THE BUSES BECOME CORRECT WHEN THE P28V LOAD IS ISOLATED, SUSPECT THE P28V SUPPLY CURRENT LIMIT TROUBLE OR PRIMARY CURRENT LIMIT TROUBLE, BUT THE P28V CURRENT LIMIT WAS CHECK IN STEP (II). THEREFORE, PRIMARY CURRENT LIMIT THAT IS EPSU, IS SUSPECT (CHANGE OUT EPSU).

(E) SYMPTOM: EPSU LIGHT EMITTING DIODE "LED" IS ON; BULK SUPPLY WILL NORMALLY BE SHUT DOWN, I.E. WILL NOT BE MAKING A NOISE (HUM).

ACTION:

(I) REMOVE POWER AND CHECK ALL POWER TRANSISTORS ON BULK SUPPLY HEAT SINK HS1, HS2.

(II) IF TRANSISTORS ARE CORRECT SUSPECT EPSU AND REPLACE

(F) SYMPTOM: BULK SUPPLY NOT HUMMING, EPSU, "LED" IS OFF, BUT INDICATING LIGHT, DS1 IS ON.

ACTION:

(I) REMOVE POWER AND CHECK FUSES; REPLACE ANY BAD FUSES.

(II) REMOVE EPSU, EPSV AND CHECK POWER TRANSISTORS ON HS1 AND HS2; REPLACE ANY BAD TRANSISTORS.

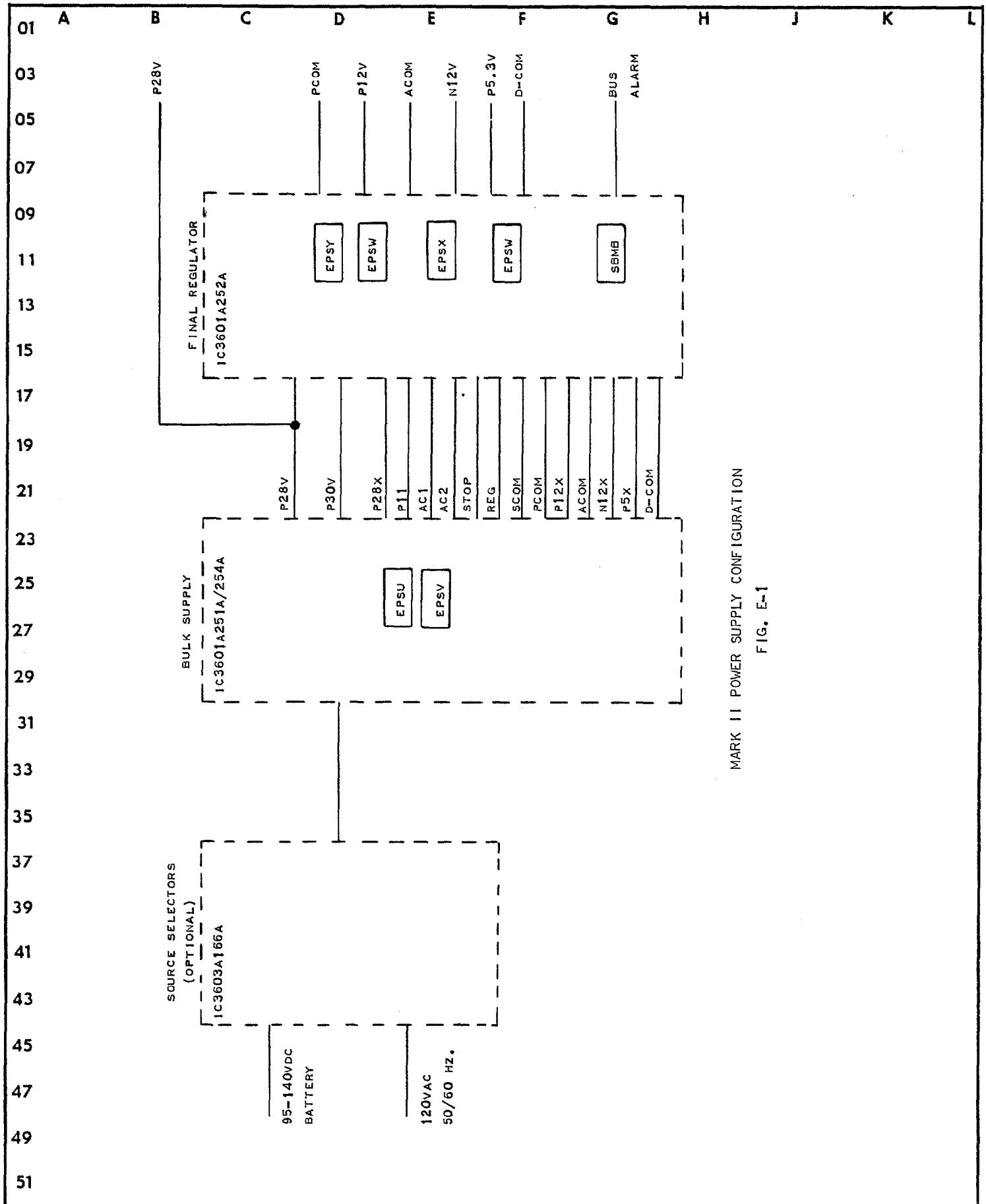
(III) APPLY POWER AND CHECK FOR BATTERY VOLTAGE FROM CA TO H-COM AND FROM CB TO H-COM (LOCATED ON TB2); BATTERY VOLTAGE MUST BE BETWEEN 95-145 VDC. IF THIS VOLTAGE IS NOT PRESENT, SUSPECT BAD POWER TRANSISTOR OR SOME OTHER LOW IMPEDANCE PATH ON THE FRONT END. REMOVE POWER, ISOLATE AND REMOVE THIS LOW IMPEDANCE PATH.

(IV) IF BATTERY VOLTAGE IS CORRECT ON CA,CB, CHECK FOR 51 ± VDC BETWEEN PINS 11, AND 51 IN THE EPSU SLOT.

(V) IF THE PROBLEM HAS NOT BEEN IDENTIFIED TO THIS POINT, REMOVE POWER, PLUG IN NEW EPSU AND OLD EPSV CARDS AND CHECK SYSTEM. IF OPERATION IS STILL NOT CORRECT, REMOVE POWER REPLACE EPSV, AND TRY AGAIN.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM POWER SUPPLY TROUBLESHOOTING GUIDE E-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 112 SH. NO. 111

FORM 1C 2390 4-74

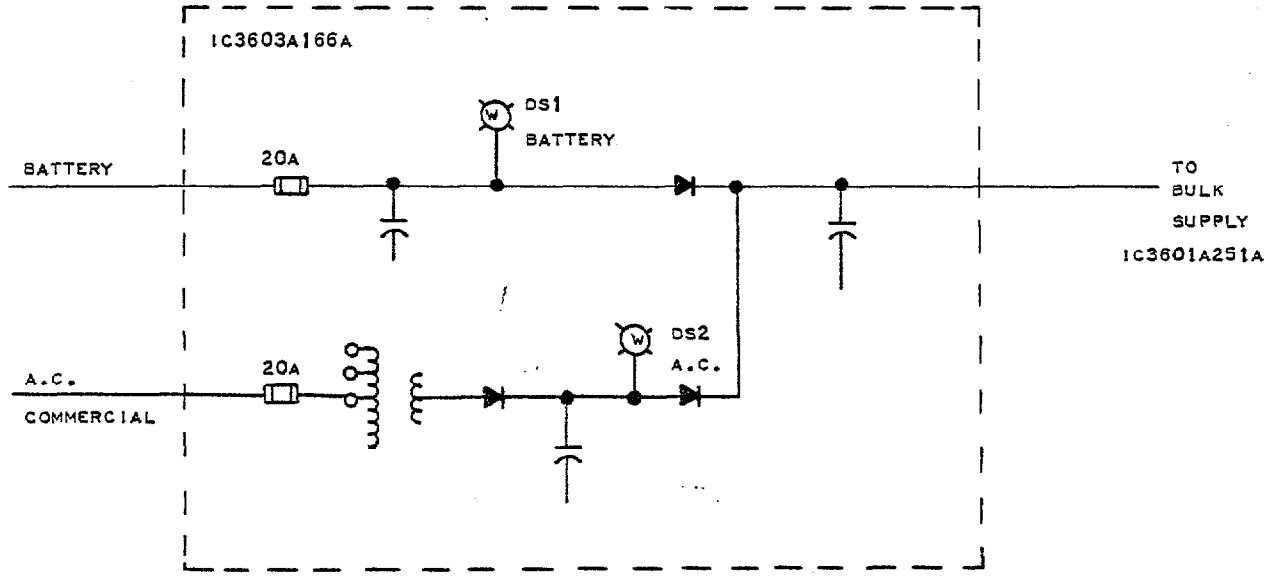


MARK II POWER SUPPLY CONFIGURATION

FIG. E-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-1
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		MK II POWER SUPPLY CONFIG.
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 113 SH. NO. 112

FORM IC 2390 4-74

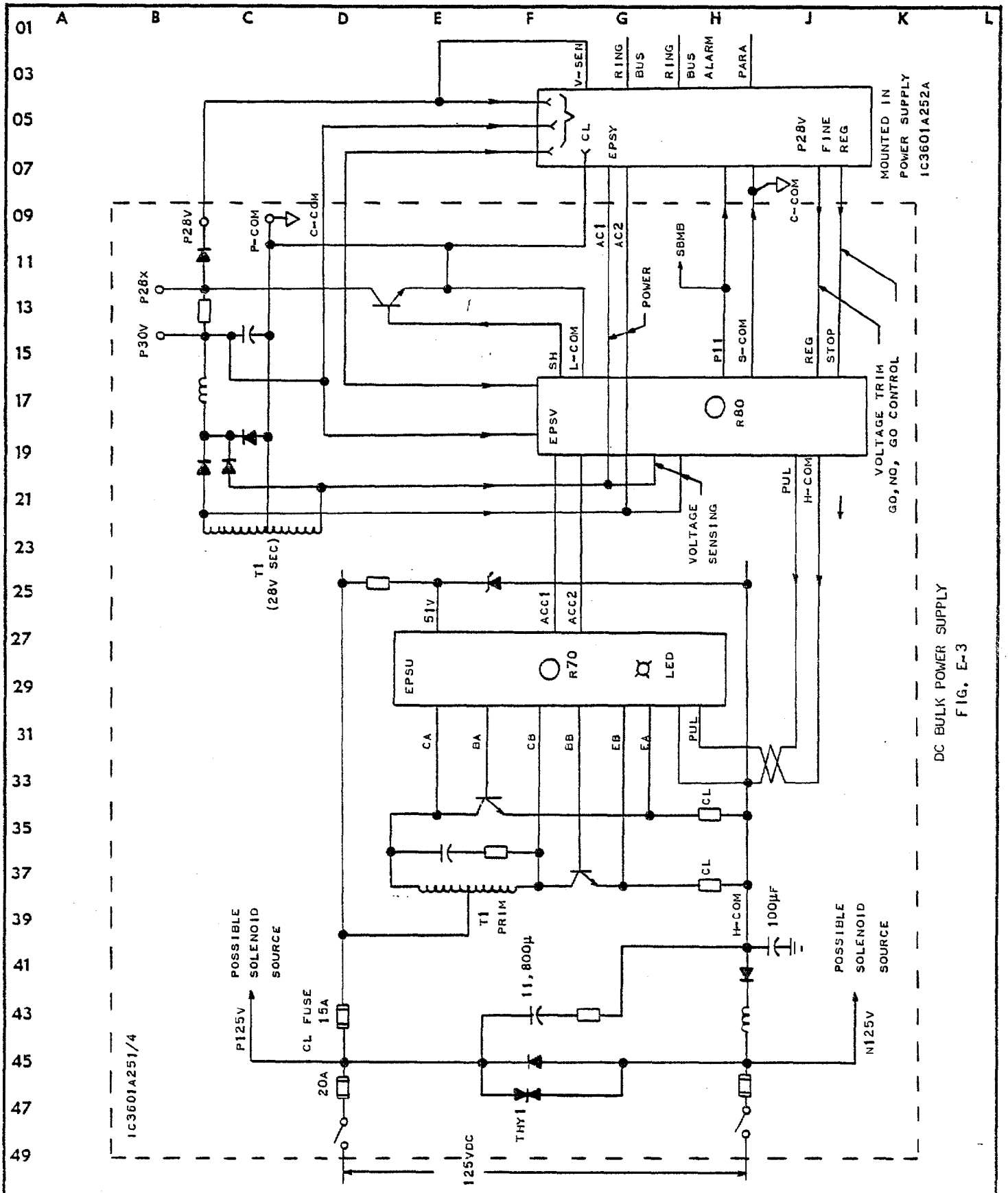


SOURCE SELECTOR FOR MARK II POWER SUPPLY (WHEN USED)

FIG. E-2

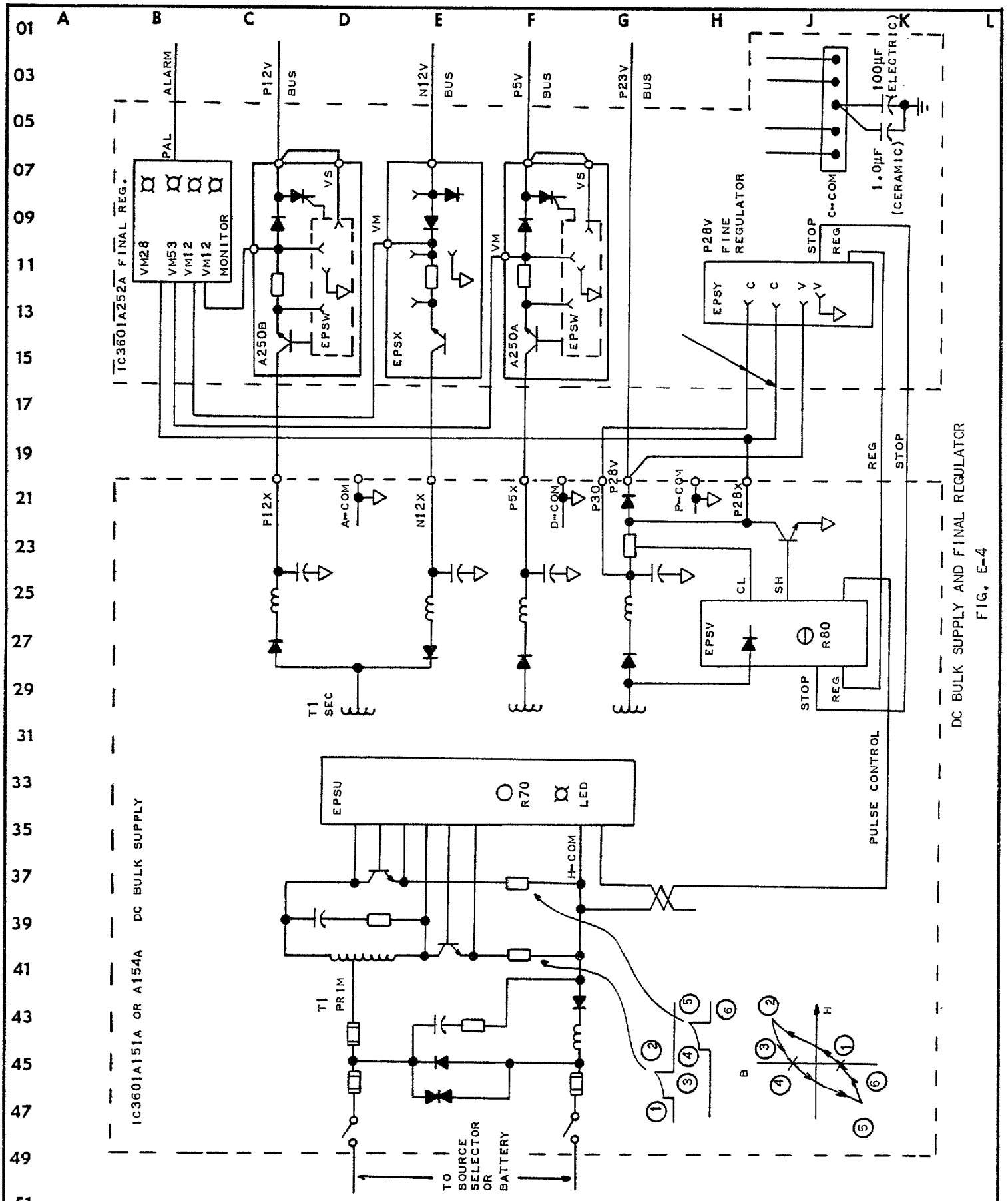
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SOURCE SELECTOR E-2
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L. M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053 CONT. ON SH. 114 SH. NO. 113

FORM 1C 2390 4-74



REV. 1	REV. 4	REV. 3	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-3
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		DC BULK POWER SUPPLY
REV. 3	MADE BY L.M. Worley			I.C. NO.	SALEM, VA. U.S.A.	GEK-41053
						CONT. ON SH. 115 SH. NO. 114

FORM 1C 2390 4-74



REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM E-4 DC BULK SUPPLY AND FINAL REGULATOR
REV. 2	ISSUED 4-9-75			FIRST MADE FOR		
REV. 3	MADE BY L.M. Worley			I.C. NO	SALEM, VA. U.S.A.	GEK-41053 CONT ON SH. FINAL SH NO. 115

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
CONTROL DEVICES OPERATION • WAYNESBORO, VIRGINIA 22980

GENERAL ELECTRIC

1/77 (750)